

When Do Stablecoins Crowd Out Bank Collateral?

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Abstract

Stablecoin issuers hold the same short-term safe assets that banks must hold to meet liquidity rules; we ask when issuance starts to crowd out bank collateral. Theoretically, we model banks that comply with the liquidity coverage ratio every day at a cost—covering shortfalls with term funding or the discount window up to their pledgeable collateral and deleveraging beyond it. The price of collateral then rises smoothly with issuance, and the regulatory threshold is the midpoint of a crowding-out zone rather than its start. Two channels operate: issuers’ bill purchases remove collateral from the system, and issuer deposits raise the precautionary demand of the banks that receive them; bills crowd out more than deposits. Because coin demand rises as bill yields fall, there can be a fold, a largest demand beyond which issuance growing from today’s stock does not settle; where it lies depends on how issuers and coin holders respond. In the US calibration the premium on safe assets is 3 basis points at today’s stock of \$311 billion, 25 at a total stock of \$661 billion and 100 at \$1,036 billion, against a regulatory threshold of \$961 billion. With the observed reserve mix the fold is at a demand of \$1,949 billion; it moves to \$2.0–3.1 trillion when issuers shift toward deposits, which pay more than bills once collateral is scarce, and it disappears when coin holders’ alternative is a bank deposit. Discount-window readiness and a Federal Reserve deposit share for issuers that is lent back to banks raise welfare most and are tied at \$1 trillion of demand; reserves held as bank deposits come next; an issuance cap lowers welfare wherever it binds; and run-off rates on issuer deposits and concentration limits mainly address the run externality. Empirically, every bank that held Circle’s cash before March 2023 was outside the liquidity coverage ratio; Silvergate held liquidity against its issuer deposits as a narrow bank would, and the model’s capital condition singles it out among partner banks; the ratios of the twenty-one bank holding companies under the rule have not fallen below 1.04 since 2018, as a probabilistic compliance cost implies; and bill-buying issuer dollars lower three-month bill yields by 10.8–19.1 basis points per \$3.5 billion for a few weeks, while the same issuers’ repo and deposit dollars do not.

JEL: E42, E44, E58, G21, G23, G28.

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1 Introduction

Stablecoins are private money backed by the assets banks use as liquidity buffers. In October 2026, dollar stablecoins outstanding reached \$311 billion, and at the end of 2025 their issuers held about \$200 billion of Treasury bills and bill-collateralized repo—1.8 percent of marketable bills—while the Treasury’s advisory committee projects \$2 trillion of stablecoins and roughly \$1 trillion of issuer bill holdings by 2028 (Treasury Borrowing Advisory Committee, 2025). The same bills sit in the high-quality liquid asset (HQLA) buffers that banks must hold against stressed outflows. This paper asks when a new class of money that claims those assets starts to crowd out the collateral banks hold for liquidity, what that does to bank credit, and which of the rules now being written—on issuers’ reserves, on the deposits they place with banks and on the banks that receive them—can prevent it.

The existing answer, in the general-equilibrium model of Hofmann et al. (2026) and in the empirical work of Ahmed and Aldasoro (2026), is a price channel: issuer demand lowers bill yields along a convenience-yield schedule, and the effect on lending is small. We show that this price channel is the early part of a crowding-out zone, once banks’ compliance with the liquidity coverage ratio (LCR) is modelled as it works in practice. A bank must meet the ratio every business day (Board of Governors of the Federal Reserve System, 2026). Its requirement moves with daily outflows, so it holds a buffer above the floor and, when a shortfall threatens, buys compliance with term funding or at the discount window—at the discount-window spread plus the stigma of borrowing—up to the collateral it can pledge, deleveraging beyond that. In a stress it can cover part of its actual outflows at the discount window and must liquidate loans for the rest. Each of these costs is a probability times a unit cost, as in the reserve-management model of Poole (1968) and the LCR model of Bech and Keister (2017), so the marginal value of a safe asset rises smoothly as the system’s slack falls. The kinked model of the earlier literature, with a regulatory threshold and a jump in the price of collateral, is the limit of this one as the daily shocks vanish.

Stablecoins crowd out bank collateral through two channels. Issuers’ bill and repo purchases remove collateral from the banking system. Issuer deposits, which the rule treats as runnable, raise the requirement of the banks that receive them, and those banks hold safe assets against them: a bank whose funding is dominated by issuer deposits becomes a narrow bank, as Lee and Tou (2026) document for the banks that hold stablecoin reserves and as Silvergate’s balance sheet shows. In the US calibration the premium on safe assets is 3 basis points at today’s stock of \$311 billion. It reaches 25 basis points at a total stock of \$661 billion, about 86 at the regulatory threshold of \$961 billion—the point at which the system’s disclosed slack is used up, which is the midpoint of the zone—and 158 with \$1 trillion of new

issuance. Lending falls by \$64 billion at that point and the expected cost of stress liquidations rises to \$4.1 billion a year. With the observed reserve mix of mostly bills, the premium is 2.4 times and the lending loss 2.4 times what they would be if the same reserves were bank deposits, because a bill purchase removes a unit of collateral while a deposit adds only its actual run-off to stress outflows.

Coin demand rises as bill yields fall, and collateral scarcity lowers bill yields. The feedback has a sharp consequence. Equilibrium requires the demand for coins at today's yield, ℓ , to equal the stock that the yield it implies would support, $h(S)$; h rises with the stock as long as one more coin lowers the yield by less than it raises demand, and falls once the deleveraging region steepens the yield. The largest demand with a normal equilibrium—the one that issuance growing from today's stock reaches—is the first peak of h , ℓ^* . In the US calibration with a fixed bill supply and the observed reserve mix, ℓ^* is about \$1,949 billion: the \$2 trillion projection for 2028, read as demand at today's yield, lies just beyond it. Beyond the fold there is no normal equilibrium: issuance growing from today's stock does not settle, and in the calibration no stable equilibrium exists at all up to a stock of \$6.3 trillion, so the stock runs into the region of forced deleveraging and collapsing bill yields. ℓ^* moves most with the collateral banks can pledge for term funding and at the discount window, from \$1,414 billion when it covers 30 percent of their requirement to \$2,417 billion at 60 percent, and a modest bill-supply response by the Treasury raises it to \$2,256 billion. These numbers hold the issuers' reserve mix fixed. Banks pay for issuer deposits what the deposits earn net of the liquidity cost they bring, which counts only their actual run-off in a stress, while a bill carries the full premium on collateral; as collateral grows scarce the bill return therefore falls below the deposit rate, by 55 basis points at the \$1 trillion demand level, and issuers that rebalance toward deposits absorb less collateral. With the reallocation elasticity we estimate for Circle the fold moves to \$3,146 billion, and with a tenth of it to \$2,022 billion; if coin holders' alternative were a bank deposit rather than a money-market yield, there would be no fold. We therefore report the fold as a result conditional on how issuers and holders respond. The crowding-out zone and the ranking of the instruments do not depend on it.

The results are propositions. Lemma 4.1 gives a unique equilibrium price for every level of issuance, with no exception at the threshold. Proposition 4.1 characterizes the crowding-out zone: an asset swap far from the threshold, a premium that rises smoothly and is half its regulatory maximum at the threshold, and the kinked model as the limit. Proposition 4.2 ranks the two channels: bill reserves crowd out more than deposit reserves whenever the regulatory run-off on issuer deposits does not exceed the collateral value of a bill, and a bank that receives issuer deposits holds safe assets against them. Proposition 4.3 gives the run-bank loop and Proposition 4.4 the issuer's reserve choice. Proposition 5.1 shows that a

recycled requirement to hold part of the reserves as deposits at the Federal Reserve—an asset the GENIUS Act already allows (U.S. Congress, 2025)—matched so that the issuer holds the same bonds and deposits as under an issuance cap, lowers the price of collateral and raises lending relative to the cap, and keeps the convenience value of the coins the cap removes. Proposition 5.2 characterizes the fold: the normal equilibrium exists if and only if demand at today’s yield is below ℓ^* , and ℓ^* rises when the yield path is flatter. Proposition 5.3 gives the run-off rate that shuts the loop and the capital condition that no run-off rate can replace.

The policy ranking follows. Against current rules, the recycled Federal Reserve deposit share and discount-window readiness raise welfare by \$1.3 and \$1.3 billion a year at \$1 trillion of demand and by \$15.2 and \$12.3 billion at \$2 trillion with a modest bill-supply response; reserves held as bank deposits come next; an issuance cap lowers welfare wherever it binds and in every calibration, by as much as \$197 billion a year at \$2 trillion with the estimated demand curve, because it rations coins that households demand. Without recycling the Federal Reserve deposit share has no value, and it beats discount-window readiness only if the Federal Reserve lends back nearly all of the deposits, more than 94.6 percent at \$2 trillion, and only if holding them costs the Federal Reserve less than 21 basis points a year; discount-window readiness needs no such condition, and at \$1 trillion the two are effectively tied. The run-off rate on issuer deposits and the 40 percent concentration limit the FDIC has proposed (Federal Deposit Insurance Corporation, 2026) do little for collateral while issuers hold bills; their case is the run externality, where the binding quantity is the receiving bank’s capital.

The evidence follows the model’s structure. First, none of the banks that held Circle’s cash before March 2023 (Silicon Valley Bank, Signature, Silvergate, Customers, New York Community and Citizens Trust) was subject to the LCR: Silicon Valley Bank because its holding company’s short-term wholesale funding was below the threshold of the 2019 tailoring rule, Signature because it had no holding company, and the others because they were below the rule’s size threshold (Congressional Research Service, 2023). Afterwards the cash moved to a bank subject to the full ratio. Second, Silvergate held cash and securities equal to 0.87 of its assets against a peer median of 0.27 and had a loan share 55 percentage points below theirs; its excess liquidity implies an internal run-off of 0.74 on its digital-asset deposits, which we use in the calibration. Banks for which such deposits were a small part of funding show no excess liquidity. Third, the partner-bank failures line up with the capital condition: Silvergate, with issuer-type deposits 10.7 times its equity and a realized fire-sale discount of 13.8 percent against a threshold of 10 percent, is predicted to fail, and no other partner bank that disclosed its digital-asset deposits violates the condition in any quarter. Fourth, the disclosures of the twenty-one bank holding companies subject to the ratio show buffers

that have not fallen below 1.04 since 2018, as a probabilistic compliance cost implies, and leave \$650–678 billion of regulatory room. Fifth, in the observed range of the ratio the model’s premium varies by a few basis points, and we find no robust lending response to issuer absorption; bill-buying issuer dollars lower three-month bill yields by 10.8–19.1 basis points per \$3.5 billion for a few weeks, while the same issuers’ repo and deposit dollars do not.

Our contribution relative to the literature is a model of stablecoin reserves and bank liquidity in which compliance with the liquidity rules is costly in the way the rules make it costly, and in which the stock of coins is set by demand at the yield the model determines. Hofmann et al. (2026) and Cova et al. (2022) compare issuer regimes in calibrated DSGE models with smooth bank funding costs; Huang and Keister (2026) and Lee and Tou (2026) study liquidity-hungry issuer deposits and partner banks without system-wide collateral scarcity; Ma et al. (2026), Bertsch (2025) and Gorton et al. (2026) model runs with the reserve-asset market exogenous. The cost of an LCR shortfall follows Bech and Keister (2017), who study its implications for the implementation of monetary policy; the nonlinear demand for reserves is estimated by Afonso et al. (2022) and López-Salido and Vissing-Jorgensen (2023); Sundaresan and Xiao (2024) show that banks meet the ratio with Federal Home Loan Bank advances, the term funding that bounds the cost of compliance here; and Banerjee and Mio (2018) and Bonner and Eijffinger (2016) show that banks respond to liquidity rules by recomposing their balance sheets. The kink in the price of bank liquidity in September 2019 is documented by Copeland et al. (2025); we place a new absorber in front of it. Coste (2025) documents that the Basel framework admits three run-off treatments for issuer deposits without a criterion for choosing among them; we give the criterion and show that for the banks where the risk materialized the question was not the run-off rate but whether any liquidity requirement applied. Table 1 sets the model beside the closest ones and lists the predictions that follow only from putting a costly liquidity rule, an endogenous market for collateral and a demand-determined stock of coins together.

The paper also draws on three older literatures. The convenience yield on Treasury bills and its slope in supply are estimated by Krishnamurthy and Vissing-Jorgensen (2012), Greenwood et al. (2015) and Nagel (2016); our premium at today’s stock is of the same order. The smooth channels of Hofmann et al. (2026), Ferrari Minesso and Siena (2026) and Altavilla et al. (2026) run through deposit rates, bank funding costs and adoption; they are complementary to the collateral channel studied here. The collateral role of safe assets in bank liquidity management is the subject of Kashyap et al. (2002), Stein (2012), Diamond and Kashyap (2016), Bianchi and Bigio (2022) and, for the safe-asset supply itself, He et al. (2019); the pecuniary externality through collateral prices is the one studied by Lorenzoni

Table 1: The closest models and what this paper adds.

Paper	Bank liquidity friction	Market for safe assets	Implication for bank collateral and credit
Hofmann et al. (2026); Cova et al. (2022)	Smooth bank funding costs	Convenience yield	A price channel: issuer demand lowers bill yields; small lending effect
Ahmed and Aldasoro (2026)	— (empirical)	Bill yields respond to issuer flows	Price effect of issuer flows
Lee and Tou (2026); Huang and Keister (2026)	Partner banks' liquidity needs	No system-wide collateral scarcity	Banks holding issuer deposits keep more liquidity and lend less
Ma et al. (2026); Bertsch (2025); Gorton et al. (2026)	—	Reserve-asset market exogenous	Run risk and peg stability
Bech and Keister (2017)	Probabilistic cost of an LCR shortfall	Interbank and term rates	A regulatory premium in term rates; no stablecoins
This paper	Daily LCR compliance at a cost, bounded by pledgeable collateral (Section 3)	Endogenous price of collateral; coin stock set by demand at that price	(1) a crowding-out zone centred on the disclosed slack; (2) bill reserves crowd out more than deposit reserves, and the receiving bank holds collateral against issuer deposits; (3) a largest demand with a normal equilibrium (the fold); (4) partner failure as a bank-specific capital condition; (5) a recycled Federal Reserve deposit share beats an issuance cap that leaves the issuer with the same holdings

(2008). On stablecoins specifically, Liao and Caramichael (2022) and Goel et al. (2026) study bank disintermediation and reserve regulation, Barthélemy et al. (2026) show that commercial-paper issuers catered to issuer demand in 2021, Lyons and Viswanath-Natraj (2023), d'Avernas et al. (2026), Routledge and Zetlin-Jones (2022) and Uhlig (2022) study peg maintenance and its failures, Gorton and Zhang (2023) and Jackson et al. (2022) the legal design, Eichengreen et al. (2025) devaluation risk, and He et al. (2026), Azar et al. (2026) and Ahmed et al. (2025) the macroeconomic and international dimensions. Our slack state is the benchmark in which stablecoins leave credit unchanged, as in Agur et al. (2026); the capital-constraint nonlinearity of Kim (2026) and the adoption nonlinearity that Altavilla et al. (2026) estimate for euro-area banks are different objects.

The paper proceeds as follows. Section 2 sets out the institutions and facts and Section 3 the model. Section 4 derives the crowding-out zone, its two channels, the run–bank loop and the issuer's reserve choice, and Section 5 the instruments, the fold and the run-off rule. Section 6 takes the model to the United States and Section 7 presents the evidence. Section 8

concludes. Proofs are in Appendix A and the data in Appendix B; Appendix C gives the run block, Appendix D further evidence, Appendix E bank ownership of the issuer and market structure, Appendix F the linear benchmark of the earlier version of this paper, and Appendix G the sensitivity of the policy comparison.

2 Institutions and facts

Who holds the bills. Dollar stablecoins outstanding were \$311 billion in October 2026 (DefiLlama). The two largest issuers disclose reserve composition in independent attestations: Tether quarterly since 2021 (Moore Cayman, later BDO), Circle monthly since 2018 (Grant Thornton, later Deloitte). Parsing all 118 reports gives the reserve mix in Figure B.1 and Table B.1. Tether held 2 percent of reserves in Treasury bills in March 2021 and 61–68 percent (plus 10–14 percent repo and 4–9 percent money funds) from 2023 onward, having liquidated \$30 billion of commercial paper after the Terra collapse. Circle held 100 percent bank deposits until mid-2021, moved to bills and deposits in 2022 (the \$3.3 billion at Silicon Valley Bank was part of the remaining deposit sleeve), and since 2023 holds reserves in the Circle Reserve Fund, a government money market fund whose split between bills and Treasury repo has ranged month to month from all repo to all bills. At end-2025 the two issuers’ bill-like holdings were about \$200 billion, 1.8 percent of marketable bills; the Treasury Borrowing Advisory Committee’s April 2025 presentation projects \$2 trillion of stablecoins and about \$1 trillion of issuer bill demand by 2028.

Where the bills come from. The bank holding companies subject to the liquidity coverage ratio disclose \$4.54 trillion of high-quality liquid assets against \$3.91 trillion of net 30-day outflows (2026Q2, twenty-one companies); the excess over the 100 percent floor, \$630 billion, is the slack in the collateral constraint of Section 3. Commercial banks as a whole hold \$4.8 trillion of Treasury and agency securities and \$2.9 trillion of reserves (H.8, H.4.1, 2026), but not all of it is eligible HQLA and not all of them are subject to the rule. Money market funds are the other holder; from 2021 to 2024 they also held up to \$2.5 trillion in the Federal Reserve’s overnight reverse repo facility, which supplied bills’ close substitute elastically at the floor rate until the facility drained to near zero in early 2025.

The run-off rule. The liquidity coverage ratio assigns outflow rates to deposits by counterparty: 3–10 percent for retail, 25 percent for operational wholesale deposits, 40 percent for non-operational deposits of non-financial corporates, 100 percent for unsecured funding from financial institutions (Basel Committee on Banking Supervision, 2013). Deposits placed by

a stablecoin issuer have no explicit category. Coste (2025) shows that three treatments are admissible under the Basel and EU texts—10 percent if the bank can identify retail holders, 25 percent as an operational deposit, 100 percent as a financial-institution deposit—and that supervisors have not chosen among them. Section 5 derives the rate that should apply.

Who is subject to the rule. The ratio does not apply to every bank. Under the 2019 tailoring rule a bank holding company with less than \$50 billion of weighted short-term wholesale funding is not subject to it, companies with \$50–75 billion meet 70 or 85 percent of it, and banks with less than \$100 billion of assets are outside the categories altogether (Board of Governors of the Federal Reserve System et al., 2019). Compliance is measured every business day: a company whose ratio falls below the minimum must notify its supervisor that day and, after three consecutive business days below it, submit a plan to restore compliance (Board of Governors of the Federal Reserve System, 2026). Banks therefore manage the ratio to a buffer. No company in our panel reported a ratio below 1.04 in any quarter since 2018, and only 0.6 percent of bank-quarters were below 1.05. A bank that needs liquidity at short notice can borrow at the discount window, at the primary credit rate, 10 basis points above interest on reserves since 2021, and against the stigma of doing so, which Armantier et al. (2011) estimate at about 44 basis points (126 after the failure of Lehman Brothers). In 2023 banks had pledged collateral with a lendable value of \$2.6 trillion at the window, \$1.8 trillion of it loans (Board of Governors of the Federal Reserve System, 2025).

Where issuers can hold reserves. The GENIUS Act requires payment-stablecoin issuers to hold reserves one for one in cash, deposits at insured banks, Treasury bills of 93 days or less, repurchase agreements backed by such bills, government money market funds, and deposits at the Federal Reserve Banks (U.S. Congress, 2025). Issuers hold almost no bank deposits today: 4.6 percent of USDT and USDC reserves, \$14.5 billion on the current stock, almost all of it Circle’s cash. Before March 2023 that cash sat at banks outside the liquidity coverage ratio; afterwards it moved to the custodian of Circle’s reserve fund, a bank subject to the full ratio (Section 7.1).

3 Model

Environment. Three dates, $t = 0, 1, 2$; risk-neutral agents; no discounting. Households have endowment e at $t = 0$ and hold it as bank deposits D^H , paying R_D at $t = 2$, or as stablecoins s , which pay no interest; $D^H = e - s$. Two banks $j = 1, 2$ (bank 1 is the issuer’s partner; the symmetric case has equal shares) take deposits, hold safe assets b_j at price q_0

and make loans L_j . Banks are price takers in the loan market: the loan rate $R(L)$, with $R' < 0$, clears at aggregate lending $L = L_1 + L_2$, and bank j earns $R(L)L_j$ on its book. The government sells $\bar{S}$ units of safe assets at $t = 0$ and repays them at par at $t = 2$; the extension $S(q_0) = \bar{S}(q_0/\bar{q})^{\varepsilon_S}$ allows elastic supply. A stablecoin issuer sells coins at par, holds a share α of reserves in safe assets and $1 - \alpha$ in deposits at bank 1 (paying $R_D^S \geq R_D$), and is not owned by banks; Appendix E lets banks own it.

Coin demand. Coins are held for the payment and settlement services they provide. Let s_0 be the stock outstanding at the start, which with its reserves is already on the initial balance sheets, and write s for issuance beyond it, so that the total stock is $S = s_0 + s$. At an opportunity cost r , the yield the marginal holder forgoes, and at an adoption level k , households demand a total stock

$$S^d(r; k) = k s_0 e^{-\varepsilon(r-r_0)}, \quad (3.1)$$

where r_0 is today's bill yield, so that $S^d(r_0; 1) = s_0$, and ε is the semi-elasticity of demand to the opportunity cost, which Section 6 estimates. The opportunity cost is the bill yield, $r = y \equiv 1/q_0 - 1$, which the model determines. The issuer mints and redeems at par on demand, so issuance is $s = \min\{\max\{S^d(y; k) - s_0, 0\}, \bar{s}\}$, where $\bar{s}$ is a cap if the regulator imposes one; because the yield depends on issuance, this is a fixed point, which Sections 6.4 and 6.5 solve. The social value of the coins issued beyond s_0 is the area under the inverse demand curve, $V(s) = \int_{s_0}^{s_0+s} r(u) du$ with $r(u) = r_0 - \ln(u/ks_0)/\varepsilon$, up to the quantity at which the marginal holder's valuation reaches zero. The issuer's remaining choice is the reserve mix α , which applies to its whole stock; Appendix E analyzes the quantity it would choose if demand did not bind, and the bank-ownership structures that would shape that choice.

Stress and collateral. With probability p a stress state occurs at $t = 1$. In it households withdraw a fraction $\tilde{\gamma}$ of their deposits, other net outflows $N_{0,j}$ (wholesale funding, committed lines) are fixed, and the issuer withdraws a share λ_a of its bank deposits to meet redemptions; λ_a is the actual 30-day run-off of issuer reserves in a stress without a run, a property of holders' behavior. The benchmark stress distribution is two-point: $\tilde{\gamma} = \gamma_{\max}$ in the stress state and zero otherwise. Banks meet outflows by pledging safe assets at their collateral value θ per unit, the haircut-adjusted value at which the central bank or the repo market lends against them; the actual shortfall

$$X_j = \tilde{\gamma} D_j^H + \lambda_a D_j^S + N_{0,j} - \theta b_j, \quad (3.2)$$

if positive, is covered by liquidating loans at a loss φ per dollar raised.

Liquidity regulation. The liquidity coverage ratio requires bank j to hold high-quality liquid assets covering a share ℓ_j of its requirement

$$\text{REQ}_j = \gamma_{\max} D_j^H + \lambda_j D_j^S + N_{0,j}, \quad (3.3)$$

where λ_j is the run-off assigned to issuer deposits and ℓ_j is the share of the requirement the rule makes the bank cover: one under the full ratio, 0.70 or 0.85 under the reduced ratios of the tailoring rule. For a bank subject to the rule $\lambda_j = \lambda_{\text{reg}}$, which the regulator sets and which need not equal λ_a . A bank outside the rule has no requirement but manages its liquidity against an internal run-off, $\lambda_j = \lambda_{\text{int}}$, which we model as a self-imposed requirement with $\ell_j = 1$. Compliance is measured every business day. The requirement is hit by a daily shock ξ_j with a continuous distribution $F(\cdot/\sigma_j)$, symmetric around zero and with full support, and a bank that falls short must cover the gap. It can do so with term funding or at the discount window, at a cost c per dollar—the window’s rate over interest on reserves plus the stigma of borrowing—up to a capacity $\kappa_j \ell_j \text{REQ}_j$ set by the collateral it can pledge; beyond that capacity it must deleverage, liquidating loans at the cost φ per dollar. In the stress state the bank must meet its actual outflow

$$\text{NEED}_j = \gamma_{\max} D_j^H + \lambda_a D_j^S + N_{0,j} : \quad (3.4)$$

a share ω_j of any actual shortfall is covered at the discount window at the crisis cost φ_{DW} and the rest by liquidating loans at φ . The bank’s expected liquidity cost is

$$\mathcal{C}_j(b) = c \mathbb{E}[(\ell_j \text{REQ}_j + \xi_j - \theta b)^+] + \varphi \mathbb{E}[((1 - \kappa_j) \ell_j \text{REQ}_j + \xi_j - \theta b)^+] + \tilde{\tau}_{a,j} \mathbb{E}[(\text{NEED}_j + \xi_j - \theta b)^+], \quad (3.5)$$

with $\tilde{\tau}_{a,j} \equiv p[\omega_j \varphi_{DW} + (1 - \omega_j) \varphi]$ the expected cost of a dollar of actual stress shortfall. The first term is the price of compliance and is paid to the Federal Reserve or the lender of term funds: it is a transfer, except for the stigma. The second and third are real: they destroy loans. The slack of the system is the excess of collateral over the sum of the requirements, $\Sigma \equiv \theta \sum_j b_j - \sum_j \ell_j \text{REQ}_j$, which is what the banks’ disclosures measure.

The run block. Coin holders can redeem at par. In a run a share λ of coins is redeemed; the issuer meets redemptions pro rata from its reserve components, selling safe assets at the stress price $q_1 = 1 - \bar{h} \leq 1$ and drawing its bank deposits. The issuer’s liquidation cost per redeemed dollar is $\phi(\lambda) = \alpha \bar{h}$ while the partner bank stands, and $\alpha \bar{h} + (1 - \alpha) \text{LGD}$ once

redemptions push the partner into failure, where LGD is the loss on uninsured deposits. The partner fails when the cash it can raise—its safe assets sold at q_1 , with the fire-sale loss $1 - q_1$ per unit charged to equity, and loans liquidated at the cost φ per dollar until equity is exhausted—falls short of the deposits it must repay; this defines the failure threshold $\hat{\lambda}$ of equation (4.5) below.¹ Redemptions follow the global game of Ma et al. (2026): holders observe noisy signals of a fundamental π and redeem if their signal is below a threshold; arbitrageurs with redemption access K absorb sales. Lemma C.1 shows that the run threshold π^* is unique for the step function $\phi(\lambda)$. The run probability is $\rho = p \Pr[\pi < \pi^*]$, with π drawn from a Beta prior calibrated to the two issuers' estimated run probabilities (Appendix C). The run covers the whole stock S , including the coins outstanding at the start, whose reserves sit on the initial balance sheets with the same mix α . The regulatory rate λ_{reg} determines how much HQLA the partner holds against issuer deposits before the stress; it does not change how much holders redeem, which is λ_a in a stress without a run and the run share λ in a run.

Timing and equilibrium. At $t = 0$ issuance is set by demand at the bill yield and any cap; the issuer chooses its reserve mix α (in the baseline calibration we hold it at the observed mix; Section 6.4 lets it respond); banks choose (b_j, L_j) taking q_0 and $R(L)$ as given; the safe-asset market clears, $b_1 + b_2 + \alpha s/q_0 = S(q_0)$, where $S(q) = \bar{S}(q/\bar{q})^{\varepsilon_s}$ allows the Treasury's bill supply to respond to the price. At $t = 1$ the daily compliance shocks and the stress state, the run game and bank failures are realized. At $t = 2$ payoffs accrue. The equilibrium objects are the price q_0 , lending L , bank holdings b_j , the run threshold π^* and the failure threshold $\hat{\lambda}$. Let q_F be the frictionless price at which $q_F R(L) = 1$, $y_F = 1/q_F - 1$ the corresponding bill yield, and $w(\alpha) \equiv \theta\alpha/q_F + \lambda_{\text{reg}}(1 - \alpha) - \gamma_{\text{max}}$ the requirement slack a coin absorbs at that price. Two assumptions carry the analysis.

Assumption 1. $\theta/q_F > \gamma_{\text{max}}$ and $\lambda_a > \gamma_{\text{max}}$: a unit of safe assets absorbed by the issuer, or a dollar of issuer deposits, raises stress outflows relative to collateral by more than the household deposit it drains.

Assumption 2. $\lambda_{\text{reg}} > \gamma_{\text{max}}$: the rule assigns issuer deposits a higher run-off than retail deposits.

What the model leaves out. Deposit rates are fixed, so the deposit-spread channel of Hofmann et al. (2026) is switched off; that channel is a smooth and separate effect of issuance

¹In the aggregate stress state banks pledge collateral at θ ; a partner facing a coin run must sell to repay and realizes the stress price q_1 . The distinction is the one Silvergate's accounts display: it sold securities at a loss rather than borrowing against them (Section 7).

on bank funding costs, and the model here complements it rather than nesting it. Section 6.4 relaxes this in two ways: the rate on issuer deposits follows the receiving bank's marginal value of them, which moves the issuer's reserve mix, and the coin holders' opportunity cost may include the deposit rate. There is one safe asset; the distinction between bills and repo in the data maps to α , and reserves and bills are not distinguished, so the intraday role of reserves documented by Copeland et al. (2025) is outside the model. The Treasury's bill supply is fixed in the baseline and responds with elasticity ε_S in a variant; the Federal Reserve's balance sheet is held fixed. The opportunity cost of holding coins is the bill yield, the money-market return their holders forgo. The issuer does not choose its quantity in the main text; Appendix E analyzes an issuer that could.

4 The price of bank collateral

This section asks how the price of the safe assets banks hold for liquidity, and with it bank lending, responds to stablecoin issuance. We first derive banks' demand for safe assets, then the crowding-out zone, then the two channels through which issuance enters it, and finally the run block and the issuer's reserve choice.

4.1 Banks' demand for safe assets

A bank enters $t = 0$ with household deposits D_j^H , issuer deposits D_j^S and equity E_j , and chooses safe assets b and loans L_j subject to $q_0 b + L_j = D_j^H + D_j^S + E_j$, taking the loan rate $R(L)$ and the price q_0 as given, to maximize

$$\Pi_{B,j} = R(L)L_j + b - R_D D_j^H - R_D^S D_j^S - \mathcal{C}_j(b), \quad (4.1)$$

with $\mathcal{C}_j$ the liquidity cost (3.5). Write $\bar{F}(z) \equiv 1 - F(z)$. The first-order condition for b is

$$q_0 R(L) = m_j(b) \equiv 1 + \theta \left[c \bar{F}\left(\frac{\theta b - \ell_j \text{REQ}_j}{\sigma_j}\right) + \varphi \bar{F}\left(\frac{\theta b - (1 - \kappa_j) \ell_j \text{REQ}_j}{\sigma_j}\right) + \tilde{\tau}_{a,j} \bar{F}\left(\frac{\theta b - \text{NEED}_j}{\sigma_j}\right) \right]. \quad (4.2)$$

The price a bank pays for a safe asset, measured in units of forgone loan return, equals one plus the expected marginal saving in compliance, deleveraging and stress costs. Each term is a probability times a unit cost, as in the reserve-management model of Poole (1968) and the liquidity-coverage model of Bech and Keister (2017). The marginal value m_j falls continuously from $1 + \theta(c + \varphi + \tilde{\tau}_{a,j})$, when the bank would be short with certainty, to one,

when its holdings far exceed every threshold. Market clearing for safe assets and the banks' and issuer's balance sheets give

$$\sum_j b_j = S(q_0) - \alpha s / q_0, \quad L = e + E - q_0 S(q_0), \quad (4.3)$$

so lending falls one for one with the value of the safe-asset stock the economy pays for. Write $\pi \equiv q_0 R(L) - 1$ for the premium on safe assets.

Lemma 4.1 (existence and uniqueness). *For every issuance s and reserve mix α there is a unique equilibrium price $q_0(s)$, and hence a unique allocation (L, b_1, b_2) ; q_0 , L and the premium π are continuous in s .*

The argument is that each bank's demand $b_j(M)$, the inverse of m_j , is continuous and strictly decreasing in $M = q_0 R(L)$ on $(1, 1 + \theta(c + \varphi + \tilde{\tau}_{a,j}))$; M is strictly increasing in q_0 ; and the supply of safe assets to banks, $S(q_0) - \alpha s / q_0$, is increasing in q_0 . Excess demand is therefore strictly decreasing in the price and has a single zero. The kinked model of the earlier literature, and of the earlier version of this paper, had an interval of equilibrium prices at the threshold when the issuer held only deposits; with daily shocks of full support the interval disappears.

4.2 The crowding-out zone

In the frictionless state the premium is zero and the price is q_F . Issuance moves the system toward its requirements: at the frictionless price a coin absorbs $w(\alpha) = \theta\alpha/q_F + \lambda_{\text{reg}}(1 - \alpha) - \gamma_{\text{max}}$ of slack, through the bond the issuer buys, the issuer deposit the rule treats as runnable and the household deposit it drains. The issuance at which the system's slack Σ_0 is used up is

$$s_1 = \frac{\Sigma_0}{\theta\alpha/q_F + \lambda_{\text{reg}}(1 - \alpha) - \gamma_{\text{max}}}. \quad (4.4)$$

Under the liquidity coverage ratio Σ_0 is the excess of HQLA over net 30-day outflows of the banks subject to the rule, and the denominator is the LCR weight of the marginal coin: one for an issuer deposit under the current 100 percent run-off or for a bill purchase ($\theta/q_F \approx 1$), minus the 3–10 percent weight of the retail deposit it replaces. In the kinked model s_1 is a threshold: below it issuance is an asset swap and above it the price of collateral jumps. With daily compliance shocks it is the midpoint of a zone.

Proposition 4.1 (the crowding-out zone). *Let banks be symmetric, with $\ell_j = 1$, $\sigma_j = \sigma$, $\kappa_j = \kappa$ and $\omega_j = \omega$, let F be symmetric, let Assumptions 1–2 hold, and let the bill supply be fixed.*

- (i) *(Asset swap far from the threshold.) The premium is bounded by $\pi(s) \leq \theta(c + \varphi + \tilde{\tau}_a) \bar{F}(\Sigma(s)/\sigma)$, where $\Sigma(s)$ is the equilibrium slack; while $\Sigma(s)$ is large relative to σ , issuance moves the price of collateral and lending by a negligible amount.*
- (ii) *(Monotonicity.) The premium $\pi(s)$ is continuous and strictly increasing in issuance, and lending $L(s)$ is continuous and strictly decreasing.*
- (iii) *(The threshold is the midpoint.) At $s = s_1$ the compliance term of the premium equals $\theta c/2$, half its maximum, up to a term proportional to the change in the price of bonds; the compliance term reaches a share η of θc where the equilibrium slack is $\Sigma = \sigma \bar{F}^{-1}(\eta)$; the zone is centred on s_1 and its width is proportional to $\sigma/w(\alpha)$.*
- (iv) *(The kinked limit.) As $\sigma \rightarrow 0$ the premium converges, at every s other than the thresholds, to the step function of the kinked model: zero below s_1 , θc between s_1 and the issuance at which the system's holdings reach its actual needs, and $\theta(c + \tilde{\tau}_a)$ beyond, with a further step of $\theta\varphi$ where the term-funding capacity is exhausted.*

Part (i) is the asset swap of the price channel: when the issuer buys bonds, a household deposit leaves a bank and the issuer's bond purchase is financed by that bank selling the bond; with the probability of a shortfall negligible, the bank is nearly indifferent between the bond and a loan, so the price is pinned by the loan return and lending barely moves. Part (ii) says the price channel and the collateral channel are one schedule, not two regimes: there is no issuance at which the price of collateral starts to respond, only one at which its response becomes large. Part (iii) locates the zone. The regulatory threshold (4.4) survives as its centre, and the daily variability of the requirement sets how far in front of it crowding out begins. Part (iv) recovers the kinked model of Appendix F as the limit without daily shocks; it is the version in which the regulatory threshold is a cliff.

4.3 Two channels and the reserve mix

Issuance enters the zone through the issuer's balance sheet and through the balance sheets of the banks that receive its deposits.

Proposition 4.2 (bills, deposits and the bank channel). *Let Assumptions 1–2 hold and let the bill supply be fixed.*

- (i) *(Bills crowd out more.) If $\lambda_{\text{reg}} \leq \theta/q_F$, then at every level of issuance the premium is at least as high, and lending at least as low, when reserves are held as bills ($\alpha = 1$) as when they are held as bank deposits ($\alpha = 0$), and strictly so whenever the deleveraging or stress terms of (4.2) are active.*
- (ii) *(The bank channel.) A bank that is small relative to the system and receives issuer deposits D_1^S holds more safe assets and a smaller share of loans the more deposits it*

receives: $\partial b_1 / \partial D_1^S > 0$ and the loan share $L_1 / (L_1 + q_0 b_1)$ falls in D_1^S , at the rate set by its run-off $\lambda_1 - \lambda_{\text{reg}}$ if it is subject to the rule, λ_{int} if it is not.

The economics of part (i) is in the two thresholds of (4.2). A bill bought by the issuer removes θ/q_F of collateral from banks; a dollar of issuer deposits adds λ_{reg} to the requirement but only λ_a to the actual stress outflow. Under the current 100 percent run-off the two reserves absorb almost the same regulatory slack, but bills absorb far more of the margin against actual stress, so they reach the deleveraging and stress-loss terms sooner. The US rule satisfies the condition, since $\theta/q_F \approx 1.04$ exceeds $\lambda_{\text{reg}} = 1$. Part (ii) is the bank channel. The requirement of the bank that receives issuer deposits rises by λ_1 per dollar, and it holds safe assets against them: the narrow banking that Lee and Tou (2026) document for the banks holding stablecoin reserves, and that Silvergate’s balance sheet shows (Section 7.2). The bank channel crowds out collateral even when the issuer buys no bills.

4.4 Runs and the loop

The crowding-out zone describes a bank-side externality of issuance. The second externality runs through the issuer’s redemptions. Coin holders can redeem at par; in a run the issuer meets redemptions pro rata from its reserves, selling bonds at the stress price q_1 and drawing its deposits. The issuer’s liquidation cost per redeemed dollar is therefore $\phi(\lambda) = \alpha \bar{h}$, with $\bar{h} = 1 - q_1$, as long as the partner bank stands, and jumps to $\alpha \bar{h} + (1 - \alpha)\text{LGD}$ once redemptions push the partner bank into failure.

Appendix C solves the run game of Section 3 for this step function (Lemma C.1): the run threshold π^* is unique, falls with redemption access K , and rises when the step moves to a smaller redemption share, which is the property the loop uses.

The partner bank fails in a run when the deposits it must repay exceed the cash it can raise before its equity is exhausted. Selling its collateral at q_1 raises $q_1 \theta b_1$ and charges $(1 - q_1) \theta b_1$ to equity, which leaves $E_1 - (1 - q_1) \theta b_1$ to absorb the cost φ per dollar of loans liquidated; the bank fails if

$$\lambda > \hat{\lambda} \equiv \frac{E_1 / \varphi + \kappa_q \theta b_1 - \gamma_{\max} D_1^H - N_{0,1}}{(1 - \alpha)s}, \quad \kappa_q \equiv q_1 - \frac{1 - q_1}{\varphi}, \quad (4.5)$$

where κ_q collects the cash a unit of collateral raises net of the loan liquidation its fire-sale loss displaces.² The step in ϕ is at $\hat{\lambda}$, and it matters only if $\hat{\lambda} < 1$, that is, only if the issuer’s deposits at the bank are large enough relative to the bank’s equity and collateral to fail it

²If the fire-sale loss on the whole stock exceeds equity, $(1 - q_1) \theta b_1 > E_1$, the bank fails on the securities alone and the numerator is $q_1 E_1 / (1 - q_1) - \gamma_{\max} D_1^H - N_{0,1}$. For the specialist partner of Section 6 this is the relevant case once its issuer deposits exceed about \$37 billion; the numerical results use it.

before they are exhausted. Far from the crowding-out zone the partner holds its requirement plus its share w_1 of the system's slack, $\theta b_1 = \text{REQ}_1 + w_1 \Sigma$ (the pre-positioning rule), and $\hat{\lambda}(s) = A_1/[(1-\alpha)s] + B_1/(1-\alpha)$, where $A_1 \equiv E_1/\varphi - (1-\kappa_q)(\gamma_{\max} D_1^H(0) + N_{0,1}) + \kappa_q w_1 \Sigma_0$ is the partner's stress surplus without issuer deposits and $B_1 \equiv (1-\kappa_q)\gamma_{\max} \text{sh}_1 + \kappa_q \lambda_1(1-\alpha)(1-w_1) + \kappa_q w_1(\gamma_{\max} - \alpha\theta/q_F)$ is the marginal cover per coin (sh_1 and w_1 are the partner's shares of deposits and of system slack).

Proposition 4.3 (the run–bank loop). *Suppose $A_1 > 0$.*

- (i) *If $B_1 \geq 1-\alpha$ the partner never fails in a coin run and the run probability is independent of s . If $B_1 < 1-\alpha$, the loop is inactive for $s \leq s_f \equiv A_1/(1-\alpha-B_1)$ and active beyond: $\hat{\lambda}(s)$ is strictly decreasing in s , π^* and ρ are increasing in s for $\hat{\lambda} \leq 1-\phi_1$, and under the analogous condition M' of Appendix A for $\hat{\lambda} \in (1-\phi_1, 1)$.*
- (ii) *A higher run-off λ_1 raises the partner's pre-positioned collateral and with it $\hat{\lambda}$, and raises the requirement slack a coin absorbs, which moves the crowding-out zone toward today's stock: the run-off is the dial between the two externalities. The run probability does not feed back into the date-0 equilibrium, so the equilibrium run probability is unique.*

Equation (4.5) is the bridge between the two externalities and the partner bank's balance sheet. Everything on its right-hand side is observable for a given bank: equity, safe-asset holdings, the stress price of those holdings and the bank's other outflows. Section 5 turns (4.5) into the rule for the run-off rate, and Section 7 evaluates it for the partner banks of 2022–23. Figure C.1 plots $\rho(s)$ for two partner types with the US calibration: a specialist bank with \$1.2 billion of equity reaches the loop at about \$32 billion of issuer deposits even under a 100 percent run-off, because beyond that amount its equity cannot absorb both the fire-sale loss on the collateral it would have to sell and the loan liquidation that the loss forces (Proposition 5.3(iv)), while a large bank with the same deposits is unaffected up to \$2 trillion.

4.5 Reserve composition

The issuer's objective is seigniorage, $s[\alpha(1/q_0 - 1) + (1-\alpha)(R_D^S - 1)]$, minus its expected run losses $\rho \phi_{\text{run}} s$; when banks own a share ω of it (Appendix E) it also weighs ω times the change in their profits, $\Delta \Pi_B$. Let $\Delta r \equiv (1/q_0 - 1) - (R_D^S - 1)$ be the spread between the bond yield and the wholesale deposit rate, and write the part of the objective that does not depend on the spread as $C(\alpha) \equiv -\rho(\alpha)\phi_{\text{run}}(\alpha)s + \omega \Delta \Pi_B(s, \alpha)$.

Proposition 4.4 (the bond–deposit boundary). *(i) The issuer's optimal bond share is nondecreasing in the spread Δr . There are thresholds $\underline{\Delta} \leq \bar{\Delta}$, determined by the*

run-loss differential between bonds and deposits and by the owner's deposit margin, such that $\alpha^ = 1$ for $\Delta r \geq \bar{\Delta}$ and $\alpha^* = 0$ for $\Delta r \leq \underline{\Delta}$.*

- (ii) *Far from the crowding-out zone, if $R > R_D^S$, the optimal bond share is nonincreasing in the ownership share ω : ownership shifts the boundary toward deposits.*

The run-loss differential is the term that makes the rule more than a yield comparison. Bond reserves expose the issuer to the fire-sale discount $\bar{h}$ in a run; deposit reserves expose it to the partner's failure (LGD) only beyond $\hat{\lambda}$. Where the partner is strong, deposits are the safer reserve and the boundary sits above the pure yield comparison; where it is weak, bonds are. The rule also prices the bank channel. A price-taking bank pays on issuer deposits what a marginal dollar earns at the loan rate less the liquidity cost it brings, $R_D^S = R(L) - \partial \mathcal{C}_1 / \partial D_1^S$, while the bill return is $1/q_0 = R(L)/M$. To first order the gap $R_D^S - 1/q_0$ is the premium on a unit of collateral less that cost. The two share the compliance term, since an issuer deposit adds λ_{reg} to the requirement; but a deposit adds only λ_a to stress outflows, while a bill removes a full unit of collateral. As collateral grows scarce, the bill return therefore falls below the rate banks pay on issuer deposits, and the spread pushes issuers toward deposits, which absorb less collateral (Proposition 4.2); Section 6.4 measures how much. The observed composition—95 percent bills, repo and funds, 4.6 percent bank deposits—is held at a spread close to zero today; actual deposit terms also reflect insurance limits, concentration rules and operational needs that the model leaves out, so we take the observed mix as the starting point, hold it fixed in the baseline policy comparison, and let it respond to changes in the spread in Section 6.4. Appendix E shows how bank ownership shifts the boundary, and Appendix D.2 that Circle's monthly allocation between bills and repo follows the sign of the bill–repo spread, as the rule predicts.

4.6 What the model says the data should show

Five implications organize the evidence. First, banks subject to the liquidity coverage ratio hold buffers above the floor and rarely approach it, because the cost of a shortfall is a probability (Section 6). Second, issuer deposits gravitate to the banks whose liquidity requirement on them is weakest, and a bank whose funding is dominated by them holds them in liquid assets (Sections 7.1 and 7.2). Third, partner-bank failure in a run is a capital condition, (4.5), that can be evaluated bank by bank (Section 7.3). Fourth, in the observed range of the liquidity coverage ratio the premium varies by a few basis points, so lending should not respond measurably to issuer absorption (Section 7.4). Fifth, any short-run price impact of issuer purchases larger than the premium's slope should be a flow effect that fades within weeks and is attributable to the dollars that actually buy bills (Section 7.5).

5 Policy: instruments, the fold and the run-off rule

5.1 Welfare and externalities

Total surplus is

$$\begin{aligned}
 W = V(s) + [\Pi(L) - L] - \sum_j \Big\{ & \varphi \mathbb{E} [((1 - \kappa_j) \ell_j \text{REQ}_j + \xi_j - \theta b_j)^+] \\
 & + \tilde{\tau}_{a,j} \mathbb{E} [(\text{NEED}_j + \xi_j - \theta b_j)^+] \Big\} - \rho F_B \mathbf{1}\{\hat{\lambda} < 1\},
 \end{aligned} \tag{5.1}$$

the households' convenience value of coins, the surplus on bank lending (with $\Pi(L) \equiv \int_0^L R$), the loan liquidations that deleveraging and stress shortfalls force, and the expected deadweight loss F_B of a partner failure in a run. The cost of compliance, c per dollar of requirement shortfall, is interest paid to the Federal Reserve or to the lenders of term funds and drops out, as do seigniorage, bond prices and deposit margins. Part of c , the stigma of borrowing, may be a real cost rather than a transfer; we treat the accounting as a scenario and report the comparison both ways (Appendix G). The fire-sale discounts at which the issuer sells its bonds in a run and the loss it takes on its deposits when the partner fails are transfers as well: the first accrues to the buyers of the bonds and the second to the other claimants on the failed bank. The deadweight loss of a failure is set to the partner's equity, $F_B = \zeta E_1$ with $\zeta = 1$.

The stock of coins is set by demand, so the planner's margin is not the issuer's. Two externalities separate private from social outcomes. The collateral externality is the loss of lending and the expected liquidation cost that issuance imposes on every bank through the price of collateral; neither the issuer nor the holders of its coins bear it. The run externality, $F_B d\rho/ds$, is the deadweight loss of the partner's failure, borne by the partner's other claimants and by the public backstop; it is active only when the partner can fail, $\hat{\lambda} < 1$. Set against both is the convenience value $V'(s)$ of the marginal coin, which households do capture.

5.2 Instruments when the stock is demand-determined

We consider an issuance cap $\bar{s}$; a requirement that issuers hold a share μ of reserves as deposits at the Federal Reserve, which the GENIUS Act already lists among eligible reserve assets and which does not compete for the collateral stock; the run-off rate on issuer deposits; a bonds-only reserve rule; a limit on the share of an issuer's deposits at any one bank; and discount-window readiness, under which every bank pre-pledges collateral equal to 90 percent of its net outflows, raising both the capacity κ_j and the coverage ω_j of (3.5), which the

calibration takes from the same pledged collateral. Each rule applies to the issuer’s whole stock: a composition rule also reallocates the reserves of the coins already outstanding, through the issuer’s and the banks’ balance sheets and the market for safe assets. Ownership rules and market structure, which act on an issuer’s choice of quantity, are analyzed in Appendix E. One condition recurs.

- (C1) The Federal Reserve lends the deposits it receives from issuers back to banks in proportion to their deposits, at its policy rate and for longer than the 30-day horizon of the rule: bank funding is not reduced by μS , the funding is not an LCR outflow, and its cost is a transfer.

Proposition 5.1 (Federal Reserve deposits versus a cap). *Let s_c be issuance under a binding cap, suppose the partner’s loop is shut under both instruments and $\lambda_j \geq \lambda_a$ for every bank, and let the share apply to the whole stock.*

- (i) *Under (C1) there is a Federal Reserve deposit share μ under which the issuer holds the same bonds and the same bank deposits as under the cap, $(1 - \mu)S_\mu = S_c$, while the demand-determined stock S_μ exceeds S_c . At that share every bank’s requirement and actual need are lower than under the cap, the price of collateral is lower and lending is higher.*
- (ii) *Welfare under that share exceeds welfare under the cap by the convenience value $V(s_\mu) - V(s_c) > 0$ of the coins the cap removes and the gain in loan surplus, less the change in the expected costs of deleveraging and stress liquidation, which has no general sign.*
- (iii) *If (C1) fails, the share removes μS of bank funding and lowers welfare relative to (i) by approximately $(R - 1)\mu S$.*

The proposition separates two ways of keeping issuance from absorbing collateral: removing coins, which the cap does, and changing what each coin absorbs, which the deposit share does. The second keeps the convenience value of the coins households demand. Matched on what the issuer holds, the share also does better on the bank side: the additional coins drain household deposits, which carry a run-off, and the recycled funds that replace them carry none, so every bank’s requirement and actual need fall. Part (ii) leaves the sign of the real costs open because banks also hold less collateral at the lower price. In the calibration they favor the share too: at the \$1 trillion level the matched share, $\mu = 0.09$, lowers the expected costs of deleveraging and stress liquidation by \$0.04 billion a year against a convenience gain of \$3.6 billion (Table A.1). Deposits at the Federal Reserve are reserves, the asset that settles payments; the model, with one safe asset, values them only as collateral and leaves out the share’s effects on the payment system and on the Federal Reserve’s balance sheet. Its weakness is (C1). If the Federal Reserve sterilizes the deposits rather than recycling

them, the share drains bank funding and does the collateral channel's work by another route; Section 6.5 measures how complete the recycling must be. The same comparison applies to any composition rule that lowers what a coin absorbs; a bonds-only rule does the opposite when $\lambda_{\text{reg}} \leq \theta/q_F$ (Proposition 4.2).

5.3 Demand feedback and the fold

Coin demand rises as the bill yield falls, and the bill yield falls as collateral becomes scarce. Write $\ell \equiv ks_0 = S^d(r_0; k)$ for the total stock demanded at today's yield and $y(s)$ for the equilibrium bill yield at issuance s , continuous by Lemma 4.1. Issuance s is an equilibrium if and only if $S^d(y(s); k) = s_0 + s$, that is,

$$\ell = h(s) \equiv (s_0 + s) e^{\varepsilon(y(s) - r_0)} : \quad (5.2)$$

$h(s)$ is the demand at today's yield that supports a stock $s_0 + s$. Call an equilibrium stable if excess demand for coins crosses zero from above, so that issuance adjusting toward demand returns to it, and call the smallest stable equilibrium, which issuance growing from today's stock reaches, the normal equilibrium.

Proposition 5.2 (the fold). *Let $\varepsilon > 0$ and let $y(s)$ be differentiable.*

- (i) $h'(s) = e^{\varepsilon(y(s) - r_0)} [1 - \varepsilon(s_0 + s)|y'(s)|]$. *An equilibrium is stable where h is increasing and unstable where it is decreasing.*
- (ii) *Let s^* be the first issuance at which $\varepsilon(s_0 + s)|y'(s)|$ reaches one, suppose it crosses one there from below with positive slope, and let $\ell^* \equiv h(s^*)$. Then h is strictly increasing on $[0, s^*)$ and has a strict local maximum at s^* . For ℓ between $h(0)$ and ℓ^* the normal equilibrium exists, is unique and lies in $[0, s^*]$; for $\ell > \ell^*$ there is no equilibrium with $s \leq s^*$, issuance growing from today's stock does not settle on $[0, s^*]$, and any equilibrium lies on a later increasing branch of h , if one exists.*
- (iii) *If a yield path $\tilde{y}$ is nowhere lower and nowhere steeper than y on $[0, s^*]$ — $\tilde{y} \geq y$ and $|\tilde{y}'| \leq |y'|$ —then $\tilde{\ell}^* \geq \ell^*$.*
- (iv) *In the kinked limit with deposit reserves the yield jumps at s_1 , h jumps down there, and the demand levels between $h(s_1^+)$ and $h(s_1^-)$ have two stable equilibria: the multiplicity band of the kinked model is the special case of (i)–(ii) in which the steep part of the yield path is a jump.*

The fold is the point at which one more coin lowers the bill yield by enough to draw in more than one further coin. The crossing condition in (ii) rules out a tangency, at which h would only flatten and could rise again; it holds at the fold of every calibration in Table 5

that has one. The normal equilibrium is defined by the branch, not by the level of the yield: with a responsive bill supply or reserve mix the bill yield at the fold is already below zero (Table 5). Below it, issuance growing from today’s stock settles where demand meets the stock. Beyond it there is no normal equilibrium: as issuance adjusts toward demand, the stock keeps growing past s^* into the region in which the banks’ term-funding capacity is exhausted, they must deleverage and the bill yield collapses. We call this process a collateral spiral. The proposition does not say whether it ends, which depends on the later branches of h ; in the calibration it does not end on the range we compute (Section 6.4). The spiral is a process rather than an equilibrium, and a supply or policy response would have to stop it. Part (iii) says which responses move the fold outward: anything that lowers the premium and flattens its path, such as a bill supply that responds to the price, more collateral pledged for term funding and at the discount window, or a reserve mix that absorbs less. Part (iv) connects the result to the earlier literature: the two equilibria of the kinked model near its threshold are the same phenomenon in the limit in which the steep part of the yield path is a cliff.

The fold needs a demand response. As $\varepsilon \rightarrow 0$, $h(s) = s_0 + s$ rises everywhere and the normal equilibrium exists at every level of demand: the stock is set by adoption, and the premium and lending are those of the crowding-out zone at that stock. The estimated semi-elasticity has a 95 percent interval that includes zero (Section 6), so the fold is a property of the calibrated demand curve, not something the data establish on their own. The ranking of the instruments does not rest on it. With the logarithmic inverse demand the marginal valuation of the coins below the demanded stock rises without bound as $\varepsilon \rightarrow 0$, so a binding cap becomes arbitrarily costly; that is a property of the functional form, and with a willingness to pay capped at two or three times today’s bill yield the cap’s loss stays bounded but remains the largest of any instrument (Appendix G). The Federal Reserve deposit share leaves the stock unchanged. Section 6.5 reports the ranking down to $\varepsilon = 0.5$.

5.4 The run-off rate, coverage and the capital condition

The liquidity coverage ratio requires a bank to hold HQLA against a share λ_{reg} of each deposit; the share is the deposit’s assumed 30-day run-off. For deposits placed by a stablecoin issuer, the Basel and EU texts admit 10, 25 or 100 percent (Coste, 2025). In the model λ_1 sets the requirement against issuer deposits (3.3), and therefore both the partner’s pre-positioned collateral, which enters the failure threshold (4.5), and the slack each coin absorbs, which locates the crowding-out zone (4.4). It does not change what holders withdraw, λ_a in a stress without a run and the run share in a run.

Write D_1^H and $N_{0,1}$ for the partner's household deposits and other outflows, w_1 for its share of the system's slack, and Σ_0 for the system's slack at zero issuance. With the pre-positioning rule $\theta b_1 = \text{REQ}_1 + w_1 \Sigma$ and $\Sigma = \Sigma_0 - [\lambda_1(1 - \alpha) - \gamma_{\max} + \alpha\theta/q_F]s$, (4.5) is linear in the run-off, and setting $\hat{\lambda} = 1$ yields the smallest run-off that shuts the loop:

$$\lambda^{\min}(s) = \frac{(1 - \alpha) - [E_1/\varphi - (1 - \kappa_q)(\gamma_{\max}D_1^H + N_{0,1}) + \kappa_q w_1(\Sigma_0 + (\gamma_{\max} - \alpha\theta/q_F)s)]/s}{\kappa_q(1 - \alpha)(1 - w_1)}. \quad (5.3)$$

Proposition 5.3 (the run-off rate). *(i) $\lambda^{\min}$ is increasing in issuance and decreasing in the partner's equity E_1 , its share of system slack w_1 and the system's slack Σ_0 .
(ii) A higher run-off raises $\hat{\lambda}$ and lowers the partner's run risk, and raises the slack a coin absorbs, which moves the crowding-out zone toward today's stock. For a bank outside the liquidity coverage ratio the internal run-off λ_{int} plays the same role, and extending a requirement to such a bank binds only if it exceeds λ_{int} .
(iii) There is a cap s_{cap} , the solution of $\lambda^{\min}(s) = 1$, above which no run-off up to 100 percent shuts the loop; beyond it the only instruments are capital, an issuance cap or a concentration limit on the partner's issuer deposits.
(iv) Full coverage shuts the loop only if the cover survives its own fire sale: with $\lambda_1 = 1$ and no other slack, the condition is $E_1 \geq (1 + \varphi)(1 - q_1)(1 - \alpha)s$, because equity must absorb both the fire-sale loss on the cover and the cost of liquidating loans to replace what the discount took. A run-off rule is therefore a joint condition on the run-off, the stress price of the cover and the partner's equity, not on the rate alone.*

For a large partner, $\lambda^{\min}$ is negative at any plausible issuance: no requirement is needed to shut the loop, and the run-off rate's only effect is through the slack each coin absorbs. For a specialist partner, $\lambda^{\min}$ is interior and rises with the issuer's deposits: the right rate is bank-specific and tracks D_1^S/E_1 . Beyond s_{cap} the specialist cannot be made safe by liquidity rules at all. In the US calibration (Figure 1) a specialist partner with \$1.2 billion of equity needs a run-off of 62 percent at \$10 billion of issuer deposits, and no rate suffices beyond \$24–32 billion, depending on the loan-liquidation cost. That ceiling is the capital condition of part (iv): with $\varphi = 0.25$ the fire-sale loss on the cover and the loan liquidation it forces exhaust the partner's equity at \$32 billion of deposits. Section 7.3 evaluates condition (iv) for the partner banks of 2022–23.

Part (iii) also speaks to the concentration limit now proposed. The FDIC's April 2026 rule would cap an issuer's exposure to any one financial institution at 40 percent of its reserves (Federal Deposit Insurance Corporation, 2026), and its Question 59 asks whether the liquidity coverage rules should be amended for stablecoin activity. The cap is written on

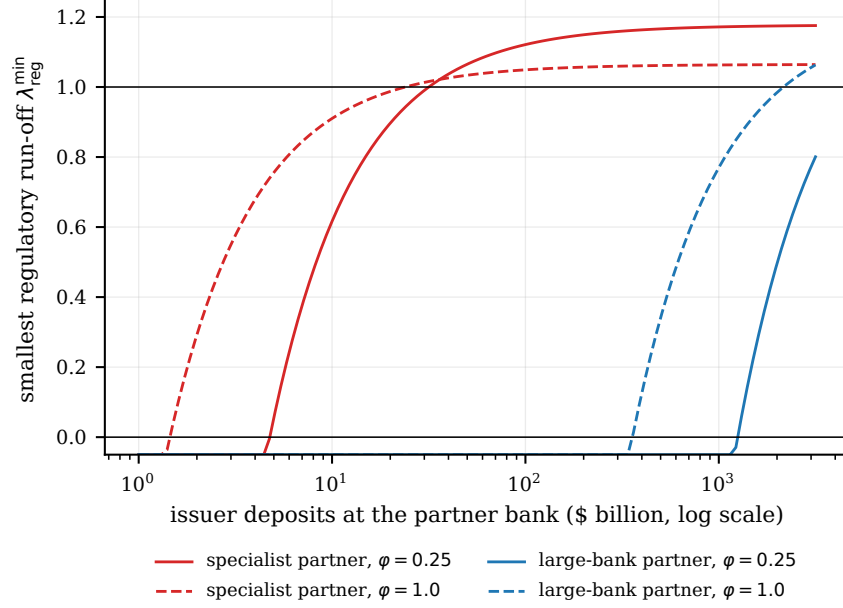


Figure 1: The smallest run-off on issuer deposits that shuts the run–bank loop. US calibration, deposit reserves. Equation (5.3) for a specialist and a large-bank partner at loan-liquidation costs $\varphi = 0.25$ and $\varphi = 1$; the loop cannot be shut where the curve exceeds one.

the issuer’s balance sheet and is the same for every bank. In the model the binding quantity is the partner’s: condition (iv) says the deposits a bank can absorb in a full redemption are $E_1/[(1 + \varphi)(1 - q_1)]$, which is \$20–32 billion for the specialist partner with no other slack (\$24–32 billion once its own slack and outflows are counted, the ceiling s_{cap} of Figure 1) and \$8.3 trillion for a large one. An issuer-relative cap of 40 percent protects a specialist partner only while $0.4s \leq s_{\text{cap}}$, that is, up to \$60–79 billion of issuance; at the size of the largest issuer today the per-bank cap that would protect such a partner is 13–17 percent, not 40. And the banks where the risk materialized in 2023 were outside the liquidity coverage ratio altogether (Section 7.1), so for them the question was not the run-off rate but whether any liquidity requirement applied. A limit that scales with the receiving bank’s equity, whatever its regulatory category, does the work that neither a uniform run-off rate nor a uniform cap can do; it is the answer the model gives to Question 59.

6 Stablecoins and bank collateral in the United States

6.1 Calibration

Table B.3 in Appendix B lists the parameters and Table 2 collects those on which the results mainly rest. Deposits, lending and bank holdings of Treasury and agency securities come from the Federal Reserve’s H.8 release; the stock of marketable bills from the Treasury’s Monthly Statement of the Public Debt; issuer holdings from the attestations of Section 2. HQLA and net cash outflows are the sums disclosed by the twenty-one bank holding companies subject to the liquidity coverage ratio (Table B.2); their aggregate ratio is 1.161 in 2026Q2, so the system’s slack is \$630 billion. The stress-withdrawal weight on household deposits is $\gamma_{\max} = 0.07$, a blend of the ratio’s 3 percent run-off on stable and 10 percent on other retail deposits, and all other 30-day outflows enter N_0 . The regulatory run-off on issuer deposits is $\lambda_{\text{reg}} = 1$, the treatment of unsecured financial-institution funding; the actual run-off is $\lambda_a = 0.13$, the largest 30-day decline of either large issuer’s stock outside the March 2023 run (13.3 percent for USDT in May 2022 and 12.6 percent for USDC in 2022). Collateral is valued at $\theta = 1$, the stress price is $q_1 = 0.97$, the loan-liquidation cost $\varphi = 0.25$ (severe case: 1) and the stress probability $p = 0.10$. The run block is calibrated to the two largest issuers’ estimated run probabilities (Appendix C).

The new parameters are those of the compliance cost, and each comes from the rules or the data. The cost of covering a requirement shortfall, $c = 54$ basis points, is the primary credit rate’s 10 basis points over interest on reserves plus the 44 basis points of stigma Armantier et al. (2011) estimate; in a crisis it is 136. The capacity for covering it, $\kappa = 46$ percent of the requirement, and the share of an actual stress shortfall covered at the window, $\omega = 0.46$, are the loans banks had pledged at the discount window, \$1.8 trillion, over the system’s net outflows; the window’s crisis cost for a three-month stress is $\varphi_{DW} = c_{\text{crisis}}/4$. The numerator covers all depository institutions in 2023 and the denominator the banks that disclose the ratio in 2026, so the value is a calibration scenario rather than an estimate of the banks’ legal capacity; Table 5 reports the fold for capacities of 30 and 60 percent. The daily shock has standard deviation $\sigma = 7.5$ percent of the requirement, which makes the median disclosed ratio, 1.157, the buffer at which a bank pays a premium of 1 basis point; the volatility of issuer deposits adds the daily standard deviation of USDC’s stock, 0.71 percent, to the part of issuer deposits not covered by the requirement. A bank outside the ratio applies the internal run-off $\lambda_{\text{int}} = 0.74$, the liquidity Silvergate held against its digital-asset deposits in excess of its peers in 2022 (Section 7.2). Issuers’ existing reserves are 95 percent bills, repo and funds

and 4.6 percent bank deposits, \$14.5 billion; new issuance keeps that mix in the current-rules baseline.

The demand curve (3.1) is estimated on monthly data for 2020–2026 (DefiLlama supply, the three-month bill yield and the bitcoin price). In first differences, a one percentage point rise in the bill yield lowers the stock by 8.5 percent within the month (standard error 2.5). An error-correction model in the levels gives a long-run coefficient of -6.6 percent per percentage point (standard error 4.7), whose 95 percent interval, $[-15.7, +2.6]$, includes zero; its impact coefficient, $+8.1$, conditions on the lagged gap from the long-run relation and is not used. The estimate is not stable: it is -7.8 without May–December 2022, -20.3 (standard error 2.9) from 2023, and -0.6 with VIX as a control; the weekly version gives -10.1 . The identifying assumptions are that issuers mint and redeem at par on demand, so that holders set the stock, and that, given bitcoin, the bill yield is not driven by shocks to crypto demand. We use $\varepsilon = 6.6$ percent per percentage point, anchored at the October 2026 stock of \$311 billion and the bill yield of 3.99 percent, as a scenario, and report the policy comparison at 2.0, 10.1 and 20.3 as well (Section 6.5). On the baseline curve the stock households would hold at a zero opportunity cost, at today’s level of adoption, is \$405 billion; larger stocks require growth in adoption, which the 2028 projection assumes.

Two partner types bracket the US cases. The large partner is a bank subject to the full ratio, like the custodian that holds Circle’s cash today, with 12 percent of system deposits, 13 percent of equity and 12 percent of slack. The specialist partner, modelled on the crypto-focused banks of 2022, is outside the ratio, holds 0.07 percent of deposits and 0.05 percent of equity (\$1.2 billion) and matches their deposit-to-equity ratios.

6.2 The crowding-out zone

Table 3 and Figure 2 trace the zone for a large partner with no policy. The premium is 3 basis points at today’s stock, where the aggregate ratio is 1.161; its slope there is 0.09 basis points per \$3.5 billion of bill purchases, the same order as the Treasury convenience-yield slope of Hofmann et al. (2026). It reaches 10 basis points at a total stock of \$486 billion, 25 at \$661 billion and 50 at \$811 billion, and is 86 at the regulatory threshold, where the disclosed slack is used up and the aggregate ratio is one. With \$1 trillion of new issuance, a total stock of \$1.3 trillion, it is 158 basis points, the bill yield is 2.47 percent, lending is \$64 billion lower and the expected cost of stress liquidations is \$4.1 billion a year. Near the \$2 trillion projection it passes 190 basis points, the sum of the cover and stress costs, and deleveraging begins. Crowding out therefore begins well in front of the regulatory threshold: by the time the stock is twice today’s, the premium is in the tens of basis points. These numbers hold the

Table 2: Core calibration. The full set of parameters and sources is in Table B.3.

Quantity	Symbol	Value
Stablecoins outstanding, October 2026; bank-deposit share of reserves	s_0	\$311 bn; 4.6%
Disclosed HQLA; net cash outflows; aggregate LCR		\$4.54 tn; \$3.91 tn; 1.161
Regulatory slack	Σ_0	\$630 bn
Run-off on issuer deposits: regulatory; actual; internal (outside the ratio)	$\lambda_{\text{reg}}; \lambda_a; \lambda_{\text{int}}$	1; 0.13; 0.74
Cost of covering a requirement shortfall (crisis)	c	54 bp (136 bp)
Daily shock to the requirement, share of requirement	σ	0.075
Term-funding and discount-window capacity; stress coverage	$\kappa; \omega$	0.46; 0.46
Stress probability; loan-liquidation cost; stress price	$p; \varphi; q_1$	0.10; 0.25; 0.97
Demand semi-elasticity, percent per percentage point	ε	6.6 (sensitivity 2.0–20.3)
Partner equity: large; specialist	E_1	\$312 bn; \$1.2 bn

reserve mix fixed; if issuers rebalance toward deposits as Section 6.4 describes, the premium at the \$1 trillion demand level is 79 rather than 112 basis points, but the zone begins where it does here, while the premium is still small.

The reserve mix matters as Proposition 4.2 says. With the same reserves held as bank deposits, the premium with \$1 trillion of new issuance is 67 basis points rather than 158, the lending loss \$26 billion rather than \$64 billion and the expected stress cost \$0.9 billion rather than \$4.1 billion: bills crowd out 2.4 times as much. Deposits do absorb collateral, through the requirement of the bank that receives them, but they add only their actual run-off to stress outflows.

Three measurement choices matter for the number. The first is whose HQLA counts. An earlier version of this paper measured the system’s HQLA as banks’ reserve balances plus their Treasury and agency securities, about \$7.8 trillion, and applied the large banks’ ratio to it; that overstates the slack, because the securities include mortgage-backed securities and holdings of banks that are not subject to the rule, and because the ratio of the largest banks is not the ratio of the rest. The disclosed sums are the right measure of the regulatory slack and give a threshold about half as large. The second is whether the existing coins have already consumed slack. The slack is measured from today’s balance sheets, so s_1 is the room for further issuance; if the existing \$311 billion had been bought out of bank HQLA, the stock at which the constraint binds would be the current stock plus s_1 . They were not. From September 2022 to June 2026 the two largest issuers added \$46 billion of bills (\$120 billion

Table 3: The crowding-out zone. Large partner subject to the ratio, current rules, fixed bill supply. Premium on safe assets in basis points a year; bill yield in percent; aggregate LCR of the banks subject to the rule; change in bank lending and expected cost of stress liquidations in \$ billion a year. “Observed” is the current reserve mix (95 percent bills, repo and funds); “deposits” holds the same reserves as bank deposits.

Stock (\$ bn)	Reserves	Premium (bp)	Bill yield (%)	LCR	Δ lending	Stress cost
311	today	3	3.97	1.161	0	0.0
461	observed	9	3.91	1.124	−2	0.1
461	deposits	8	3.92	1.121	−2	0.1
611	observed	22	3.79	1.087	−8	0.2
611	deposits	19	3.81	1.084	−7	0.2
961	observed	86	3.16	1.002	−34	1.3
961	deposits	51	3.50	1.006	−20	0.6
1,311	observed	158	2.47	0.917	−64	4.1
1,311	deposits	67	3.34	0.938	−26	0.9
1,986	observed	195	2.12	0.752	−79	12.1
1,986	deposits	154	2.51	0.830	−62	4.0

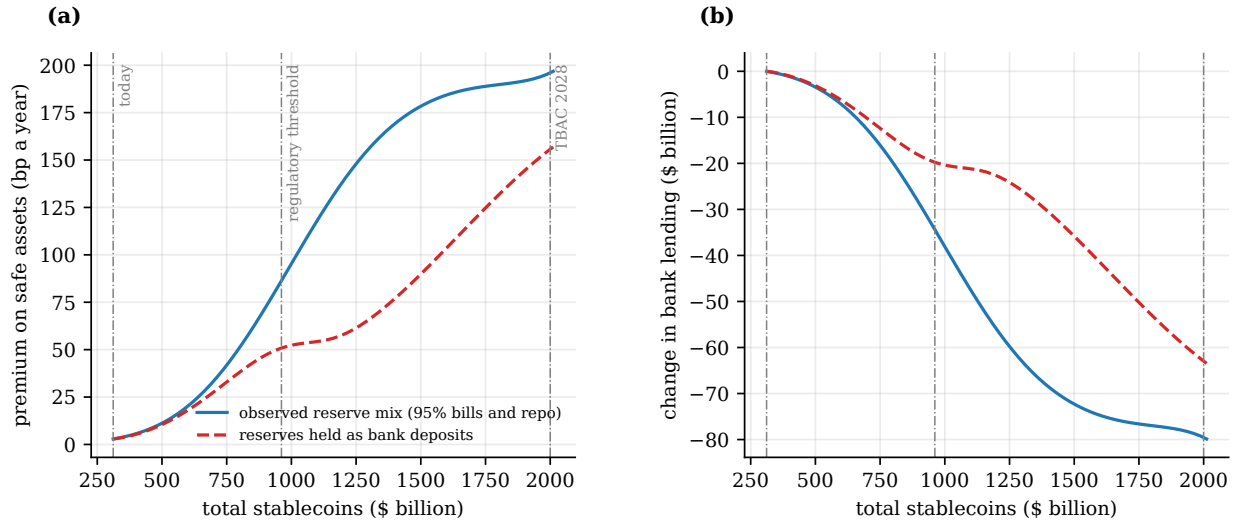


Figure 2: The crowding-out zone, US calibration. (a) Premium on safe assets against the total stock of stablecoins, for the observed reserve mix and for reserves held as bank deposits. (b) Change in bank lending. Vertical lines: today’s stock, the regulatory threshold (the stock at which the disclosed slack is used up) and the 2028 projection.

with bill repo) while the Treasury issued \$3,042 billion of net new bills; money funds absorbed \$2,025 billion of Treasury securities, financed by running down their reverse repos at the Federal Reserve by \$2,399 billion, and banks added \$339 billion of non-mortgage Treasury and agency securities, with reserves almost unchanged (H.8, H.4.1 and the Financial Accounts, Z.1). The issuers’ purchases were a small part of new supply and banks were net buyers, so

we read s_1 as the distance from here.³ The absorbers have changed since the reverse-repo facility emptied. From the end of 2024 reverse repos fell by only \$447 billion, banks bought \$265 billion of securities and reserves fell by \$296 billion: new bills are increasingly paid for out of bank reserves, which is the case in which further issuance draws on bank HQLA, as the model assumes.

The third is whether slack moves between banks. Equation (4.4) pools it, which presumes that the repo and interbank markets carry HQLA to wherever the issuer’s absorption lands. The slack is not concentrated: the three largest holders, Goldman Sachs, JPMorgan Chase and Citigroup, have 41 percent of it, and its distribution is equivalent to 10 banks of equal size (Table B.2). It is uneven relative to outflows, however: banks with 69 percent of the system’s net outflows run ratios below 1.15. If slack did not move at all and further issuance drew on each bank in proportion to its outflows, PNC, at a ratio of 1.056, would reach its floor after \$225 billion of bill-backed issuance, banks with a quarter of the outflows after \$410 billion and banks with half after \$484 billion, against \$650 billion for the pooled threshold. The pooled number is therefore the optimistic scenario in a second sense: it assumes a market for collateral among banks that works without friction, which is what failed in September 2019. The same assumption runs through the model, whose banks trade collateral in one market. If frictions in moving collateral between banks made the yield path steeper where the slack is thin, Proposition 5.2(iii) would place the fold closer; the model does not establish that they would. The numbers of Sections 6.2–6.5 are those of the frictionless case, and the rows of Table 4 are scenarios rather than bounds.

Figure 3 shows how the slack has moved. Panel (a) plots the aggregate LCR, total HQLA over total net cash outflows, of the five largest bank holding companies, which report amounts in every quarter since 2018, and of all disclosing banks. The five-bank ratio has stayed between 1.14 and 1.22: it was 1.17 in the third quarter of 2019 and is 1.14 today, at the bottom of that range. Panel (b) plots the disclosed slack itself. For the five banks it was \$283 billion in September 2019, peaked at \$436 billion in 2024 and is \$399 billion now; for all disclosing banks it was \$422 billion in 2019 (eighteen banks), \$607 billion in 2022 (nineteen) and \$630 billion in 2026 (twenty-one). The room has grown with the slack, but the stock of coins has grown faster: negligible against the room in 2019, it is now 32 percent of the total stock at which the constraint binds.

³Weekly local projections of these quantities on bill-buying issuer dollars, with the design of Table 11 and controls for the Treasury General Account and the Federal Reserve’s securities holdings, are too imprecise to attribute the purchases: their standard errors are \$4–32 per dollar of issuer flow at four weeks.

Table 4: The threshold under alternative measurement assumptions. Room for further issuance and the total stock at which the collateral constraint binds, \$ billion, from the 2026Q2 disclosures; bill reserves unless stated.

Assumption	Room (\$ bn)	Total stock (\$ bn)
Regulatory floor, slack pooled across banks, bill reserves	650	961
deposit reserves	678	989
Effective floor 1.05	448	759
Effective floor 1.10	246	558
No redistribution: lowest-ratio bank at its floor	225	536
banks with half of net outflows at their floor	484	796
Run-off on issuer deposits 50 percent, deposit reserves	1,465	1,777

The managed buffer. The September 2019 repo spike, the one observed kink in the price of bank liquidity (Afonso et al., 2021; Copeland et al., 2025), occurred with a regulatory slack of roughly \$283 billion and an aggregate ratio of 1.17, not at zero. The model gives two reasons a kink need not sit at the regulatory minimum. Banks manage the ratio to a buffer, because a shortfall is a probability rather than a certainty: with the calibrated shocks, a bank at the median disclosed ratio, 1.157, already pays a premium of 1 basis point, and the premium rises well before the ratio reaches one. And the 2019 episode was a shortage of the one component of HQLA that settles payments, reserves, which is the reading of Copeland et al. (2025) and which the model, with one safe asset, does not distinguish. Table 4 collects the regulatory room under the measurement choices of this section; the zone of Table 3 is centred on the first row.

A zone that moves with the state. The compliance cost is not a structural constant. In a system-wide stress the stigma of borrowing rises and banks hoard liquidity; with the crisis cover cost of 136 basis points the premium at the \$1 trillion demand level is 177 rather than 112 basis points. Issuance that sits in the early part of the zone in normal times can therefore be deep in it in a crisis, without any change in the stock of coins, and the crossing would coincide with the stress state in which issuer redemptions are most likely (Section 4.4). The response can also run the other way: supervisors can announce that buffers may be used, as the Federal Reserve did in March 2020, and the disclosed ratio rose rather than fell through 2020 as reserves expanded.

6.3 Partner types and the loop

Figure C.1 reports run probabilities. With a 100 percent run-off rate and a large partner the loop is shut at every size considered: the run probability stays at its baseline of 1.5 percent

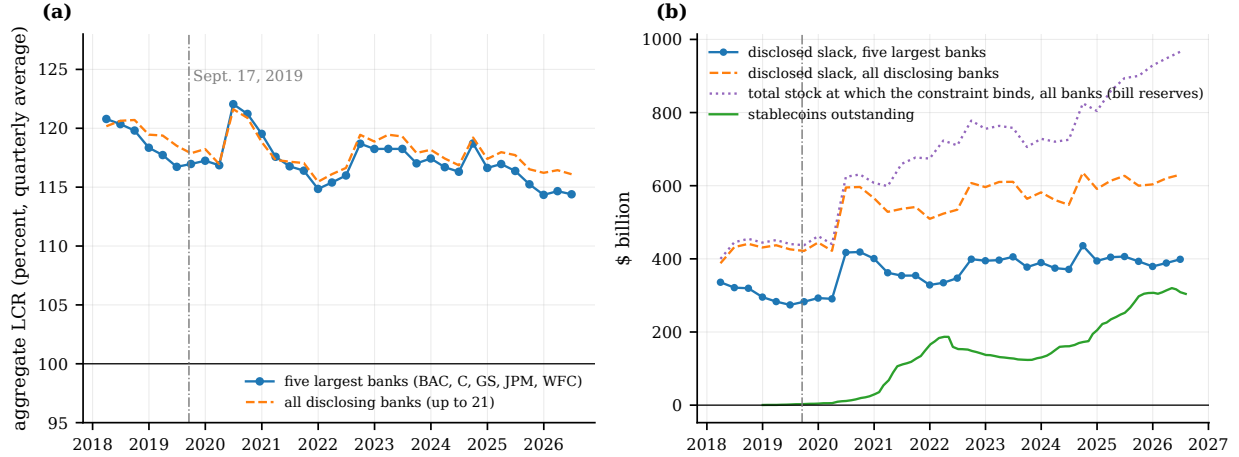


Figure 3: Disclosed liquidity coverage and the threshold over time, 2018–2026.

(a) Aggregate LCR, total HQLA over total net cash outflows, of the five largest US bank holding companies and of all banks that report amounts (up to twenty-one), from the quarterly averages in their 10-Q and 10-K filings or their public LCR disclosure reports (Table B.2). (b) Disclosed regulatory slack, HQLA minus net cash outflows, for the five largest banks and for all disclosing banks (the latter an unbalanced set whose size is shown in Table B.2); the total stock at which the constraint binds, the stock outstanding plus the room $\Sigma/(\theta/q_0 - \gamma_{\max})$ for bill reserves from the all-bank slack; and stablecoins outstanding (DefiLlama). The vertical line is September 17, 2019.

for deposit reserves (2.3 percent for bond reserves, which carry the fire-sale discount), and at 50 and 25 percent run-off rates it stays there up to \$1 trillion of issuer deposits, because the partner’s equity covers any fire sale of its collateral. For the specialist partner the rate does not matter beyond \$32 billion of issuer deposits: its equity cannot absorb the fire-sale loss on the cover it would have to sell together with the loan liquidation that the loss forces, the failure threshold falls below one, and the run probability rises to the stress probability itself, 10.0 percent at \$50 billion under any run-off rate. Below that amount the rate does matter: at \$10 billion the failure threshold is 1.33 at a 100 percent rate and falls below one at 50 percent. The specialist’s ceiling s_{cap} , beyond which no run-off rate protects it, is \$32 billion at $\varphi = 0.25$ and \$24 billion at $\varphi = 1$ (Figure 1). These magnitudes are those of 2022: Silvergate held \$11.9 billion of issuer-type deposits against \$1.11 billion of equity, within the ceiling at the 3 percent discount of the calibration but far beyond it at the 14 percent discount it realized on the securities it actually held (Section 7.3).

6.4 The fold

Table 5 and Figure 4 apply Proposition 5.2. The function h rises one for one with the stock until the premium steepens and then turns down. With a fixed bill supply its first peak, the

largest demand at today's yield that has a normal equilibrium, is $\ell^* = \$1,949$ billion, at a stock of \$2,361 billion and a bill yield of 1.06 percent. Read as demand at today's yield, the \$2 trillion projection lies just beyond it. A direct search over demand levels in steps of \$50 billion agrees: the normal equilibrium exists up to \$1,900 billion and not from \$1,950 billion. Beyond the fold the calibration has no stable equilibrium at all on the range we compute, which extends to a stock of \$6.3 trillion: the stock does not settle, which is the collateral spiral of Section 5.3. Below the fold, for demand between \$1,100 and \$1,600 billion, a second stable equilibrium coexists with the normal one: a deleveraging equilibrium at a stock of \$4.3–6.3 trillion and a bill yield of about -17 percent, in which banks have exhausted their term-funding capacity. Issuance growing from today's stock does not reach it, it is not the outcome beyond the fold, and the upper end of its band is set by the range we compute.

The fold moves most with the capacity of term funding and the discount window: from \$1,414 billion when banks can cover 30 percent of their requirement to \$2,417 billion at 60 percent. A bill supply that responds to the price with elasticity $\varepsilon_S = 2$ raises it to \$2,256 billion; a reserve mix of bank deposits to \$3,015 billion; and discount-window readiness, under which every bank pre-pledges collateral equal to 90 percent of its net outflows, to \$3,640 billion. Across the remaining variants it lies between \$1,750 and \$2,300 billion. The direction is the one Proposition 5.2(iii) predicts: every change that lowers and flattens the premium path moves the fold outward. The numbers are conditional on the deleveraging cost beyond capacity, which is a modelling choice, and on the reserve mix, to which we now turn.

The fold when issuers choose their reserves. Section 4.5 shows that competitive banks pay more for issuer deposits than bills return once collateral is scarce: at the \$1 trillion demand level with the observed mix, the bill return is 55 basis points below the rate on issuer deposits. We let the bill share respond to that spread, $\alpha = \alpha_0 + \eta(\text{spread} - \text{spread}_0)$, applied to the whole stock, with η the reallocation semi-elasticity: 0.7 percentage points of the bill share per basis point, the response of Circle's monthly allocation between bills and repo to their spread (Appendix D), and a half and a tenth of it, since bank deposits are poorer substitutes for bills than repo is. The fixed point is unique, because more bills raise the premium and lower the spread. With the full response the bill share falls to 0.81 at the \$1 trillion demand level, the premium there is 79 basis points rather than 112, and the fold moves to \$3,146 billion; with half of it, to \$2,861 billion; with a tenth, to \$2,022 billion (Table 5). With the larger responses the issuers hold only bank deposits by the time the fold is reached, and the bill yield at the peak of h is already below zero, as it is with a responsive bill supply (Table 5): the economy absorbs more demand before the fold, but only by passing through very low bill yields. The coin holders' side matters as much. The demand curve is

Table 5: The fold. Largest total demand at today’s yield with a normal equilibrium, the one reached by issuance growing from today’s stock; large partner, observed reserve mix, \$ billion. Grid: demand levels in steps of \$50 billion. ℓ^* : first peak of $h(S)$ in (5.2), computed on a fine grid; bill yield at the peak. The last rows let the issuer choose its reserve mix (η : percentage points of the bill share per basis point of the bill-minus-deposit spread) or put weight on the deposit rate in the coin holders’ opportunity cost (Section 6.4); the grid search is reported for the observed mix.

Calibration	Normal eq. up to	None from	ℓ^*	Yield at ℓ^* (%)
Baseline	1,900	1,950	1,949	1.06
Shock s.d. 0.05	2,100	2,150	—	—
Shock s.d. 0.10	1,800	1,850	—	—
No discount-window cover ($\omega = 0$)	1,800	1,850	—	—
Discount-window cover 0.9	2,000	2,050	—	—
$\varepsilon = 2.0$	2,300	2,350	—	—
$\varepsilon = 10.1$	1,750	1,800	—	—
Cover capacity 30%	1,400	1,450	1,414	0.64
Cover capacity 60%	2,400	2,450	2,417	1.31
Elastic bill supply, $\varepsilon_S = 2$	2,250	2,300	2,256	-0.81
$\varepsilon_S = 2, \omega = 0$	2,150	2,200	—	—
Reserves held as bank deposits	3,000	3,050	3,015	0.58
Reserve mix chosen, $\eta = 0.07$	—	—	2,022	0.81
Reserve mix chosen, $\eta = 0.35$	—	—	2,861	-3.80
Reserve mix chosen, $\eta = 0.7$	—	—	3,146	-0.27
Opportunity cost half deposit rate	—	—	2,198	1.36
Opportunity cost the deposit rate	—	—	no fold	—
DW readiness ($\kappa = \omega = 0.9$)	—	—	3,640	2.62

estimated against the bill yield, the return on a money-market alternative. If part of the holders’ alternative were a bank deposit, whose competitive rate barely moves with collateral scarcity, the feedback would weaken: with half the weight on the deposit rate the fold lies at \$2,198 billion, and with the whole weight there is none. The fold is therefore a conditional result. It is near the 2028 projection if issuers keep their mix and holders compare coins with money-market yields, and further out otherwise; what does not change is that the premium rises steeply well before it, and that the instruments that lower and flatten that path move it outward.

6.5 The instruments in numbers

Table 6 and Figure 5 compare the instruments with current rules for a large partner. Issuance solves the demand curve at the model’s own bill yield under each instrument. At the regulatory threshold and at the \$1 trillion demand level, with a fixed bill supply, every effect is small: the recycled Federal Reserve deposit share and discount-window readiness raise welfare by

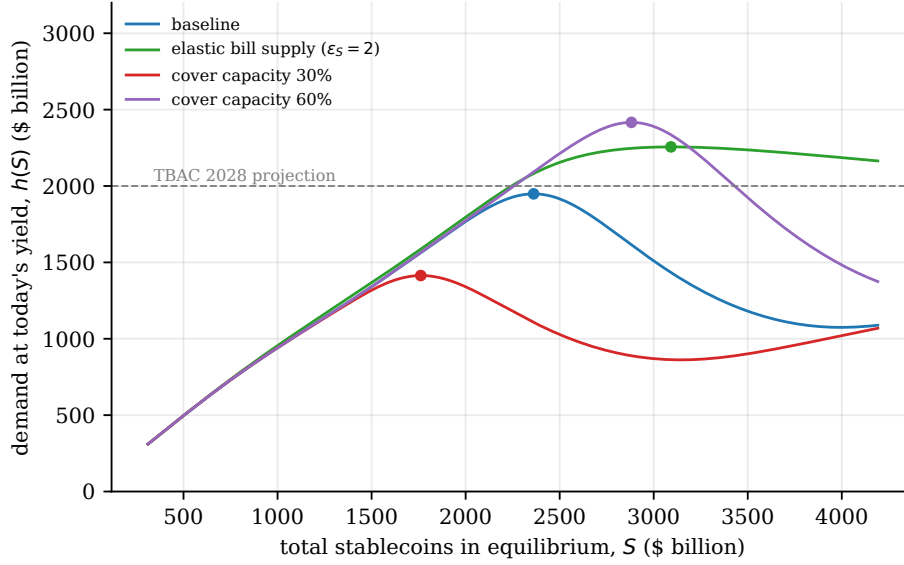


Figure 4: The fold. $h(S)$, the total demand at today's bill yield that supports an equilibrium stock S , from (5.2). Normal equilibria lie on the rising branch; the dots are the first peaks, ℓ^* , beyond which no normal equilibrium exists. Large partner, observed reserve mix.

\$1.3 and \$1.3 billion a year at \$1 trillion, reserves held as bank deposits by \$1.2 billion, and a cap at the threshold lowers it by \$3.0 billion. The run-off rate on issuer deposits (+0.0) and the 40 percent concentration limit (+0.0) do nothing for collateral, because the issuer holds almost no deposits. Without recycling the best Federal Reserve deposit share is zero (+0.0).

At the \$2 trillion level current rules with a fixed bill supply have no normal equilibrium. With the Treasury's bill supply responding ($\varepsilon_S = 2$), the gains are \$15.2 billion a year for the Federal Reserve deposit share, \$12.3 billion for discount-window readiness and \$11.3 billion for deposit reserves, while a cap loses \$81 billion. Across the six calibrations with a normal equilibrium at \$2 trillion the Federal Reserve deposit share gains \$1.4–29 billion—least when the discount window already covers 90 percent of stress shortfalls—and the cap loses \$69–197 billion. When reserves are held as bank deposits, the concentration limit gains \$3.3 billion at \$2 trillion, slightly more than the Federal Reserve deposit share (\$2.7 billion), because it spreads the requirement across banks. Appendix G reports the comparison under each variant, and Appendix F the comparison in the kinked model.

Two qualifications bound the ranking (Table G.2). The Federal Reserve deposit share needs nearly complete recycling: it beats discount-window readiness only if the Federal Reserve lends back more than 94.6 percent of the deposits it receives at the \$2 trillion level and 99.96 percent at \$1 trillion, because every dollar it keeps is a dollar of bank funding withdrawn; discount-window readiness needs no such condition. Nor can the share bear much cost: if holding the deposits cost the Federal Reserve more than 21 basis points a year,

Table 6: Instruments against current rules. Large partner, welfare gains over current rules in \$ billion a year. Demand is the total stock demanded at today’s bill yield; “stock” is the current-rules equilibrium and “premium” its premium in basis points. Fed deposits: the best share μ (in parentheses) of reserves held at the Federal Reserve, recycled to banks or not; DW readiness: pre-pledged collateral raised to 90 percent of net outflows for every bank (capacity κ and stress coverage ω); requirement: best run-off on issuer deposits; 40% limit: at most 40 percent of issuer deposits at the partner. Compliance costs are transfers; F_B equals the partner’s equity.

Demand	Stock	Premium	Cap	Fed deposits		DW ready	Deposits	Bonds	Req.	40%	
				recycled (μ)	not rec.						
<i>Fixed bill supply</i>											
962	1,026	101	−1.6	+1.1 (0.6)	0.0	+1.1	+0.9	−0.3	0.0	0.0	
1,000	1,074	112	−3.0	+1.3 (0.5)	0.0	+1.3	+1.2	−0.4	0.0	0.0	
2,000	no normal equilibrium (beyond the fold, Section 5.3)										
<i>Bill supply responds ($\varepsilon_S = 2$)</i>											
962	1,008	82	−1.1	+3.2 (1.0)	0.0	+2.1	+1.6	−0.4	+0.1	0.0	
1,000	1,053	90	−2.3	+3.6 (1.0)	0.0	+2.3	+1.9	−0.5	+0.1	0.0	
2,000	2,244	204	−81.3	+15.2 (1.0)	0.0	+12.3	+11.3	−1.5	+0.2	0.0	

through credit risk on the recycled lending or operating costs, discount-window readiness would be better at \$2 trillion. At \$1 trillion the two are effectively tied (\$1.332 and \$1.328 billion), and 0.09 basis points would reverse the order. And at \$1 trillion the two instruments trade places in six of thirty-two combinations of the demand semi-elasticity (0.5 to 20.3) and the cover cost (27 to 136 basis points), by at most \$0.6 billion. The cap lowers welfare in every combination.

The ranking also survives the issuer’s choice of reserves (Table G.3). With the full reallocation response, discount-window readiness and the Federal Reserve deposit share gain \$0.57 and \$0.57 billion at the \$1 trillion level; at \$2 trillion, where current rules now have a normal equilibrium even with a fixed bill supply, they gain \$6.9 and \$7.7 billion, ahead of deposit reserves (\$6.0 billion), while the cap loses \$3.1 and \$88 billion. The gains are smaller because the issuers’ own rebalancing does part of the work.

7 Evidence

The model makes five predictions that public data can check. The first, that banks subject to the liquidity coverage ratio manage it to a buffer, is in Section 6. This section takes the other four: where issuer deposits sit and how the banks that receive them hold them; when a partner bank fails in a run; whether bank lending responds to issuer absorption in the observed range; and which issuer dollars move bill prices. The specifications were fixed before estimation; where the model’s prediction is a zero, we report the estimates as a test of that

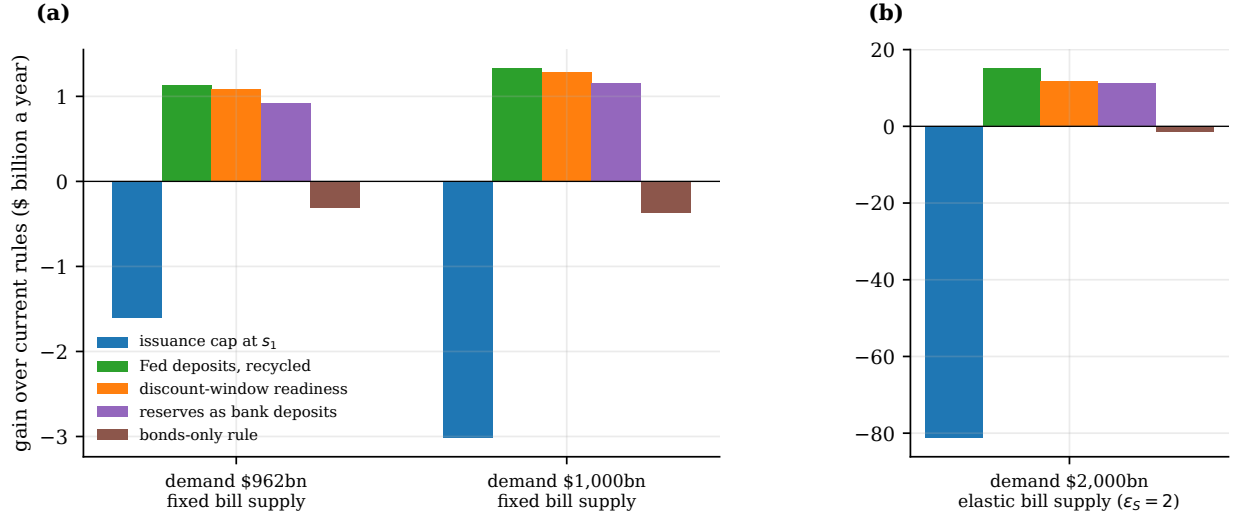


Figure 5: Instruments against current rules, US calibration. Welfare gain over current rules by instrument: (a) at the regulatory threshold and at the \$1 trillion demand level, fixed bill supply; (b) at the \$2 trillion demand level with a bill supply that responds to the price ($\varepsilon_S = 2$), the fixed-supply economy having no normal equilibrium there.

prediction and state what the confidence interval rules out. Issuers' deposits by bank are public only for the few banks that disclosed them, so a design that interacts each bank's prior exposure with common issuer shocks, as Lee and Tou (2026) do with confidential payment data, is outside our reach; the bank-level evidence below measures where issuer deposits sat and how they were held, and we do not give it a causal reading.

7.1 Where issuer deposits sat

Table 7 lists the banks that held Circle's cash in March 2023, when Circle named them, and their status under the liquidity coverage ratio. None of the banks that held the cash before the failures was subject to it. Silicon Valley Bank was a Category IV holding company with less than \$50 billion of weighted short-term wholesale funding, and Signature Bank was not part of a holding company, so neither met the ratio (Congressional Research Service, 2023); Silvergate, Customers, New York Community Bank and Citizens Trust were below the \$100 billion size threshold. Cross River, added on 12 March, was too. After the failures Circle moved its cash, about a quarter of its reserves, to the custodian of its reserve fund, a Category I bank subject to the full ratio. Tether reports almost no bank deposits.

The pattern is the one Proposition 4.4 predicts once the requirement on issuer deposits enters the price a bank will pay for them: a bank that must hold a dollar of HQLA against each dollar of issuer deposits can pay at most the HQLA yield, while a bank outside the ratio can lend part of them and pay more. Issuer deposits therefore gravitate to the banks

Table 7: Where issuer deposits sat. Banks holding Circle’s cash reserves in March 2023 and the custodian to which the cash moved, with total assets (\$ billion, call reports, 2022–23) and status under the liquidity coverage ratio. Sources: Circle’s statements of 10–13 March 2023, CRS Insight IN12130, FDIC call reports and the 2019 tailoring rule.

Bank	Role	Assets	Liquidity coverage ratio
Silicon Valley Bank	Circle cash (\$3.3bn at failure)	209–218	not subject (Category IV, wSTWF < \$50bn)
Signature Bank	Circle cash; digital-asset deposits \$17.8–23.5bn	110–122	not subject (no holding company)
Silvergate Bank	Circle cash; digital-asset deposits \$11.9–13.3bn	11–16	not subject (< \$100bn)
Customers Bank	Circle cash; CBIT deposits \$1.9–2.3bn	20–22	not subject (< \$100bn)
New York Community Bank (Flagstar)	Circle cash	90 (2022Q4)	not subject (< \$100bn)
Citizens Trust Bank	Circle cash	0.8	not subject (< \$100bn)
Cross River Bank	added 12 March 2023	9	not subject (< \$100bn)
BNY Mellon	Reserve Fund custodian; Circle cash after March 2023	–	full LCR (Category I)

with the weakest requirement on them—and those are the banks whose failure the run–bank loop makes likely. The concentration of risk in 2023 was a consequence of where the liquidity rules applied, not of the run-off rate they assign.

7.2 How partner banks held issuer deposits

Table 8 compares the balance sheets of the partner banks that disclosed their digital-asset deposits with those of all banks with assets above \$10 billion in the same quarter. Silvergate held cash and securities equal to 0.87 of its assets against a peer median of 0.27, and its loan share was 55 percentage points below its peers’: it ran as a narrow bank. Its liquidity in excess of a peer of its size, per dollar of digital-asset deposits, implies an internal run-off of 0.74 on those deposits, which is the value of λ_{int} in the calibration. Signature, whose digital-asset deposits were a quarter of its funding, held liquidity implying a run-off of 0.33 in the quarter shown and 0.23 in the next; Customers and Metropolitan, for which such deposits were a small part of funding, held less liquidity than their peers. Weighted by deposits, the implied run-off is 0.41. Because the calibration takes λ_{int} from Silvergate’s two quarters, they are not independent evidence for the bank channel; the independent support is Silvergate’s balance sheet before it disclosed those deposits—cash and securities equal to 0.88 of its assets at the end of 2021, against a peer median of 0.33—together with Signature and the evidence of Lee and Tou (2026).

Table 8: How partner banks held digital-asset deposits. Quarter-end balance sheets, \$ billion and shares of total assets; peers: median of all banks with assets above \$10 billion in the same quarter. Liquid assets: cash and balances due plus securities. Implied run-off: liquid assets in excess of the peer median share, per dollar of digital-asset deposits.

Bank, quarter	Assets	Digital deposits	Liquid share		Loan share		Implied run-off
			Bank	Peers	Bank	Peers	
Silvergate, 2022-06	15.7	13.3	0.87	0.29	0.09	0.63	0.69
Silvergate, 2022-09	15.3	11.9	0.87	0.26	0.09	0.65	0.79
Signature, 2022-09	114.5	23.5	0.33	0.26	0.65	0.65	0.33
Customers, 2022-06	20.2	2.1	0.19	0.29	0.77	0.63	−0.92
Customers, 2022-09	20.4	1.9	0.21	0.26	0.75	0.65	−0.56
Metropolitan, 2022-06	6.9	1.2	0.34	0.29	0.63	0.63	0.29
Metropolitan, 2022-09	6.4	0.8	0.26	0.26	0.71	0.65	−0.02

This is the bank channel of Proposition 4.2(ii), measured with public call reports. It matches the evidence of Lee and Tou (2026), who link on-chain and interbank payment data and find that banks holding stablecoin reserves keep substantially larger reserve balances and have loan shares about 14 percentage points below their peers. In the calibrated model a specialist partner outside the ratio that receives \$30 billion of issuer deposits sees its loan share fall from 0.50 to 0.35. The comparison is not causal: banks chose to serve crypto firms, and the sample is four banks. It measures how the deposits were held.

7.3 Partner banks in March 2023

The events. Three banks with issuer or crypto-platform deposits failed or closed within five days in March 2023. Silvergate held \$11.9 billion of digital-asset customer deposits against \$1.11 billion of equity in September 2022, met a 68 percent run in the fourth quarter by selling \$5.2 billion of securities at a \$718 million loss, and announced voluntary liquidation on March 8. Silicon Valley Bank, where Circle held \$3.3 billion, failed on March 10 after a general run; Signature Bank, with \$17.8 billion of digital-asset deposits against \$8.01 billion of equity, was closed on March 12. The coins exposed to them lost their pegs that weekend in the order of their exposure (Appendix D.4).

The capital condition. Equation (4.5) says a partner bank fails in a full redemption when the losses it takes to raise the cash exhaust its equity. Let the bank use its cash C first, then sell securities of book value Sec at the discount $d = 1 - q_1$, and liquidate loans at the cost φ

Table 9: Partner banks and the capital condition. \$ billion; one pre-run quarter per bank, all bank-quarters with cash, securities and sources in Table D.2. D^S is the bank’s digital-asset deposits; ^bCircle’s deposit at Silicon Valley Bank (SVB) only. Silvergate announced its liquidation, and Signature and SVB were closed, in March 2023. The threshold is the fire-sale discount d^* at which repaying all digital-asset deposits, cash first and securities next, with loans liquidated at $\varphi = 0.25$ beyond the securities, exhausts equity (equation (7.1)); “cash covers” means cash exceeds the deposits. The next column applies Silvergate’s realized discount of 0.138.

Bank, quarter	Equity E	Deposits D^S	D^S/E	d^*	Fails at 0.138	Outcome
Silvergate, 2022Q3	1.11	11.87	10.7	0.10	yes	liquidated
Signature, 2022Q3	7.69	23.50	3.1	0.39	no	closed
Customers, 2022Q3	1.48	1.90	1.3	0.50	no	survived
Metropolitan, 2022Q3	0.59	0.76	1.3	0.92	no	survived
SVB, 2022Q4	15.46	3.30 ^b	0.2	cash covers	no	failed

per dollar raised only once the securities are gone. Raising $D^S - C$ then costs

$$\ell(d) = \begin{cases} \frac{d}{1-d} (D^S - C) & \text{if } D^S - C \leq (1-d) \text{ Sec,} \\ d \text{ Sec} + \varphi [D^S - C - (1-d) \text{ Sec}] & \text{otherwise,} \end{cases} \quad (7.1)$$

and the bank fails if and only if $\ell(d) > E$. The discount itself enlarges the sale: to raise a dollar at discount d the bank sells $1/(1-d)$ of book value. The threshold discount d^* solves $\ell(d^*) = E$; when the securities suffice it is $E/(E + D^S - C)$. Table 9 evaluates (7.1) for every partner bank that disclosed its digital-asset deposits, one pre-run quarter each, and Table D.2 at each quarter-end from June 2022 to March 2023 (13 bank-quarters), with FDIC call reports for equity, cash and securities and $\varphi = 0.25$. The condition fails only for Silvergate, in June and September 2022, at any discount above 0.09 and 0.10. Signature’s threshold never falls below 0.39; Customers and Metropolitan, the other two banks with crypto deposit businesses, would need discounts of at least 0.44 or held cash that covered those deposits outright. By December 2022 Silvergate’s own cash covered what remained of its digital-asset deposits, after the outflows had already forced the sales. The condition separates the one bank whose failure ran through its digital-asset deposits from the others, using only balance-sheet data that were public before the run.

Silvergate’s digital-asset deposits were 10.7 times its equity; the threshold discount was 10 percent and the discount it realized when it sold \$5.2 billion of securities was 13.8 percent, matching the 14 percent unrealized loss on its held-to-maturity book in its XBRL filings. The model predicts failure in a full run, and the bank survived the 68 percent run of the fourth quarter only by borrowing \$4.3 billion from the Federal Home Loan Bank, with equity

reduced by half, before liquidating in March. Signature’s issuer-type deposits were two to three times its equity in the second half of 2022 and its threshold discount never below 0.39; no plausible fire sale fails it through those deposits, and the model attributes its closure to the run on its other deposits, which the supervisors’ account also does. Silicon Valley Bank held \$3.3 billion of Circle’s reserves against \$15.46 billion of equity; the issuer deposit is irrelevant to its failure, which ran from its securities losses—16.5 percent on a \$91 billion held-to-maturity book, equal to its equity—through a general run to the issuer, the reverse direction of transmission. Written as a limit, (7.1) says a bank whose securities suffice can carry issuer-type deposits net of cash up to $E(1 - d)/d$: at the discount each bank realized, Silvergate’s limit was \$6.9 billion against an exposure of \$10.0 billion, Signature’s \$72 billion and Silicon Valley Bank’s \$78 billion. The FDIC’s proposed 40 percent cap, set on the issuer’s side, would have bound in none of the three cases: Silvergate’s deposits came from many crypto firms, and Circle’s deposit at Silicon Valley Bank was 7.6 percent of its reserves.

7.4 Lending in the observed range

Proposition 4.1(i) predicts, for an economy far below s_1 , a negligible lending response to issuer absorption, whatever a bank’s liquidity headroom. Two pieces of evidence bear on it, and each identifies something different. The first is the aggregate time series. In the weekly local projections of Section D.1, an issuer inflow of \$3.5 billion changes bank loans (H.8) by +8.1 billion dollars at four weeks (standard error 7.8) outside the 2022 episode, and deposits by +19.4 (16.4); neither is distinguishable from zero. The second is a panel of 160 banks with assets above \$10 billion over 26 quarters of 2020–2026 (Table 10). The quarterly change in log loans is regressed on issuer absorption interacted with an indicator for below-median liquid-asset ratios, with bank and time fixed effects and standard errors clustered by bank. The time fixed effects absorb the common component of absorption, so the panel identifies only the differential response of low-headroom banks relative to high-headroom banks; a response common to all banks would not appear in it, which is why the aggregate series is reported alongside. The interaction is -0.47 percentage points per \$10 billion of absorption (standard error 0.25) and the additional interaction with bill-scarce quarters is $+0.64$ (0.29), the opposite of the sign a binding constraint would produce. With errors clustered by bank the 95 percent interval would exclude a differential decline larger than one percentage point per \$10 billion; with errors clustered by bank and quarter, which allow for common quarterly shocks, it is $[-1.11, +0.18]$, so the direction and the size of the differential response are uncertain. With absorption measured from the attestations, as the issuers’ bill and repo holdings, the headroom interaction is $+0.34$ (0.21) and the bill-scarce interaction -0.55

Table 10: Bank lending and issuer absorption by liquidity headroom. Panel of 160 US banks with assets above \$10 billion, 2020–2026 quarterly, bank and time fixed effects, standard errors clustered by bank (quarter-clustered and bootstrap inference in Table D.3). Coefficients in percentage points per \$10 billion of absorption. Low headroom is a liquid-asset ratio (securities plus cash over deposits) below the 2021Q4 median; bill-scarce quarters are those in the bottom tercile of the bill share of marketable debt.

Dependent variable	Absorption measure	Absorb $\times$ low headroom	Absorb $\times$ low headroom $\times$ bill-scarce	<i>n</i>
$\Delta \ln$ net loans	stablecoin supply	−0.47 (0.25)	+0.64 (0.29)	3736
$\Delta \ln$ net loans	attestation bills and repo	+0.34 (0.21)	−0.55 (0.30)	3035
$\Delta \ln$ securities	stablecoin supply	−0.03 (0.53)	−0.60 (0.61)	3736
$\Delta \ln$ securities	attestation bills and repo	−0.59 (0.57)	+0.84 (0.68)	3035
$\Delta \ln$ deposits	stablecoin supply	+0.19 (0.20)	−0.68 (0.27)	3736
$\Delta \ln$ deposits	attestation bills and repo	+0.18 (0.36)	−0.23 (0.41)	3035
$\Delta \ln$ net loans, 2016 money-fund reform	government-fund inflows	−0.05 (0.50)		

(0.30). The regressions interact a common series with a bank characteristic, so errors may be correlated across banks within a quarter. With standard errors clustered by quarter, or by bank and quarter, and a wild cluster bootstrap by quarter, which is reliable with 21–26 clusters (Table D.3), the supply measure’s interactions are insignificant (p of 0.15 and 0.11), but the attestation measure’s bill-scarce interaction becomes significant at the 5 percent level ($p = 0.035$, and 0.049 without the deposit control). That coefficient has the sign a binding constraint would produce and is not small; it has the opposite sign under the supply measure. The difference has two sources. The supply measure’s estimates rest on the five quarters before the attestation series begins: on the 21 common quarters its interactions are +0.14 and +0.02. And within those quarters the bill-like part of absorption and the rest of coin growth enter with opposite signs, with bill-scarce interactions of −0.51 and +0.57 when both enter one regression, so a measure that adds them nets out. The bill-like part has the sign the bill channel predicts, but with errors clustered by bank and quarter its interval includes zero (standard error 0.31). The regressions control for the bank’s same-quarter deposit growth; dropping that control, which would absorb part of a deposit-outflow channel, leaves the picture unchanged (−0.34 (0.30) for the supply measure’s headroom interaction and −0.72 (0.39) for the attestation measure’s bill-scarce interaction). The same design applied to the money-fund reform of October 2016, when about \$1 trillion moved into government funds that absorbed bills, agencies and repo, gives −0.05 (0.50). The evidence does not show a lending response to issuer absorption, in the aggregate or across banks by headroom, but it is imprecise: the aggregate interval includes declines of several billion dollars of loans per \$3.5 billion of inflow, and one cross-sectional interaction points the way a binding constraint would.

The steep part of the zone—a sharp rise in the price of bank liquidity as buffers approach the floor—is not in our sample. A kink of this kind is documented for the aggregate reserve

buffer in September 2019, when reserves had fallen to \$1.4 trillion and the repo rate reached 10 percent on a day of tax payments and Treasury settlements (Afonso et al., 2021; Copeland et al., 2025). Our contribution is to place a new absorber in front of it and to give the location of the zone in issuance; the direct test requires a sample that enters its steep part.

7.5 Which dollars buy bills

The fifth implication of Section 4.6 is that, far from the threshold, the permanent price effect of issuer purchases is the convenience slope only, and that any larger short-run impact comes from the dollars that actually buy bills. We test it by weighting each issuer’s weekly flow by its bill share at the previous attestation, which splits issuer dollars into bill-buying and non-bill dollars; Appendix D gives the data, the specification and the robustness checks. Table 11 reports four-week local projections of the three-month bill yield. Bill dollars lower the yield by 19.1, 13.7 and 10.8 basis points per \$3.5 billion in the samples from July 2022, January 2023 and July 2023; non-bill dollars do not lower it in any sample, and the difference is significant at the 5 percent level in the two longer samples (t of 2.3 and 2.8) and at the boundary in the shortest (2.0). It survives lagging the composition by an extra quarter and controlling for the spread that drives the composition, and only 4 of 1,000 placebo re-timings of Circle’s shares reproduce it. The flow-weighted dollars are a proxy for net purchases, not exogenous purchase shocks, so the evidence is compatible with the model’s implication rather than a test of it.

The effect is gone within six to twelve weeks, and moving an existing stock between bills and repo leaves yields unchanged (+0.3 basis points per \$3.5 billion reallocated, standard error 0.5). New dollars that buy bills move prices for a few weeks, as a flow in a segmented market; the stock of issuer purchases has no detectable level effect, against a model slope of 0.09 basis points per \$3.5 billion. That is the asset-swap pattern of Proposition 4.1(i): money funds holding reverse repos replace a departing buyer and, within weeks, accommodate a new one. The reserve choice itself follows Proposition 4.4: a 10 basis point rise in the bill–SOFR spread moves Circle’s bill share by +7.0 percentage points (standard error 3.0), Tether’s composition is event-driven, and after March 2023 Circle moved its bank deposits to its custodian rather than out of deposits, the margin the run-loss term points to (Appendix D).

8 Conclusion

Stablecoins are private money whose reserves are the assets banks hold for liquidity. We have shown that, once banks’ compliance with the liquidity coverage ratio is modelled as the rule

Table 11: Bill-buying versus non-bill issuer dollars and the three-month bill yield. Weekly local projections; coefficients in basis points per \$3.5 billion with Newey–West standard errors in parentheses; the last column is the t -statistic of the difference between the two coefficients, with the HAC covariance of the two estimates in its standard error. Robustness in Table D.4.

Horizon, sample	Bill dollars	Non-bill dollars	Difference (t)
4 weeks, 2022.7–	−19.1 (7.5)	+11.3 (6.1)	−2.3
4 weeks, 2023.1–	−13.7 (4.3)	+9.0 (4.3)	−2.8
4 weeks, 2023.7–	−10.8 (5.1)	+8.7 (5.0)	−2.0
2 weeks, 2023.7–	−6.5	+4.4	−2.1
Shares lagged one extra quarter, 4 weeks, 2023.1–	−12.6		−2.7
Constant issuer-mean shares (between issuers only), 2023.7–	−9.3 (4.4)		−1.8
Permutation placebo, 1,000 re-timings of Circle’s monthly shares (45 months), 2023.1–: mean t −0.35, s.d. 0.70; 4 draws reach −2.8			
Permutation placebo, 2022.7–: mean t 0.15, s.d. 0.98; 6 draws reach −2.3			

and the discount window make it costly, stablecoin issuance crowds out bank collateral along a smooth schedule rather than at a cliff, and through two channels: the issuer’s purchases of bills and the requirement its deposits impose on the banks that receive them. The answer to the question in the title has three parts. Crowding out begins well before the regulatory threshold: in the US calibration the premium on safe assets reaches 25 basis points at a total stock of \$661 billion. It is about half-way to its maximum at the threshold, \$961 billion, where the disclosed slack is used up. And because coin demand rises as collateral scarcity lowers the bill yield, there is a largest demand with a normal equilibrium: about \$1,949 billion with the observed reserve mix, just short of the 2028 projection, and \$2.0–3.1 trillion if issuers rebalance toward bank deposits as competitive deposit pricing leads them to; beyond it issuance does not settle—in the calibration there is no stable equilibrium at all on the range we compute—and the stock runs into a collateral spiral that a supply or policy response would have to stop.

Policy follows from the two channels and from the fact that the stock of coins is set by demand. Instruments that change what a coin absorbs dominate instruments that remove coins. Requiring issuers to hold part of their reserves at the Federal Reserve, an asset the GENIUS Act already allows, removes their claim on bank collateral without removing the coins households demand, provided the Federal Reserve lends nearly all of the funds back to banks; it is the most valuable instrument at the stocks projected for 2028 if more than 94.6 percent are recycled and holding them costs the Federal Reserve less than 21 basis points a year, and worthless if they are sterilized. At \$1 trillion it and discount-window readiness are effectively tied. Discount-window readiness, which raises the collateral banks can use to cover shortfalls, is nearly as valuable without that condition and moves the fold outward, to about \$3.6 trillion when banks pre-pledge collateral equal to 90 percent of their net outflows. Issuance caps lower welfare wherever they bind, in every calibration. The run-off rate on

issuer deposits and the proposed 40 percent concentration limit do little for collateral while issuers hold bills; their role is the run externality, where the binding quantity is the receiving bank’s capital, and where the evidence says the gap is in coverage: the banks that held issuer deposits before March 2023 were outside the liquidity coverage ratio altogether, and the one whose failure ran through those deposits fails the capital condition the model derives.

The evidence available is compatible with an economy in the early part of the zone: banks keep their ratios well above the floor, bill-buying issuer dollars move bill prices for a few weeks and other issuer dollars do not, and bank lending does not respond measurably to issuer absorption. It does not test the fold, which no economy has yet approached. The quantities that locate it—the collateral banks can pledge, the stigma of borrowing against it, the Treasury’s response to bill demand, the issuers’ response to the spread between bills and bank deposits, and what coin holders treat as the alternative—are observable, and an economy whose collateral buffer is within reach of planned issuance is the natural place to measure them.

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Appendix A Proofs

Notation is as in Sections 3–5. Write $M \equiv q_0 R(L)$ for the marginal return a bank forgoes on a safe asset, $B(q) \equiv S(q) - \alpha s/q$ for the safe assets available to banks at price q , and $b_j(M)$ for bank j 's demand, the inverse of m_j in (4.2).

A.1 Lemma 4.1

Proof. Each term of m_j is a positive constant times $\bar{F}$ evaluated at an increasing affine function of b ; with F continuous and of full support, m_j is continuous and strictly decreasing in b , with limits $1 + \theta(c + \varphi + \tilde{\tau}_{a,j})$ as $b \rightarrow -\infty$ and 1 as $b \rightarrow \infty$. Its inverse $b_j(M)$ is therefore continuous and strictly decreasing on $(1, \bar{M}_j)$, $\bar{M}_j \equiv 1 + \theta(c + \varphi + \tilde{\tau}_{a,j})$, with $b_j \rightarrow \infty$ as $M \downarrow 1$ and $b_j \rightarrow -\infty$ as $M \uparrow \bar{M}_j$. By (4.3), $L(q) = e + E - qS(q)$ is decreasing in q , so $R(L(q))$ is increasing and $M(q) = qR(L(q))$ is continuous and strictly increasing; $B(q)$ is increasing. Excess demand $Z(q) \equiv \sum_j b_j(M(q)) - B(q)$ is therefore continuous and strictly decreasing on the set of prices with $M(q) \in (1, \min_j \bar{M}_j)$, positive near the lower end and negative near the upper end, and has exactly one zero. Continuity in s follows because Z is jointly continuous in (q, s) and strictly monotone in q . $\square$

A.2 Proposition 4.1

Proof. With symmetric banks write the aggregate first-order condition as $M - 1 = \theta[c \bar{F}(\Sigma/\sigma) + \varphi \bar{F}((\Sigma + \kappa \text{REQ})/\sigma) + \tilde{\tau}_a \bar{F}(\Sigma_a/\sigma)]$, where $\Sigma \equiv \theta B - \text{REQ}$ is the requirement slack and $\Sigma_a \equiv \theta B - \text{NEED}$ the actual slack, both at the equilibrium price, and REQ, NEED are system totals.

(i) Since $\lambda_a \leq \lambda_{\text{reg}}$, $\text{NEED} \leq \text{REQ}$, so $\Sigma_a \geq \Sigma$ and $\Sigma + \kappa \text{REQ} \geq \Sigma$; $\bar{F}$ is decreasing, so each term is at most its coefficient times $\bar{F}(\Sigma/\sigma)$, which gives the bound. The lending change between two issuance levels is $-\bar{S} \Delta q_0$, and Δq_0 is bounded by the change in M divided by $\partial M / \partial q > 0$.

(ii) At a given price, issuance lowers both slacks: $d\Sigma/ds = -[\theta\alpha/q + \lambda_{\text{reg}}(1 - \alpha) - \gamma_{\text{max}}] < 0$ and $d\Sigma_a/ds = -[\theta\alpha/q + \lambda_a(1 - \alpha) - \gamma_{\text{max}}] < 0$ by Assumptions 1–2 (for q near q_F), and

$d\text{REQ}/ds = \lambda_{\text{reg}}(1 - \alpha) - \gamma_{\text{max}}$ is bounded. Excess demand $Z(q; s)$ is therefore increasing in s at every q , and with Z strictly decreasing in q (Lemma 4.1) the equilibrium price, and with it M and π , is strictly increasing in s ; lending $L = e + E - q_0\bar{S}$ is strictly decreasing. Continuity is Lemma 4.1.

(iii) At s_1 the requirement slack at the frictionless price is zero by (4.4); at the equilibrium price q_0 it is $\theta\alpha s_1(1/q_F - 1/q_0)$, proportional to the change in the price of bonds and small relative to σREQ when that change is small. With F symmetric, $\bar{F}(0) = 1/2$, so the compliance term is $\theta c/2$. The compliance term equals $\eta\theta c$ where $\bar{F}(\Sigma/\sigma) = \eta$, i.e. $\Sigma = \sigma\bar{F}^{-1}(\eta)$; since Σ falls with issuance at rate $w(\alpha)$ near q_F , the issuance at which this happens is $s_1 - \sigma\bar{F}^{-1}(\eta)/w(\alpha)$ to first order, symmetric around s_1 and with width proportional to $\sigma/w(\alpha)$.

(iv) As $\sigma \rightarrow 0$, $\bar{F}(x/\sigma) \rightarrow \mathbf{1}\{x < 0\}$ for $x \neq 0$, so the right-hand side of the first-order condition converges to $\theta[c\mathbf{1}\{\Sigma < 0\} + \varphi\mathbf{1}\{\Sigma < -\kappa\text{REQ}\} + \tilde{\tau}_a\mathbf{1}\{\Sigma_a < 0\}]$, the correspondence of the kinked model (with the value at a threshold any point between the adjacent steps). Equilibria converge to those of the kinked model, which has the stated step structure. $\square$

A.3 Proposition 4.2

Proof. (i) Compare $\alpha = 1$ and $\alpha = 0$ at the same issuance s and the same price q . Requirement slack falls by $s[\theta/q - \gamma_{\text{max}}]$ with bills and by $s[\lambda_{\text{reg}} - \gamma_{\text{max}}]$ with deposits; with $\lambda_{\text{reg}} \leq \theta/q_F$ the fall is at least as large with bills near the frictionless price. Actual slack falls by $s[\theta/q - \gamma_{\text{max}}]$ with bills and by $s[\lambda_a - \gamma_{\text{max}}]$ with deposits, which is smaller since $\lambda_a < \lambda_{\text{reg}} \leq \theta/q_F$. The capacity threshold $(1 - \kappa)\text{REQ}$ moves with REQ , which rises by $(1 - \alpha)\lambda_{\text{reg}}s$ less than with bills while B falls by $\alpha s/q$, so the deleveraging slack $\theta B - (1 - \kappa)\text{REQ}$ also falls at least as much with bills. Each of the three arguments of $\bar{F}$ in (4.2) is therefore no larger with bills at every q , so aggregate demand for safe assets relative to supply is at least as high, and by the monotonicity of Lemma 4.1 the equilibrium premium is at least as high and lending at least as low. The inequality is strict wherever a term with a strictly smaller argument has positive density, that is, whenever the deleveraging or stress terms are active.

(ii) For a bank small relative to the system, M is given. Its first-order condition $m_1(b_1; D_1^S) = M$ has thresholds $\ell_1\text{REQ}_1$, $(1 - \kappa)\ell_1\text{REQ}_1$ and NEED_1 , which rise in D_1^S at rates $\ell_1\lambda_1$, $(1 - \kappa)\ell_1\lambda_1$ and λ_a ; holding M fixed, θb_1 rises with them, so $\partial b_1/\partial D_1^S > 0$, at a rate between λ_a/θ and $\ell_1\lambda_1/\theta$ depending on which term is active. Its assets $A_1 = L_1 + q_0 b_1$ rise one for one with its deposits. Writing $\beta \equiv q_0 \partial b_1/\partial D_1^S$, its loans rise by $1 - \beta$ per dollar of deposits, so

$$\frac{d(L_1/A_1)}{dD_1^S} = \frac{(1 - \beta)A_1 - L_1}{A_1^2} = \frac{q_0 b_1/A_1 - \beta}{A_1},$$

which is negative whenever β exceeds the bank's liquid share $q_0 b_1 / A_1$. With λ_1 at $\lambda_{\text{reg}} = 1$ or at the calibrated λ_{int} , β is close to $\ell_1 \lambda_1$ and exceeds the liquid share of a bank whose requirement binds, so the loan share falls. $\square$

A.4 Proposition 4.3

Proof. Under the pre-positioning rule the partner holds $\theta b_1 = \text{REQ}_1 + w_1 \Sigma$ with $\text{REQ}_1 = \gamma_{\max} D_1^H + N_{0,1} + \lambda_1(1 - \alpha)s$ and $\Sigma = \Sigma_0 - [\lambda_1(1 - \alpha) - \gamma_{\max} + \alpha\theta/q_F]s$, where $D_1^H = \text{sh}_1(e - s)$. Substituting into (4.5),

$$(1 - \alpha)s \hat{\lambda} = E_1/\varphi - (1 - \kappa_q)(\gamma_{\max} D_1^H + N_{0,1}) + \kappa_q \lambda_1(1 - \alpha)s + \kappa_q w_1 \Sigma = A_1 + B_1 s,$$

with A_1 and B_1 as defined in the text. (i) $\hat{\lambda} \geq 1$ if and only if $A_1 \geq (1 - \alpha - B_1)s$. If $B_1 \geq 1 - \alpha$ this holds for every s (given $A_1 > 0$) and $\phi(\lambda) = \alpha \bar{h}$ on $[0, 1]$, so π^* and ρ do not depend on s . If $B_1 < 1 - \alpha$ it holds exactly for $s \leq s_f = A_1/(1 - \alpha - B_1)$. Beyond s_f , $\hat{\lambda}(s) = A_1/[(1 - \alpha)s] + B_1/(1 - \alpha)$ is strictly decreasing in s because $A_1 > 0$; by Lemma C.1(iii), π^* is strictly increasing in s while $\hat{\lambda} \leq 1 - \phi_1$. For $\hat{\lambda} \in (1 - \phi_1, 1)$, $dS/d\hat{\lambda} = 1 - (1 - \phi_1)/\hat{\lambda} > 0$ and $dH/d\hat{\lambda} = \text{hold}(\hat{\lambda}; \phi_0) > 0$, and $d\pi^*/d\hat{\lambda} < 0$ if and only if $H[1 - (1 - \phi_1)/\hat{\lambda}] < S \text{ hold}(\hat{\lambda}; \phi_0)$ (condition M'), which holds in all our calibrations. Since $\rho = p \Pr[\pi < \pi^*]$ with a continuous prior, ρ is nondecreasing in π^* .

(ii) θb_1 rises with λ_1 by $(1 - w_1)(1 - \alpha)s$ per unit; since $\kappa_q > 0$ the numerator of (4.5) rises, and so does $\hat{\lambda}$. The slack a coin absorbs, $\theta\alpha/q_F + \lambda_1(1 - \alpha) - \gamma_{\max}$, rises with λ_1 for $\alpha < 1$, so by (4.4) and Proposition 4.1(iii) the zone moves toward today's stock. The date-0 equilibrium does not depend on ρ , so $\hat{\lambda}$, π^* and $\rho = p \Pr[\pi < \pi^*]$ are determined in sequence and are unique. $\square$

A.5 Proposition 5.1

Proof. (i) Under (C1) the Federal Reserve lends μS back to banks in proportion to their deposits, so the identity (4.3) for aggregate lending is unchanged, and the funding is neither an LCR outflow nor a stress outflow. Let $S(\mu)$ be the stock in the normal equilibrium under the share. At $\mu = 0$ it is the stock without a cap, which exceeds S_c because the cap binds, and $(1 - \mu)S(\mu) \rightarrow 0$ as $\mu \rightarrow 1$; by continuity there is a μ with $(1 - \mu)S(\mu) = S_c$, and then $S_\mu = S_c/(1 - \mu) > S_c$. Because the share applies to the whole stock, the issuer holds $\alpha(1 - \mu)S_\mu = \alpha S_c$ in bonds and $(1 - \alpha)S_c$ in bank deposits, as under the cap, so the supply of safe assets to banks, $B(q)$, and every D_j^S are the same. Households hold $S_\mu - S_c$ more coins and as many fewer deposits, which the recycled funds $\mu S_\mu = S_\mu - S_c$ replace in the

Table A.1: Proposition 5.1 in numbers. Large partner. A cap at s_1 against the Federal Reserve deposit share μ that leaves the issuer with the same bonds and bank deposits, $(1 - \mu)S_\mu = S_c$, with full recycling. Stocks and lending in \$ billion; premium in basis points; changes (share minus cap) in \$ billion a year. “Real costs” is the fall in the expected costs of deleveraging and stress liquidation. In every row both banks’ requirement and actual need, the price of collateral and bank holdings are lower and lending is higher under the share.

Demand	μ	Stock, cap/share	Premium, cap/share	ΔL	ΔV	Δ loan surplus	Real costs	ΔW
Threshold	0.10	962 / 1,068	86.5 / 84.7	+0.8	+4.2	+0.03	+0.05	+4.3
\$1tn	0.09	962 / 1,055	86.5 / 84.9	+0.7	+3.6	+0.03	+0.04	+3.7
\$2tn, $\varepsilon_S = 2$	0.53	962 / 2,066	72.9 / 57.1	+17.6	+94.7	+0.72	+0.31	+95.7

same shares: bank funding is unchanged, and REQ_j and NEED_j are both lower by $\gamma_{\max}$ times bank j ’s share of $S_\mu - S_c$.

Next, m_j in (4.2) is strictly increasing in REQ_j at given D_j^S and $b \geq 0$. Write $\sigma_j = (\sigma_N^2 \text{REQ}_j^2 + v_j^2)^{1/2}$ with $v_j = (1 - \lambda_j)\sigma_S D_j^S$, and the three arguments of $\bar{F}$ as $z_k = (\theta b - T_k)/\sigma_j$ with $T_1 = \ell_j \text{REQ}_j$, $T_2 = (1 - \kappa_j)\ell_j \text{REQ}_j$ and $T_3 = \text{NEED}_j = \text{REQ}_j - (\lambda_j - \lambda_a)D_j^S$. Then

$$\begin{aligned} \frac{\partial z_1}{\partial \text{REQ}_j} &= -\frac{\ell_j v_j^2 + \theta b \sigma_N^2 \text{REQ}_j}{\sigma_j^3}, & \frac{\partial z_2}{\partial \text{REQ}_j} &= -\frac{(1 - \kappa_j)\ell_j v_j^2 + \theta b \sigma_N^2 \text{REQ}_j}{\sigma_j^3}, \\ \frac{\partial z_3}{\partial \text{REQ}_j} &= -\frac{v_j^2 + \sigma_N^2 \text{REQ}_j (\text{REQ}_j - \text{NEED}_j + \theta b)}{\sigma_j^3}, \end{aligned}$$

all negative because $\theta b \geq 0$ and $\text{REQ}_j \geq \text{NEED}_j$. Since $\bar{F}$ is decreasing, every term of m_j rises with REQ_j . Hence each bank’s demand $b_j(M)$ is lower under the share at every M ; with $B(q)$ unchanged, excess demand is lower at every price, and by the argument of Lemma 4.1 the equilibrium price and M are lower. With full recycling $L = e + E - q_0 S(q_0)$, so lending is higher.

(ii) With the loop shut the run terms of (5.1) vanish. V is higher because $S_\mu > S_c$. $\Pi(L) - L$ rises with L wherever $R(L) \geq 1$, which holds in equilibrium because $q_0 R(L) = M \geq 1$ and $q_0 \leq 1$. The expected deleveraging and stress shortfalls $\mathbb{E}[(T_k + \xi_j - \theta b_j)^+]$ fall with the lower thresholds but rise with the lower holdings, since $B(q)$ is increasing in q ; their net change has no general sign.

(iii) Without recycling bank funding falls by μS , lending by approximately μS at given prices, and $\Pi(L) - L$ by approximately $(R - 1)\mu S$. $\square$

A.6 Proposition 5.2

Proof. (i) Coin demand at the equilibrium yield is $S^d(y(s); k) = \ell e^{-\varepsilon(y(s)-r_0)}$, so excess demand is $g(s) = \ell e^{-\varepsilon(y(s)-r_0)} - (s_0 + s) = e^{-\varepsilon(y(s)-r_0)}[\ell - h(s)]$. Its sign is that of $\ell - h(s)$, so g crosses zero from above where h crosses ℓ from below, that is, where h is increasing. Differentiating h with $y' < 0$ gives the expression for h' .

(ii) On $[0, s^*)$, $\varepsilon(s_0 + s)|y'| < 1$, so h is strictly increasing and $h(s^*)$ is its maximum on $[0, s^*]$. For $\ell \in [h(0), \ell^*]$ the intermediate value theorem gives exactly one $s \in [0, s^*]$ with $h(s) = \ell$, and it is stable by (i); it is the smallest stable equilibrium. For $\ell > \ell^*$, $h < \ell$ on $[0, s^*]$, so $g > 0$ there: issuance growing from today's stock does not stop on $[0, s^*]$. Because $\varepsilon(s_0 + s)|y'|$ crosses one at s^* from below with positive slope, h' changes sign from positive to negative there, so s^* is a strict local maximum and $h < \ell^*$ on a right neighbourhood of s^* ; without the crossing condition h could only flatten at s^* and rise again, and the normal equilibrium could lie beyond it. Beyond s^* , h decreases until $\varepsilon(s_0 + s)|y'|$ falls back below one, and any further equilibrium lies where h rises again to ℓ .

(iii) If $\tilde{y} \geq y$ on $[0, s^*]$ then $\tilde{h}(s) = (s_0 + s)e^{\varepsilon(\tilde{y}(s)-r_0)} \geq h(s)$ there. If $|\tilde{y}'| \leq |y'|$ on $[0, s^*]$ then $\varepsilon(s_0 + s)|\tilde{y}'| < 1$ on $[0, s^*)$, so $\tilde{s}^* \geq s^*$. Hence $\tilde{\ell}^* = \max_{s \leq \tilde{s}^*} \tilde{h}(s) \geq \max_{s \leq s^*} \tilde{h}(s) \geq \max_{s \leq s^*} h(s) = \ell^*$.

(iv) In the kinked limit with $\alpha = 0$, Proposition 4.1(iv) gives a yield that is constant on each side of s_1 and jumps down there by $y_I - y_{III}$, so $h(s_1^+) = h(s_1^-)e^{-\varepsilon(y_I - y_{III})} < h(s_1^-)$, and h rises with slope one on both sides. For $\ell \in (h(s_1^+), h(s_1^-))$ there is one solution below s_1 and one above, both on increasing branches and so stable by (i), and an unstable one at s_1 at the yield in the interval of equilibrium prices of the kinked model at which demand equals $s_0 + s_1$. $\square$

A.7 Proposition 5.3

Proof. With the pre-positioning rule, substitution into (4.5) gives

$$\hat{\lambda} = \kappa_q \lambda_1 (1 - w_1) + \frac{E_1/\varphi - (1 - \kappa_q)(\gamma_{\max} D_1^H + N_{0,1}) + \kappa_q w_1 (\Sigma_0 + (\gamma_{\max} - \alpha \theta / q_F) s)}{(1 - \alpha) s},$$

linear in λ_1 with positive slope. Setting $\hat{\lambda} = 1$ and solving gives (5.3). (i) The bracketed term is divided by s , so $\lambda^{\min}$ increases in s and decreases in E_1 , w_1 and Σ_0 . (ii) The first statement is Proposition 4.3(ii). For a bank outside the ratio the requirement (3.3) is the internal one with $\lambda_1 = \lambda_{\text{int}}$; a regulatory requirement λ_{reg} raises its threshold only where $\lambda_{\text{reg}} > \lambda_{\text{int}}$. (iii) s_{cap} solves $\lambda^{\min}(s) = 1$; for $s > s_{\text{cap}}$, $\hat{\lambda} < 1$ at $\lambda_1 = 1$. (iv) With

$\lambda_1 = 1$, $w_1 = 0$ and no other slack, $\hat{\lambda} \geq 1$ if and only if $E_1/\varphi + \kappa_q(1 - \alpha)s \geq (1 - \alpha)s$, i.e. $E_1 \geq \varphi(1 - \kappa_q)(1 - \alpha)s = (1 + \varphi)(1 - q_1)(1 - \alpha)s$. $\square$

The proofs that follow, of Lemma C.1, Proposition 4.4 and the results of Appendix E, are carried over from the earlier version of this paper. They use the kinked benchmark of Appendix F, in which the slack regime is the region below s_1 where every compliance term vanishes—the limit of the asset-swap region of Proposition 4.1(i)—and the constrained regime the region in which the premium is at its maximum.

A.8 Lemma C.1

Proof. (i) In the vanishing-noise limit the marginal holder assigns a uniform distribution to the redemption share λ and is indifferent between selling and holding: $\pi^*\mathcal{H} = \mathcal{S}$ (Ma et al., 2026, Propositions 1–2); the argument there uses only that the payoff gain from holding is linear in π and that the price impact does not depend on the fundamental, both of which hold when ϕ depends on λ but not on π . For a nondecreasing step function with finitely many jumps the integrands are bounded and piecewise continuous, so $\mathcal{S}$ and $\mathcal{H}$ exist. $\mathcal{H} > 0$: on $[0, 1 - \phi_0)$, with ϕ_0 the first step value, the holding payoff $(1 - \phi_0 - \lambda)/[(1 - \phi_0)(1 - \lambda)]$ is strictly positive, and $\eta \geq 0$ elsewhere. The payoff difference $D(\pi) = \pi\mathcal{H} - \mathcal{S}$ is linear and strictly increasing in π , so $\pi^* = \mathcal{S}/\mathcal{H}$ is unique. For a constant ϕ ,

$$\mathcal{S} = \int_0^{1-\phi} (1 - K\lambda) d\lambda + \int_{1-\phi}^1 \left[\frac{1-\phi}{\lambda} - K\lambda \right] d\lambda = (1 - \phi)[1 - \ln(1 - \phi)] - \frac{K}{2},$$

and, writing the holding payoff as $1/(1 - \phi) - \phi/[(1 - \phi)(1 - \lambda)]$,

$$\mathcal{H} = \int_0^{1-\phi} \left[\frac{1}{1-\phi} - \frac{\phi}{(1-\phi)(1-\lambda)} + \eta \right] d\lambda = 1 + \frac{\phi \ln \phi}{1-\phi} + \eta(1 - \phi),$$

which is equation (4.10) of Ma et al. (2026).

(ii) The selling payoff contains $-K\lambda$ on all of $[0, 1]$, so $\partial\mathcal{S}/\partial K = -\int_0^1 \lambda d\lambda = -1/2$, while $\mathcal{H}$ does not depend on K .

(iii) Let $\phi(\lambda) = \phi_0$ on $[0, \hat{\lambda})$ and $\phi_1 > \phi_0$ on $[\hat{\lambda}, 1]$. By Leibniz's rule, $d\mathcal{S}/d\hat{\lambda} = \text{sell}(\hat{\lambda}; \phi_0) - \text{sell}(\hat{\lambda}; \phi_1)$ and $d\mathcal{H}/d\hat{\lambda} = \text{hold}(\hat{\lambda}; \phi_0) - \text{hold}(\hat{\lambda}; \phi_1)$. For $\hat{\lambda} \leq 1 - \phi_1 \leq 1 - \phi_0$ both selling payoffs equal $1 - K\hat{\lambda}$, so $d\mathcal{S}/d\hat{\lambda} = 0$, while $\partial\text{hold}/\partial\phi = -\lambda/[(1 - \lambda)(1 - \phi)^2] < 0$ for $\lambda > 0$, so $d\mathcal{H}/d\hat{\lambda} > 0$. Hence $d\pi^*/d\hat{\lambda} = -\mathcal{S}(d\mathcal{H}/d\hat{\lambda})/\mathcal{H}^2 < 0$.

(iv) On the interval I_k where $\phi = \phi_k$, raising ϕ_k lowers the selling payoff by $1/\lambda$ per unit on $I_k \cap (1 - \phi_k, 1]$ and leaves it unchanged elsewhere (the selling payoff is continuous at $\lambda = 1 - \phi_k$), so $\partial\mathcal{S}/\partial\phi_k = -\int_{I_k \cap (1 - \phi_k, 1]} d\lambda/\lambda \leq 0$; it lowers the holding payoff by $\lambda/[(1 - \lambda)(1 - \phi_k)^2]$

on $I_k \cap [0, 1 - \phi_k]$ and removes η at the boundary if $1 - \phi_k \in I_k$, so $\partial \mathcal{H} / \partial \phi_k \leq 0$. Then $\partial \pi^* / \partial \phi_k = [\mathcal{H} \partial \mathcal{S} / \partial \phi_k - \mathcal{S} \partial \mathcal{H} / \partial \phi_k] / \mathcal{H}^2 > 0$ if and only if $\mathcal{S} |\partial \mathcal{H} / \partial \phi_k| > |\partial \mathcal{S} / \partial \phi_k| \mathcal{H}$, which is condition M. For a constant ϕ , $\partial \mathcal{S} / \partial \phi = \ln(1 - \phi)$ and $\partial \mathcal{H} / \partial \phi = [\ln \phi + 1 - \phi] / (1 - \phi)^2 - \eta$, and condition M reads $K < g(\phi) \equiv 2(1 - \phi)[1 - \ln(1 - \phi)] - 2|\ln(1 - \phi)| \mathcal{H}(\phi) / |\mathcal{H}'(\phi)|$, condition (4.11) in Proposition 2(ii) of Ma et al. (2026). As $\phi \rightarrow 0$ the right-hand side tends to 2, so the condition holds for all small ϕ and $K < 2$; on the grid $\phi \leq 0.6$, $K \leq 0.25$, $\eta \leq 0.15$ the margin $\mathcal{S} |\mathcal{H}'| - |\ln(1 - \phi)| \mathcal{H}$ is at least 0.23. $\square$

A.9 Propositions E.1 and E.2

Proof. For the consortium, $\Delta \Pi_B(s, \alpha)$ is the sum over banks of profits at the equilibrium (b_j, L_j) minus profits at $s = 0$. By the envelope theorem applied to each bank's choice of b_j , $d\Pi_B/ds$ equals the partial derivative holding b_j fixed plus the price effect. Holding b_j and q_0 fixed, $L_j = D_j^H + D_j^S + E_j - q_0 b_j$ changes by $-\alpha$ per unit of s summed over banks (household deposits fall by one, issuer deposits rise by $1 - \alpha$), so the partial derivative is $R \cdot (-\alpha) + R_D - R_D^S(1 - \alpha) - \tau \partial(\bar{b} - b)^+ / \partial s$ with $\theta \partial \bar{b} / \partial s = \lambda_{\text{reg}}(1 - \alpha) - \gamma_{\text{max}} \geq 0$ under Assumption 2. The price effect is $(\partial \Pi_B / \partial q_0)(dq_0/ds) = -Rb dq_0/ds \leq 0$ since $dq_0/ds \geq 0$ by Proposition 4.1. With price-taking banks there is also a loan-rate effect, $L_j R'(L) dL/ds$, which is zero in the slack regime, where $dL/ds = 0$ at $\psi = 0$, and positive beyond s_1 , where lending contracts; the proposition is therefore stated for the slack regime, and Appendix E shows the sign reversal that the loan-rate effect can produce beyond it. Hence $d\Delta \Pi_B/ds \leq -\alpha R + R_D - (1 - \alpha)R_D^S = -(R - R_D) + (1 - \alpha)(R - R_D^S) \leq 0$ when $R_D^S \geq R_D$, because $(1 - \alpha)(R - R_D^S) \leq R - R_D^S \leq R - R_D$. Then $\partial^2 \Phi / \partial s \partial \omega = d\Delta \Pi_B/ds \leq 0$ and, by the monotone comparative statics of Milgrom and Shannon (1994), the set of maximizers $s^*(\omega)$ is nonincreasing in ω in the strong set order, for each α . With α chosen jointly, $\Phi^*(s, \omega) = \max_{\alpha} \Phi(s, \alpha; \omega)$ need not have increasing differences, because $\partial^2 \Phi / \partial s \partial \alpha$ carries the sign of the bond-deposit spread; the joint comparative static is therefore verified numerically.

For a single owner receiving all issuer deposits, the partial derivative of its profit is $-(R - R_D)/N + (1 - \alpha)(R - R_D^S)$ in the slack regime (it loses $1/N$ of the household drain and gains all issuer deposits); the sign is that of (E.3). Beyond s_1 the collateral terms are negative and the capture incentive is bounded. $\square$

A.10 Proposition 4.4

Proof. (i) Write the issuer's objective as $\Phi(\alpha; \Delta r) = s\alpha \Delta r + s(R_D^S - 1) + C(\alpha) - cs^2/2$ with $C(\alpha) = -\rho(\alpha)\phi_{\text{run}}(\alpha)s + \omega \Delta \Pi_B(s, \alpha)$. Since $\partial^2 \Phi / \partial \alpha \partial \Delta r = s > 0$, Φ has increasing

differences in $(\alpha, \Delta r)$ and $\alpha^*(\Delta r)$ is nondecreasing in the strong set order (Milgrom and Shannon, 1994). Define $\bar{\Delta} \equiv \sup_{\alpha < 1} [C(\alpha) - C(1)]/[s(1 - \alpha)]$ and $\underline{\Delta} \equiv \inf_{\alpha > 0} [C(0) - C(\alpha)]/[s\alpha]$. For $\Delta r \geq \bar{\Delta}$, $\Phi(1) - \Phi(\alpha) = s(1 - \alpha)\Delta r - [C(\alpha) - C(1)] \geq 0$ for every α , so $\alpha^* = 1$; for $\Delta r \leq \underline{\Delta}$, $\Phi(0) - \Phi(\alpha) = -s\alpha\Delta r + [C(0) - C(\alpha)] \geq 0$, so $\alpha^* = 0$. The thresholds are the run-loss differential between bonds and deposits (through $\rho\phi_{\text{run}}$) and the owner's deposit margin (through $\omega\Delta\Pi_B$).

(ii) $\partial^2\Phi/\partial\alpha\partial\omega = \partial\Delta\Pi_B/\partial\alpha$. By the envelope theorem, holding b_1 and q_0 fixed, the owner's deposits fall by s per unit of α (times its share of issuer deposits, which is one for a single owner and sums to one across a consortium), so $\partial L_1/\partial\alpha = -s$ and the direct effect is $-s(R - R_D^S) + \tau\lambda_{\text{reg}}s\chi(b_1)$; in the slack regime $\chi(b_1) = 0$ and the direct effect is $-s(R - R_D^S) < 0$ when $R > R_D^S$. The price effect is $-Rb_1 dq_0/d\alpha \leq 0$ since more bonds absorbed weakly raise q_0 . Hence Φ has decreasing differences in (α, ω) and $\alpha^*(\omega)$ is nonincreasing. $\square$

A.11 Proposition E.3

Proof. (i) is Lemma C.1(ii) together with the secondary-market price $1 - K\lambda$ while the issuer is solvent. (ii) $|\partial\pi^*/\partial K| = 1/(2\mathcal{H})$ and $\partial\mathcal{H}/\partial\phi_k \leq 0$ (proof of Lemma C.1(iv)). (iii) Let the issuer choose K to maximize $\Psi(K; \phi) = -pG(\pi^*(K, \phi))c_r - \kappa(K)$, where G is the prior's distribution function, $c_r > 0$ the loss per run and $\kappa' > 0$ the cost of price instability. Then $\partial\Psi/\partial K = pc_r G'(\pi^*)/(2\mathcal{H}) - \kappa'(K)$ and

$$\frac{\partial^2\Psi}{\partial K \partial\phi_k} = \frac{pc_r}{2\mathcal{H}} \left[G''(\pi^*) \frac{\partial\pi^*}{\partial\phi_k} + G'(\pi^*) \frac{|\partial\mathcal{H}/\partial\phi_k|}{\mathcal{H}} \right] \geq 0$$

when $G'' \geq 0$ on the relevant range and $\partial\pi^*/\partial\phi_k \geq 0$ (condition M). By Topkis's theorem K^* is nondecreasing in ϕ_k . The Beta(a, b) density is increasing on $(0, 1)$ whenever $a \geq 1$ and $b \leq 1$, which the calibrated prior satisfies. $\square$

A.12 Proposition E.4

Proof. (i) With a par bid for the first H of sales and the bank holding the coins, the selling payoff is 1 on $[0, H]$ and unchanged beyond, and the holding payoff is unchanged because the issuer's liquidation is unchanged (Ma et al., 2026, Appendix C.2). If $H \leq \min\{\hat{\lambda}, 1 - \phi_0\}$ the baseline selling payoff on $[0, H]$ is $1 - K\lambda$, so $\Delta\mathcal{S} = \int_0^H K\lambda d\lambda = KH^2/2$ and $\Delta\pi^* = KH^2/(2\mathcal{H}) > 0$. (ii) In stress the partner can pay for coins only from the collateral value left after its own outflows and the cover it holds against the issuer's deposits, which a run draws on, $\theta q_1 b_1 - \gamma_{\text{max}} D_1^H - N_{0,1} - \lambda_{\text{reg}}(1 - \alpha)s$, which with the pre-positioning rule equals $q_1 w_1 \Sigma(s) - (1 - q_1)[\gamma_{\text{max}} D_1^H + N_{0,1} + \lambda_{\text{reg}}(1 - \alpha)s]$ and is therefore bounded by $q_1 w_1 \Sigma(s)$,

which falls to zero as $s \rightarrow s_1$. A backstop that removes the run requires $H \geq 1 - \phi$ (then the issuer is never insolvent in the secondary market), i.e. $(1 - \phi)s \leq q_1 w_1 \Sigma_0$, which fails whenever the issuer is larger than the partner's share of the system's slack. $\square$

A.13 Proposition E.5

Proof. With $\alpha = 1$ the issuer holds no deposits, $\phi = \bar{h}$ is constant and ρ does not depend on s . Its profit is $\Pi_S(s) = s[1/q_0(s) - 1] - \rho\bar{h}s - cs^2/2$. Under a two-state stress distribution the partially binding regime is the set of s at which the price makes the collateral constraint hold with equality, $\bar{S} - s/q_0 = \bar{b}(s) = [\gamma_{\max}(e - s) + N_0]/\theta$, so $s/q_0 = \bar{S} - \bar{b}(s)$ and $s(1/q_0 - 1) = \bar{S} - \bar{b}(s) - s$, whose derivative is $\gamma_{\max}/\theta - 1 < 0$. Hence (i). In the slack regime $q_0 = q_I$ and $\Pi'_S(s_1^-) = y_I - \rho\bar{h} - cs_1 \geq 0$ exactly when $c \leq \hat{c}$. In the fully binding regime $q_0 = q_{III} > q_I$ is constant, $\Pi_S(s) = s(y_{III} - \rho\bar{h}) - cs^2/2$, and its supremum over $s \geq s_2$ is at most $(y_{III} - \rho\bar{h})^2/(2c)$ when $y_{III} > \rho\bar{h}$ (and is attained below s_2 , hence dominated by $\Pi_S(s_1)$ through (i), otherwise). The issuer prefers s_1 to every point of the constrained regime if and only if $s_1(y_I - \rho\bar{h}) - cs_1^2/2 \geq (y_{III} - \rho\bar{h})^2/(2c)$, a quadratic inequality in c whose solution set is $[\underline{c}, \bar{c}]$ with $\underline{c}, \bar{c} = [(y_I - \rho\bar{h}) \mp \sqrt{(y_I - \rho\bar{h})^2 - (y_{III} - \rho\bar{h})^2}]/s_1$; since $\hat{c}$ lies in this interval, the issuer stops at s_1 for $c \in [\underline{c}, \hat{c}]$. $\square$

A.14 Proposition E.6

Proof. (i) ω does not appear in (5.1). In the slack regime with the loop shut the planner's condition is $V'(s^{SP}) = cs^{SP}$. The issuer's marginal objective at s^{SP} is $\sigma' - cs^{SP} + \omega \partial\Delta\Pi_B/\partial s$, where σ' is marginal seigniorage net of run losses and $\partial\Delta\Pi_B/\partial s \leq -(R - R_D) + (1 - \alpha)(R - R_D^S) < 0$ by (E.2); it is negative whenever $\sigma' \leq V'(s^{SP})$, and with Φ concave in s (linear seigniorage, convex cost) the issuer's optimum then lies below s^{SP} .

(ii) Let $(s^\omega, \alpha^\omega)$ be the allocation under the floor and $\alpha^p \leq \alpha^\omega$ the issuer's bond share under the pair. Choose $\hat{\mu} < 1$ large enough that $\hat{\lambda} \geq 1$ (possible since $\hat{\lambda} = A_1/d + \text{const} \rightarrow \infty$ as $d \rightarrow 0$) and that $(1 - \alpha^p)(1 - \hat{\mu}) \leq 1 - \alpha^\omega$. Under the pair the cap fixes $s = s^\omega$, the partner cannot fail (the F_B term is zero), the bonds absorbed, $\alpha^p(1 - \hat{\mu})s^\omega \leq \alpha^\omega s^\omega$, and the deposits placed, $(1 - \alpha^p)(1 - \hat{\mu})s^\omega \leq (1 - \alpha^\omega)s^\omega$, are both no larger than under the floor, so the safe assets available to banks are weakly larger at every price, lending is weakly higher and liquidation losses weakly lower, and the fire-sale loss $\rho(1 - \hat{\mu})\alpha^p\bar{h}s^\omega$ is no larger than $\rho^\omega\alpha^\omega\bar{h}s^\omega$ because ρ is nondecreasing in the liquidation cost (condition M) and $(1 - \hat{\mu})\alpha^p \leq \alpha^\omega$. Every term of (5.1) is therefore weakly higher under the pair, for every $F_B \geq 0$. $\square$

A.15 Propositions E.7 and E.8

Proof. Write $W^B - W^{LF} = \Lambda - \Xi + F_B \rho^{LF} \mathbf{1}\{\hat{\lambda}^{LF} < 1\}$, where Λ collects the differences in convenience value, lending surplus, liquidation losses and issuance cost between the two allocations and $\Xi \equiv \rho^B \bar{h} s^B - \rho^{LF} \phi_{\text{run}}^{LF} s^{LF}$ is the difference in fire-sale losses. The rule dominates if and only if $F_B \rho^{LF} \mathbf{1}\{\cdot\} \geq \Xi - \Lambda$, which defines $\bar{F}_B$. If $\hat{\lambda}^{LF} \geq 1$ the indicator is zero: then $\phi^{LF}(\lambda) = \alpha^{LF} \bar{h} \leq \bar{h}$ for all λ , so by Lemma C.1(iv) π^* and ρ are weakly higher under the rule, $\Xi \geq 0$ whenever $s^B \geq s^{LF} \alpha^{LF} \rho^{LF} / \rho^B$, and the rule lowers welfare unless $\Lambda > \Xi$; a monopolist's s^B is s_1 by Proposition E.5. For Proposition E.8, a price-taking issuer's marginal condition is $y(s) - \rho\phi = cs$, a monopolist's $y(s) + sy'(s) - \rho\phi = cs$ with $y' \leq 0$, so the competitive quantity is weakly larger; with bonds and a two-state stress distribution, $y(s_1^+) = y_I$ while the competitive condition at s_1 reads $y_I - \rho\bar{h} > cs_1$, so the competitive industry passes s_1 whenever that inequality holds. $\square$

A.16 Corollary E.1

Proof. The denominator of (4.4) is the change in the system's measured headroom per coin: the HQLA the coin removes from the measured stock plus the stress outflow it adds, minus the retail outflow of the household deposit it replaces (γ_{max}). (iii) is the non-bank case of Section 4.2: bonds bought from banks leave the measured stock at θ/q_0 per unit of value and deposits at the partner add λ_{reg} . (i) If the subsidiary's reserves count in the group's HQLA, bonds the issuer buys from other banks stay in the system's measured stock, so the first term is zero; deposits at the parent are intra-group and eliminated; the coins add the outflow λ_c . The denominator is $\lambda_c - \gamma_{\text{max}}$, independent of α . (ii) If the reserves are excluded but the outflows are counted, the bonds leave the measured stock at $\theta\alpha/q_0$ and the coins add λ_c ; deposits at the parent remain intra-group. Since $\theta\alpha/q_0 \geq 0$ and $\lambda_c \geq \lambda_{\text{reg}}(1 - \alpha)$ whenever $\lambda_c \geq \lambda_{\text{reg}}$, the denominator in (ii) is the largest of the three, so the threshold is the smallest. $\square$

Appendix B Data

Issuer flows and reserve composition. Circulating supply by coin, daily, from DefiLlama (USDT, USDC and ten smaller dollar coins); weekly flows are Friday-to-Friday changes. Reserve composition from 96 monthly Circle examination reports (October 2018 to August 2026; Grant Thornton to December 2022, Deloitte thereafter) and 22 Tether attestations (February 2021 to June 2026; Moore Cayman to 2021, BDO thereafter). Reports were parsed from the PDF text with word-coordinate line reconstruction; the four Circle report formats

Table B.1: Issuer reserve composition, selected dates. Shares of total reserves; totals in \$ billion. Sources: BDO and Moore Cayman attestations (Tether), Grant Thornton and Deloitte examination reports (Circle).

Date	USDT bills	USDT repo	USDT CP	USDT total	USDC bills	USDC repo	USDC cash	USDC total
2021-06	0.24	0.02	0.49	62.8	0.13	—	0.46	25.2
2022-06	0.43	0.05	0.13	66.4	(interp.)	—	—	—
2022-12	0.59	0.05	0.00	67.0	0.76	0.00	0.24	44.7
2023-06	0.65	0.11	0.00	86.5	0.20	0.67	0.13	27.4
2024-06	0.68	0.10	0.00	118.4	0.33	0.53	0.14	32.3
2025-06	0.65	0.11	0.00	162.6	0.42	0.43	0.15	61.4
2026-06	0.61	0.14	0.00	187.8	0.12	0.71	0.17	73.3

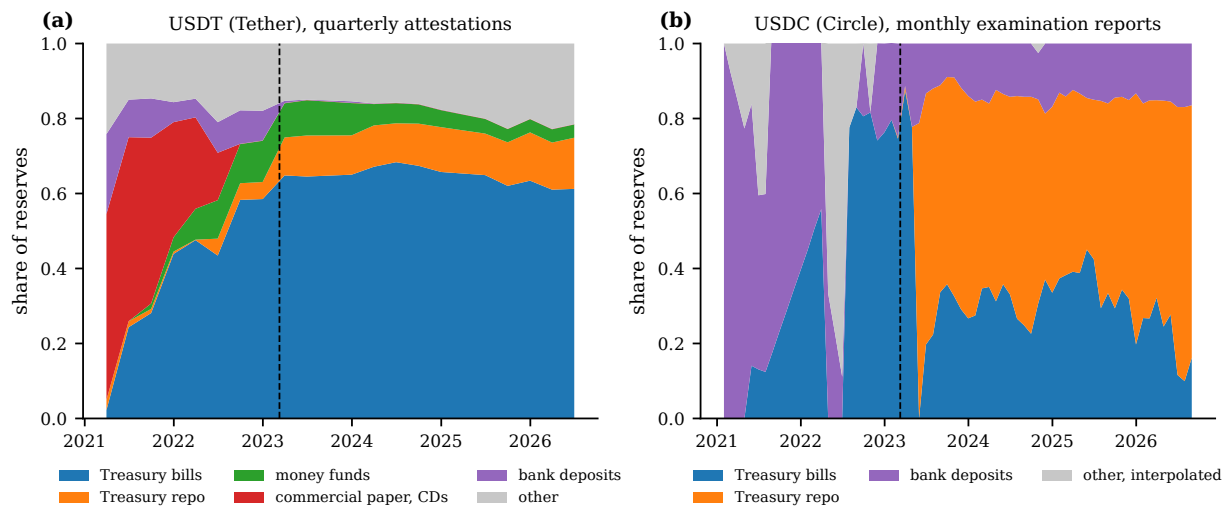


Figure B.1: Reserve composition of USDT and USDC, 2021–2026. Shares of reserves by asset class from the issuers’ attestations: (a) Tether, quarterly; (b) Circle, monthly (November 2021 to June 2022 interpolated). The dashed line marks March 10, 2023.

and the wrapped-cell Tether tables each required their own parsing rules. Circle reports for November 2021 to June 2022 are encoded with a font that defeats text extraction and are interpolated. For Tether, components sum to the reported subtotal in every parsed quarter; the March 2025 report is unparsed and interpolated. Shares used in Section D.1 are those at the previous month-end (Circle) or quarter-end (Tether). Table B.1 gives selected dates and Figure B.1 the full series.

Rates and supply. Constant-maturity bill and note yields, the effective federal funds rate, SOFR, interest on reserves, overnight reverse repo balances, reserve balances, the Treasury General Account, VIX and financial and nonfinancial commercial paper rates from FRED. Bill supply from the Treasury’s auction results (gross issuance and maturities by issue and maturity date, excluding SOMA) and from the Monthly Statement of the Public Debt (bills as a share of marketable debt).

Table B.2: Banks in the liquidity coverage panel. Number of quarters with HQLA, net cash outflows and LCR, 2018Q1–2026Q2; the LCR and the slack (HQLA minus net cash outflows, \$ billion) are 2026Q2 quarterly averages. Report: the public disclosure required by Regulation WW. ^aCore five banks, used for the core aggregate in Figure 3.

Bank holding company	Source	Quarters	First	Last	LCR	Slack
JPMorgan Chase ^a	10-Q, 10-K	34	2018Q1	2026Q2	1.10	89.9
Bank of America ^a	10-Q, 10-K	34	2018Q1	2026Q2	1.12	73.3
Citigroup ^a	10-Q, 10-K	34	2018Q1	2026Q2	1.13	73.7
Wells Fargo ^a	10-Q, 10-K	34	2018Q1	2026Q2	1.19	67.2
Goldman Sachs ^a	10-Q, 10-K	34	2018Q1	2026Q2	1.26	94.8
Morgan Stanley	10-Q to 2019, then report	34	2018Q1	2026Q2	1.29	72.5
BNY Mellon	report	34	2018Q1	2026Q2	1.11	14.3
State Street	report (narrative)	34	2018Q1	2026Q2	1.07	6.9
Northern Trust	report	31	2018Q4	2026Q2	1.07	4.5
U.S. Bancorp	report	33	2018Q2	2026Q2	1.08	8.7
PNC	report	30	2019Q1	2026Q2	1.06	5.8
Truist	report	27	2019Q4	2026Q2	1.13	11.0
Capital One	report	33	2018Q2	2026Q2	1.65	26.0
Charles Schwab	10-Q, 10-K	31	2018Q4	2026Q2	1.35	13.4
American Express	10-Q, Table 21	2	2026Q1	2026Q2	1.96	9.3
TD Group US	report	33	2018Q2	2026Q2	1.40	26.0
HSBC North America	report (narrative)	33	2018Q2	2026Q2	1.16	7.0
Barclays US LLC	report	33	2018Q2	2026Q2	1.58	7.7
DB USA	report	33	2018Q2	2026Q2	1.68	7.6
UBS Americas Holding	report	21	2021Q2	2026Q2	1.24	5.9
BMO Financial Corp	report (summary box)	7	2024Q4	2026Q2	1.09	5.0
RBC US Group	report	3	2019Q1	2019Q3	—	—

Banks. Quarterly call-report variables (assets, deposits, equity, securities, cash and balances due, net loans, noninterest-bearing deposits) for all banks with assets above \$10 billion, 2014–2026, from the FDIC’s BankFind API, and for the eight partner banks of Section 7.3. XBRL company facts from the SEC for Silvergate Capital, SVB Financial, Customers Bancorp and Metropolitan Bank Holding (fair value and amortized cost of available-for-sale and held-to-maturity securities, unrealized losses, equity, deposits); 10-Q and 10-K text for the same companies 2020–2024. Signature Bank’s digital-asset deposits are from its company updates and the FDIC’s 2023 report on its supervision (it filed with the FDIC, not the SEC); Circle’s deposit at Silicon Valley Bank is from Circle’s statement of March 10, 2023.

Liquidity coverage disclosures. Quarterly averages of the HQLA amount, total net cash outflows and the LCR for the bank holding companies in Table B.2, 2018–2026, at most twenty-one in any quarter. Six US companies (JPMorgan Chase, Bank of America, Citigroup, Wells Fargo, Goldman Sachs and Charles Schwab), and Morgan Stanley before 2020, are

Table B.3: US parameters. Values used in Sections 4–6; ranges are those reported as sensitivity. *Implied* entries are computed from the rows above; *assumption* entries are not targeted to data.

Parameter	Value	Source
<i>Banking system</i>		
Deposits e	\$19.6 tn	Federal Reserve H.8, all commercial banks, September 2026 (system-wide)
Bank equity E	\$2.4 tn	FDIC Quarterly Banking Profile, total equity capital, 2026Q2
HQLA $\bar{S}$	\$4.54 tn	Sum of the HQLA amounts disclosed by the twenty-one LCR-subject bank holding companies, 2026Q2 (Table B.2)
Aggregate LCR	1.161	Total HQLA over total net cash outflows of the same banks, 2026Q2 (1.16 on average over 2025Q4–2026Q2)
Slack Σ_0	\$630 bn	Disclosed HQLA minus disclosed net cash outflows
Retail run-off $\gamma_{\max}$; other outflows N_0	0.07; \$2.54 tn	Blend of the LCR’s 3% (stable) and 10% (other) retail rates (Basel Committee on Banking Supervision, 2013); $N_0 = \text{NCO} - \gamma_{\max}e$
Run-off on issuer deposits: regulatory λ_{reg} ; actual λ_a ; internal λ_{int}	1; 0.13; 0.74	LCR rate for financial-institution funding; largest 30-day decline of USDT or USDC outside the March 2023 run; Silvergate’s excess liquidity per dollar of digital-asset deposits, 2022Q2–Q3
Cost of covering a requirement shortfall c	54 bp (crisis 136 bp)	Primary credit rate minus IORB (FRED) plus discount-window stigma (Armantier et al., 2011)
Daily shock to the requirement σ	0.075 of requirement	Median disclosed LCR (1.157) held at a premium of 1 bp
Term-funding and window capacity κ ; stress coverage ω	0.46; 0.46	Loans pre-pledged at the discount window (\$1.8 tn, 2023) over net outflows
Issuer-flow volatility	0.71% a day	USDC circulating supply, daily, from July 2023
<i>Stress and liquidation</i>		
Collateral value θ	1	HQLA valued after regulatory haircuts
Stress price of safe assets q_1	0.97	Bill and short-Treasury discounts, March 2020 and October 2022
Loan-liquidation cost φ	0.25 (severe case: 1)	Assumption; discount of about 20% on loans sold in stress
Stress probability p	0.10	Assumption
<i>Rates</i>		
Loan return at L^* ; household deposit rate; issuer deposit rate	4.0%; 1.5%; 4.0%	3-month bill yield and SOFR (FRED); FDIC national deposit rate
Convenience term ψ , b_c (linear benchmark)	0.025; \$2.4 tn	Matched to the Treasury convenience yield in Hofmann et al. (2026)

Table B.3: US parameters (continued).

Parameter	Value	Source
<i>Run block</i>		
Redemption access K	0.18 (USDC 0.156, USDT 0.209)	Ma et al. (2026)
Holding premium η	0.07	Lowest value consistent with both issuers' run probabilities
Prior on fundamental, Beta(a, b)	(1.01, 0.087)	Fitted to run probabilities of 2.13% (USDC), 2.50% (USDT) in Ma et al. (2026)
Loss on uninsured deposits LGD; fire-sale discount $\bar{h}$	0.10; 0.03	FDIC resolution losses; $\bar{h} = 1 - q_1$
<i>Stablecoins</i>		
Stablecoins outstanding	\$311 bn	DefiLlama, October 2, 2026
Existing issuer bank deposits	\$14.5 bn (4.6%)	Circle and Tether reserve reports
Issuers' bill and repo holdings	≈\$200 bn	Tether quarterly attestations (BDO); Circle monthly examination reports (Deloitte)
<i>Coin demand and failure cost</i>		
Semi-elasticity of demand ε	6.6% per pp (95% interval $[-15.7, +2.6]$; 2.0–20.3 as sensitivity)	Monthly regression of the coin stock on the bill yield and bitcoin, 2020–2026
Anchor (s_0, r_0)	\$311 bn; 3.99%	Stock in October 2026 (DefiLlama); 3-month bill yield
Deadweight loss of a partner failure F_B	ζE_1 , $\zeta = 1$	FDIC resolution losses of 2023 relative to equity

read from the liquidity sections of their 10-Q and 10-K filings (SEC EDGAR); the others are read from the LCR disclosure reports that Regulation WW requires them to publish, in the standard template or, for HSBC North America and State Street, from the narrative that accompanies it. State Street reports both its total HQLA and the amount eligible for the ratio's numerator; we use the eligible amount. BMO Financial Corp.'s tables are images and were read by hand from the summary box; American Express's figures are from Table 21 of its 10-Q. Every bank-quarter is kept only if the ratio of the two amounts reproduces the disclosed LCR to within 3 percent (1.5 percent for the disclosure reports). Credit Suisse Holdings (USA), merged into UBS Americas in 2024, is missing, as are the quarters before UBS Americas and BMO Financial Corp. began publishing amounts on their current sites; Category IV companies are not subject to the ratio and publish nothing. The core aggregate in Figure 3 uses the five banks that report in every quarter.

Issuer banks and the liquidity coverage ratio. Circle's banks in March 2023 are those it named in its statements of 10–13 March 2023 and those reported by the press on those dates; total assets are from the FDIC's call reports (BankFind API, certificates of each bank), and the status under the ratio is from the 2019 tailoring rule and, for Silicon Valley Bank and

Signature Bank, from Congressional Research Service (2023). The existing issuer deposits, \$14.5 billion, are the cash lines of Circle’s monthly and Tether’s quarterly reports (4.6 percent of their combined reserves), applied to the total stock.

Partner banks’ liquidity. Liquid assets are cash and balances due plus securities in the call reports; peers are all banks with total assets above \$10 billion in the same quarter, excluding the partner banks. The implied run-off on digital-asset deposits is the bank’s liquid assets in excess of the peer median share times its assets, divided by its period-end digital-asset deposits (Table 8).

Compliance cost. The primary credit rate and interest on reserve balances are FRED series DPCREDIT and IORB; their difference has been 10 basis points since July 2021. The stigma premium is from Armantier et al. (2011). Pre-pledged collateral at the discount window is from the Federal Reserve’s discount-window readiness data for 2023 (Board of Governors of the Federal Reserve System, 2025). The daily standard deviation of issuer flows is that of the log change in USDC’s circulating supply from July 2023 (DefiLlama). The shock to the requirement is calibrated so that the median disclosed ratio in 2026Q2, 1.157, is the buffer at which the premium is 1 basis point; the within-bank standard deviation of the quarterly change in the disclosed ratio is a smoothed measure of the daily variability and is of the same order.

Prices in the March 2023 run. One-minute OHLC prices from Bitstamp (USDC/USD, USDT/USD, March 9–14, 2023) and Binance (BUSD/USDT for the full window; USDC, TUSD and USDP against USDT from March 11, 14:00 UTC, when Binance relisted them; DAI via the inverse of USDT/DAI).

Crypto-demand instruments. Bitcoin prices (FRED, Coinbase) and the eight-hourly BTC perpetual funding rate (Binance), the instruments in Section D.1.

Appendix C The run block and elastic supply

C.1 The run game

We take the run game from Ma et al. (2026): holders observe noisy signals of a fundamental π and redeem below a threshold; arbitrageurs with redemption access K absorb secondary-market sales at the price $1 - K\lambda$ while the issuer is solvent. In the vanishing-noise limit the

threshold π^* equates the expected payoff of selling and holding for a marginal holder with a uniform belief over the redemption share λ ,

$$\pi^* = \frac{\mathcal{S}}{\mathcal{H}}, \quad \mathcal{S} \equiv \int_0^1 \text{sell}(\lambda) d\lambda, \quad \mathcal{H} \equiv \int_0^1 \text{hold}(\lambda) d\lambda, \quad (\text{C.1})$$

where $\text{sell}(\lambda) = 1 - K\lambda$ for $\lambda \leq 1 - \phi(\lambda)$ and $(1 - \phi(\lambda))/\lambda - K\lambda$ otherwise, and $\text{hold}(\lambda) = (1 - \phi(\lambda) - \lambda)/[(1 - \phi(\lambda))(1 - \lambda)] + \eta$ for $\lambda \leq 1 - \phi(\lambda)$ and zero otherwise, with η the holding premium. Their payoff functions assume a constant ϕ ; ours is a step function.

Lemma C.1 (run threshold with a step liquidation cost). *Let $\phi(\lambda)$ be a nondecreasing step function with finitely many jumps and values in $[0, 1)$.*

- (i) *The run threshold $\pi^* = \mathcal{S}/\mathcal{H}$ exists and is unique. For a constant ϕ , $\mathcal{S} = (1 - \phi)[1 - \ln(1 - \phi)] - K/2$ and $\mathcal{H} = 1 + \phi \ln \phi / (1 - \phi) + \eta(1 - \phi)$, which is the closed form of Ma et al. (2026).*
- (ii) *π^* is strictly decreasing in K , with $\partial\pi^*/\partial K = -1/(2\mathcal{H})$.*
- (iii) *For a single step from ϕ_0 to ϕ_1 at $\hat{\lambda} \leq 1 - \phi_1$, π^* is strictly decreasing in $\hat{\lambda}$.*
- (iv) *π^* is increasing in each step value of ϕ whenever $\mathcal{S} |\partial\mathcal{H}/\partial\phi_k| > |\partial\mathcal{S}/\partial\phi_k| \mathcal{H}$ (condition M), which for a constant ϕ is condition (4.11) of Ma et al. (2026, Proposition 2(ii)) and holds on the whole range we use ($\phi \leq 0.6$, $K \leq 0.25$, $\eta \leq 0.15$).*

Uniqueness needs no condition: the marginal holder's payoff difference is linear and increasing in π . Part (iii) is the property the loop uses, and it is unconditional: moving the step to the left leaves the selling payoff unchanged at the step and lowers the holding payoff there. The run probability is $\rho = p \Pr[\pi < \pi^*]$; we draw π from a Beta prior calibrated so that the model reproduces the estimated run probabilities of the two largest issuers (Section 6).

Calibration. The run block uses the redemption-access parameters of Ma et al. (2026) ($K = 0.156$ for USDC and 0.209 for USDT; 0.18 for the representative issuer), a holding premium $\eta = 0.07$, and a Beta(1.01, 0.087) prior on the fundamental chosen so that the model reproduces their estimated run probabilities for the two issuers (2.13 and 2.50 percent) at their reported illiquidity shares; $\eta = 0.03$ cannot match both, and $\eta \in \{0.05, 0.10, 0.15\}$ are reported as sensitivity. Figure C.1 reports the run probabilities that result, by partner type and regulatory run-off rate; Section 6 discusses them.

C.2 Elastic supply of safe assets

In the kinked benchmark of Appendix F, with $S(q_0) = \bar{S}(q_0/\bar{q})^{\varepsilon_S}$ the issuer's purchases call forth new supply as the price rises, which widens the transition regime and lowers the lending

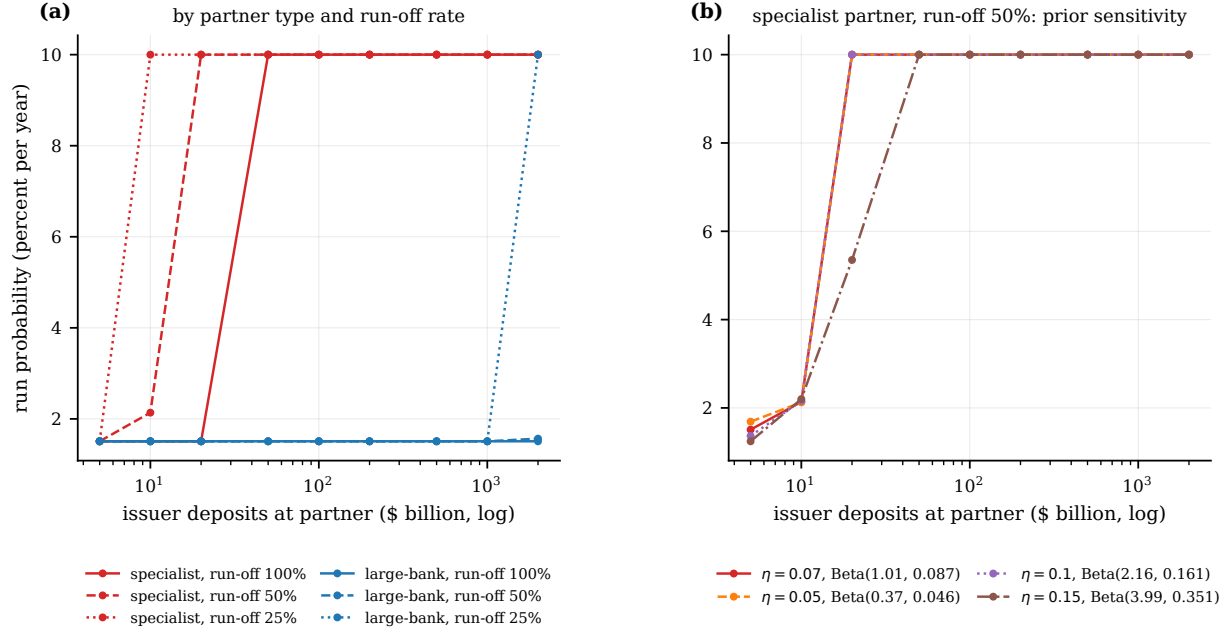


Figure C.1: Run probability and issuer deposits by partner-bank type. US calibration, deposit reserves ($\alpha = 0$), Beta prior on the fundamental. (a) Baseline prior ($\eta = 0.07$) for a specialist and a large-bank partner at regulatory run-off rates of 100, 50 and 25 percent. (b) Specialist partner at a 50 percent run-off under the baseline and three alternative holding premia, each with the Beta prior re-fitted to the issuers' estimated run probabilities.

loss per unit of issuance within it: in the US calibration the loss falls from 6.3 units of lending per unit of issuance at $\varepsilon_S = 0$ to 1.3 at $\varepsilon_S = 2$ and 1.0 at $\varepsilon_S = 10$, while its end, s_2 , moves from \$666 billion to \$1,400 billion. This is a property of the calibration rather than a theorem: with elastic supply the value of the stock the economy pays for, $q_0 S(q_0)$, rises with the price through both the price and the quantity, and the sign of the net effect on the lending loss per unit depends on the slope of loan demand as well. In the model of Section 3 the same response flattens the yield path and moves the fold outward (Section 6.4).

Appendix D Further evidence on bill prices and reserve composition

D.1 Bill-buying dollars: data, specification and robustness

Data. Weekly changes in the circulating supply of each dollar stablecoin (DefiLlama) give issuer flows. The monthly and quarterly attestations of Circle and Tether, parsed into the shares of reserves held in bills, repo, money funds, deposits and other assets (Figure B.1, Table B.1), give the composition of each issuer's marginal dollar. We define bill-buying

dollars as $\sum_i f_{i,t} s_{i,t-1}$, the sum over issuers of flows weighted by the issuer’s bill share at the previous report date, and non-bill dollars as $\sum_i f_{i,t}(1 - s_{i,t-1})$. Yields are the three-month and four-week constant-maturity bill rates and, as expectation-free spreads, the three-month bill rate relative to SOFR, to the federal funds rate and to financial commercial paper.

Specification. Local projections of $y_{t+h} - y_{t-1}$ on bill dollars and non-bill dollars, with other issuers’ flows, lagged changes in y and in flows, the change in VIX, and the changes over the horizon in the effective federal funds rate and the two-year Treasury yield as controls; Newey–West standard errors with $h + 1$ lags. The hypothesis is that the coefficient on bill dollars is negative and that it exceeds the coefficient on non-bill dollars in absolute value.

Results. Table 11 reports the four-week coefficients per \$3.5 billion. Bill dollars lower the three-month yield by 19.1 basis points in the sample from July 2022, 13.7 from January 2023 and 10.8 from July 2023; non-bill dollars do not lower it in any sample. The difference between the two coefficients, tested with the HAC covariance of the two estimates, is significant at the 5 percent level in the samples from July 2022 and January 2023 (t of 2.3 and 2.8) and at the 5 percent boundary in the sample from July 2023 (t of 2.0). The same ordering holds at two weeks in the recent sample and in the spreads to SOFR and to the federal funds rate, where the expectation component cancels (t of 2.7–2.9). Lagging the composition by an additional quarter, so that the share cannot respond to the month’s rates, leaves the estimates unchanged (bill-dollar coefficients -13.1 and -12.6 , t 2.7–2.6). Because Circle’s composition follows the bill–repo spread (Section D.2), we also add the lagged level of the SOFR–bill spread, so that the predictable, rate-driven part of the composition cannot carry the estimate; the coefficients do not change (-14.4 in the sample from January 2023, t of the difference -2.88). A placebo that keeps the distribution of Circle’s monthly bill shares but permutes their timing within the sample window does not reproduce the result: in 1,000 permutations the t -statistic of the difference has mean -0.35 and standard deviation 0.70 in the sample from January 2023, and 4 of 1,000 draws reach the actual value of -2.8 (6 of 1,000 reach -2.3 in the sample from July 2022). From July 2023 the identification changes source. The two issuers’ mean shares then differ (USDC 0.29, USDT 0.65); the between-issuer difference alone, with each issuer’s share held at its sample mean, produces the same sign and size (-9.3 , t 1.8) but is not significant at conventional levels on its own, and re-timing Circle’s shares accordingly leaves the estimate in place. The two sources of variation point the same way; the within-issuer timing is the sharper one. The effect is confined to bills: it is present at three months and one year (-11.4 , t 2.3), weaker at six months (-5.1), too

noisy to sign at four weeks (standard error 17.8), and the two-year yield is held fixed by construction.

Persistence and interpretation. The effect peaks at two to four weeks and is gone by six to twelve: the four-week coefficient of -13.7 in the 2023 sample becomes -0.9 at six weeks and $+0.8$ at eight. It is a price impact in a segmented bill market, of the order found by Ahmed and Aldasoro (2026) for aggregate stablecoin flows, not a change in the level of yields. The model’s permanent effect at today’s stock, 0.09 basis points per \$3.5 billion, is consistent with the absence of a level effect. What the cross-section identifies is the price impact of the issuer’s reserve choice α of Section 4.5 at the margin: new dollars move bill prices in the weeks after they arrive when the issuer buys bills with them, and not when it lends them in repo. Most of the identifying variation is Circle’s, whose bill share changes from month to month; Tether’s share changes little between reports, and its flows alone give a precise estimate only in the sample from July 2023 (Table D.4, rows E).

The attestations also give each month’s net purchases, which include the reallocations within a reserve that the flow-weighted measure misses, and they qualify the reading. Regressing the monthly change in the bill–SOFR spread on Circle’s reserve growth and on its reallocation from repo into bills, with Tether’s bill purchases, net bill issuance, the change in reverse-repo balances, the lagged change and the lagged spread level as controls (October 2022 to June 2026, $n = 46$), reserve growth enters at -4.0 basis points per \$3.5 billion (standard error 2.6) and the reallocation at $+0.3$ (0.5), $+0.3$ without March and April 2023. The largest reallocation, \$23.7 billion out of bills into repo in May 2023, moved the spread by -0.5 basis points. Moving an existing stock between bills and repo does not move bill yields; new dollars that buy bills do, for a few weeks. That is the asset-swap pattern of Proposition 4.1(i), in which money funds holding reverse repos replace a departing buyer and, within weeks, accommodate a new one (Section 6).

The level effect can be bounded directly. Regressing the monthly three-month bill–SOFR spread on the issuers’ stock of bills and Treasury repo, with the bill share of marketable debt and reverse-repo balances as controls, gives $+0.057$ basis points per \$1 billion of stock (Newey–West standard error 0.063, $n = 63$), while the bill share of marketable debt enters at -6.4 basis points per percentage point (2.8). The lower end of the 95 percent band, -0.067 , rules out a permanent lowering of the yield larger than 0.07 basis points per billion, or 13 basis points at the current stock of \$202 billion, and the model’s convenience slope (3.2 basis points at that stock) lies inside the band. The persistent yield declines that Cerutti et al. (2026) find after stablecoin demand shocks are therefore not in conflict with the flow effect above: their shocks are news about market capitalization, which carry expectations of future

purchases, whereas the realized stock of purchases has no detectable level effect once bill supply is controlled for. The difference also matters for the reading of Kim (2026), whose high-frequency price response to Tether issuance is of the same expectational kind.

Two further facts complete the picture. Aggregate stablecoin flows, without the composition weights, have no robust effect on bill yields once the 2022 tightening-and-redemption episode is excluded (-0.4 basis points with standard error 0.9 at four weeks), and the apparent state dependence of that effect on the bill-supply regime disappears with the same exclusion. Instrumenting total flows with crypto-demand shifters gives estimates of the same sign and larger size (-6.8 with standard error 4.1 using the perpetual-futures funding rate, first-stage F of 15.5), consistent with measurement error and reverse causality biasing the least-squares estimate toward zero; the instruments are too weak for the bill split. Quantities move where prices do not: outside 2022 an inflow of \$3.5 billion raises balances in the Federal Reserve’s reverse repo facility by \$40 billion with a standard error of \$17 billion, the displacement of money funds from bills that the asset-swap logic of Proposition 4.1(i) implies, while bank deposits, loans and Treasury holdings do not respond.

D.2 The reserve-composition rule

Proposition 4.4 says the bond share of reserves follows the sign of the spread between bill yields and the rate on deposit-like reserves. Circle’s allocation does (Table D.1). Circle held 69 percent of its reserves in bills in 2022, when bills yielded 39 basis points more than overnight repo, and its Reserve Fund held 31–35 percent in 2024–25, when they yielded 18 basis points less, with the balance in repo. In first differences over 2023–2026, a 10 basis point rise in the bill–SOFR spread raises the bill share by +7.0 percentage points (standard error 3.0); the spread’s range in the sample is about 60 basis points, so the estimate spans the observed variation without extrapolation. A dummy for the period after the GENIUS Act, which named repo and money funds as eligible reserves, is -25 percentage points (13). Tether’s composition is event-driven rather than rate-driven: it liquidated \$30 billion of commercial paper after the Terra collapse of 2022 and has held 61–68 percent of reserves in bills since, so that the two issuers sit on opposite sides of the boundary of Proposition 4.4 with the same market rates—a difference the model attributes to their redemption access, holder base and regulatory domicile.

The partner-bank failures of March 2023 offer a sharper test of the run-loss term in Proposition 4.4: they made the failure risk of a weak partner salient overnight. Circle did not move out of bank deposits. Its deposit share averaged 14 percent in the eight months before March 2023 and 14 percent in the twelve after, and a post-March 2023 indicator in the

Table D.1: Reserve composition and the bill–repo spread. Monthly (Circle) and quarterly (Tether) bill shares regressed on the three-month bill yield minus SOFR, the bill yield minus interest on reserves, the bill yield level and a post-GENIUS Act dummy; coefficients are share changes per basis point, standard errors in parentheses.

Issuer, sample	Form	bill–SOFR	bill–IOR	bill level	post-GENIUS	<i>n</i>	<i>R</i> ²
Circle, 2021.6–2026.8	levels	−0.004 (0.006)	+0.010 (0.006)	+0.000 (0.000)	−0.167 (0.094)	60	0.44
Circle, 2023.1–	levels	+0.004 (0.006)	−0.002 (0.006)	−0.001 (0.001)	−0.253 (0.132)	44	0.24
Circle, 2023.1–	first differences	+0.007 (0.003)	−0.005 (0.004)	−0.005 (0.003)		43	0.12
Tether, 2021.6–2026.8	levels	−0.005 (0.001)	+0.006 (0.001)	+0.001 (0.000)	−0.011 (0.018)	19	0.88
Tether, 2021.6–2026.8	first differences	−0.004 (0.002)	+0.005 (0.003)	+0.000 (0.000)		18	0.29

composition regression is −6 percentage points with a standard error of 3. What changed was where the deposits sat. Before the failures \$3.3 billion of it was at Silicon Valley Bank; from April 2023 the reserve reports state that more than 90 percent of the cash was at the Bank of New York Mellon, the fund’s custodian. That is the margin the proposition points to: deposit reserves are safe at a strong partner and dangerous at a weak one. Inside the fund, bills fell by 49 percentage points and repo rose by 54 after March 2023; that shift coincides with the debt-ceiling standoff of May and June 2023, when bills maturing near the projected default date traded at elevated yields, and cannot be attributed to the bank failures. Tether, whose bank deposits were 4 percent of reserves throughout, had little to move.

D.3 Partner banks and bank lending: further tables

Table D.2 gives every bank-quarter behind Table 9, and Table D.3 the inference for the bank-lending panel of Section 7.4.

D.4 Stablecoin prices in the March 2023 run

The cross-section of coins the next weekend orders by exposure to the three banks (Figure D.1). USDC, with 7.6 percent of its reserves at Silicon Valley Bank, traded at \$0.805 on Bitstamp at 07:14 UTC on March 11 and stayed below \$0.99 for 46 hours; DAI, collateralized in part by USDC, reached \$0.874; Paxos’s USDP, with deposits at Signature, \$0.940; TUSD \$0.961; BUSD, with no exposure, \$0.987 for thirty minutes; USDT \$0.998. The USDC path—\$0.99 at the close of SVB’s last day, \$0.93 by 05:00, \$0.805 at the trough, back to \$0.90 after Circle announced it would cover any shortfall from its own capital, and to \$0.99 after the Treasury and the Federal Reserve guaranteed uninsured deposits on the evening of March 12—separates the effect of an issuer backstop (Proposition E.4: partial, and only briefly effective) from that of a sovereign guarantee.

Table D.2: Partner banks and the capital condition: all bank-quarters, 2022Q2–2023Q1. \$ billion. Digital-asset deposits are the period-end totals the banks disclosed (Silvergate and Customers: 10-Q and 10-K; Metropolitan: investor presentations; Signature: company update for September 2022 and the FDIC’s report on its supervision for December 2022). Equity, cash and securities from FDIC call reports. The threshold is the fire-sale discount d^* at which repaying all digital-asset deposits, cash first and securities next, with loans liquidated at $\varphi = 0.25$ beyond the securities, exhausts equity (equation (7.1)); “cash covers” means cash exceeds the deposits. The last-but-one column applies Silvergate’s realized discount of 0.138. Western Alliance, First Republic and Cross River disclosed no digital-asset deposit totals; for Silicon Valley Bank only Circle’s deposit is known.

Bank, quarter	Equity	Cash	Securities	Digital-asset deposits D^S	D^S/E	Threshold d^*	Fails at $d = 0.138$	Outcome
Silvergate, 2022-06	1.23	1.89	11.82	13.30	10.8	0.09	yes	liquidation announced Mar 8, 2023
Silvergate, 2022-09	1.11	1.89	11.42	11.87	10.7	0.10	yes	liquidation announced Mar 8, 2023
Silvergate, 2022-12	0.57	4.57	5.73	3.83	6.7	cash covers	no	liquidation announced Mar 8, 2023
Signature, 2022-09	7.69	11.58	25.81	23.50	3.1	0.39	no	closed Mar 12, 2023
Signature, 2022-12	8.01	6.06	26.15	17.80	2.2	0.41	no	closed Mar 12, 2023
Customers, 2022-06	1.47	0.26	3.62	2.10	1.4	0.44	no	survived
Customers, 2022-09	1.48	0.42	3.81	1.90	1.3	0.50	no	survived
Customers, 2022-12	1.50	0.46	3.81	2.30	1.5	0.45	no	survived
Customers, 2023-03	1.54	2.02	3.77	1.90	1.2	cash covers	no	survived
Metropolitan, 2022-06	0.59	1.34	1.00	1.25	2.1	cash covers	no	survived; exited crypto 2023
Metropolitan, 2022-09	0.59	0.71	0.95	0.76	1.3	0.92	no	survived; exited crypto 2023
Metropolitan, 2022-12	0.58	0.26	0.96	0.49	0.8	0.71	no	survived; exited crypto 2023
Metropolitan, 2023-03	0.61	0.30	0.95	0.28	0.5	cash covers	no	survived; exited crypto 2023
Silicon Valley Bank, 2022-12	15.46	12.51	117.30	3.30 (Circle only)	0.2	cash covers	no	failed Mar 10, 2023

Table D.3: Bank-lending panel: inference with common quarterly shocks. Loan regressions of Table 10; coefficients in percentage points per \$10 billion of absorption. Standard errors clustered by bank, by bank and quarter, and by quarter; p is the wild cluster bootstrap by quarter with the null imposed (Webb weights, 1,999 draws).

Absorption	Deposit control	Coefficient	Estimate	SE bank	SE two-way	SE quarter	Bootstrap p
stablecoin supply	yes	low headroom	−0.5	0.25	0.33	0.31	0.155
	yes	low headroom × bill-scarce	+0.6	0.29	0.36	0.36	0.108
	no	low headroom	−0.3	0.30	0.37	0.35	0.374
	no	low headroom × bill-scarce	+0.2	0.34	0.40	0.40	0.709
attestation bills and repo	yes	low headroom	+0.3	0.21	0.16	0.18	0.143
	yes	low headroom × bill-scarce	−0.6	0.30	0.25	0.21	0.035
	no	low headroom	+0.5	0.34	0.09	0.14	0.058
	no	low headroom × bill-scarce	−0.7	0.39	0.25	0.28	0.049

D.5 The price of LCR slack

Proposition 4.1 implies that the premium on safe assets should rise, nonlinearly, as the aggregate liquidity coverage ratio falls, and Bech and Keister (2017) that the term premium on unsecured bank funding beyond 30 days should rise with it. Table D.5 regresses quarterly averages of three spreads on the premium the calibrated model implies at the disclosed aggregate ratio of the five largest banks, 1.14–1.22 over 30 quarters, with and without controls for overnight reverse repo balances, the policy rate and the bill share of marketable debt. Over that range the model’s premium moves by only 0.3–5.4 basis points. The bill-fed funds spread falls with it, as the model predicts, with a coefficient of −3.0 (t of −1.6) without controls and −1.1 (t of −0.5) with them, against a model value of −1; the commercial-paper

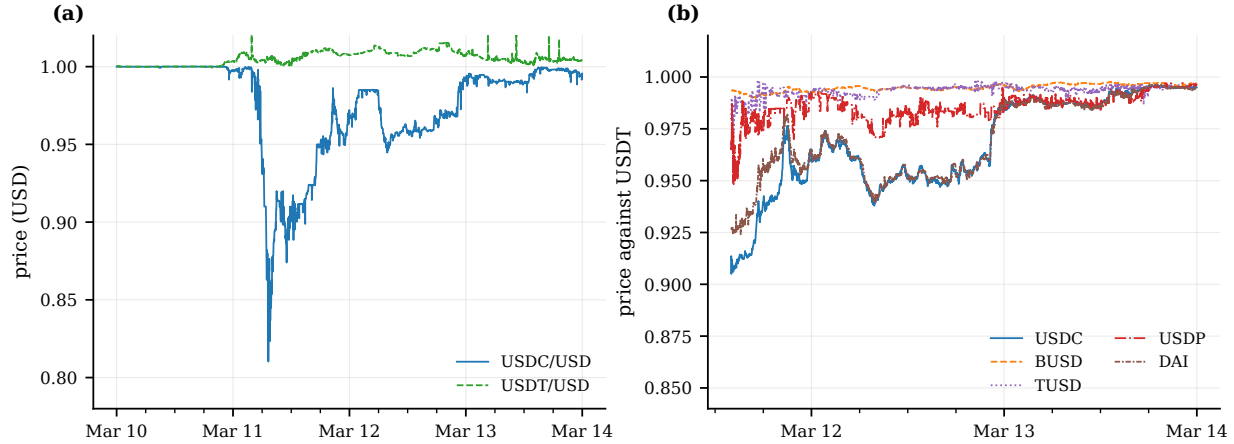


Figure D.1: Stablecoin prices in the March 2023 run. One-minute closing prices. (a) USDC/USD and USDT/USD on Bitstamp, March 10–13. (b) USDC, BUSD, TUSD, USDP and DAI against USDT on Binance from March 11, 14:00 UTC, when the pairs were relisted.

Table D.4: Robustness of the bill-dollar / non-bill-dollar local projections. Weekly local projections as in Table 11 (three-month bill unless stated, $h = 4$ weeks, basis points per \$3.5 billion, Newey–West standard errors in parentheses; the last column is the t -statistic of the difference between the two coefficients, with their HAC covariance in its standard error).

Row	Sample	Bill dollars	Non-bill dollars	Diff. t	n
A. Full attestation sample	2021.4–	−14.1 (6.0)	+4.8 (3.9)	−2.02	228
A. Excluding May–December 2022	2021.4–	−6.4 (3.8)	+4.8 (3.2)	−1.69	207
A. Shares lagged one extra quarter	2022.7–	−13.1 (4.0)	+6.1 (3.9)	−2.57	172
A. Plus lagged SOFR – 3M bill level	2023.1–	−14.4 (4.3)	+9.0 (4.2)	−2.88	162
B. 4-week bill	2023.1–	+5.3 (17.8)	−8.8 (15.3)	0.43	162
B. 6-month bill	2023.1–	−5.1 (4.0)	+3.6 (4.3)	−1.09	162
B. 1-year bill	2023.1–	−11.4 (4.2)	+6.3 (3.8)	−2.31	162
C. Financial CP – 3M bill	2022.7–	+12.2 (6.6)	−10.1 (4.7)	2.05	155
C. SOFR – 3M bill	2023.1–	+12.6 (3.8)	−8.7 (3.9)	2.90	162
C. 3M bill – EFFR	2023.1–	−11.6 (3.4)	+6.8 (3.8)	−2.74	162
C. 3M bill – ex-post OIS proxy	2022.7–	−11.2 (4.6)	+4.7 (3.4)	−2.14	168
C. ON RRP balances (\$bn)	2023.1–	−9.8 (37.6)	+56.5 (41.5)	−0.93	162
D. Inflows only / outflows only	2023.1–	−13.5 (6.4) / −16.2 (5.9)	+10.6 (6.2) / +6.2 (6.1)		162
E. USDC flows only, Circle monthly shares	2023.1–	−17.7 (4.8)	+10.7 (4.2)	−3.45	190
E. USDT flows only, Tether quarterly shares	2023.7–	−44.1 (19.6)	+75.7 (39.3)	−2.04	136
F. Horizon 2 / 6 / 8 / 12 weeks	2023.1–	−1.0 / −0.9 / +0.8 / +2.6	−0.9 / +2.3 / +0.2 / +0.6		154–164
G. Monthly data, $h = 1$ month	2022.7–	−16.7 (9.1)	+15.5 (8.6)	−1.85	41
G. Monthly data, $h = 1$ month	2023.1–	−3.2 (4.0)	+4.4 (3.3)	−1.09	35
H. Aggregate flow, OLS	2020.3–	−4.7 (1.1)			338
H. Aggregate flow, 2SLS, funding-rate instrument (F 15.5)	2020.3–	−6.8 (4.1)			338
H. Aggregate flow, 2SLS, BTC return and funding rate (F 7.4)	2020.3–	−8.7 (3.7)			338

The two-year yield enters every regression as a control and is therefore not a dependent variable. Rows H instrument total issuer flow, not the bill split, with crypto-demand instruments; the instruments are too weak for the split. Monthly aggregation leaves 35–41 observations; the sign holds and the precision does not.

spreads do not move. The evidence is consistent with the small premium the model implies in the observed range and has no power to measure it.

Table D.5: The price of LCR slack. Quarterly averages, 2019–2026. Spread (basis points) on the premium implied by the calibrated model at the disclosed aggregate LCR of the five largest bank holding companies. Controls: overnight reverse repo balances, the effective federal funds rate, bills as a share of marketable debt. Newey–West standard errors (four lags) in parentheses.

Spread	Controls	Coefficient	s.e.	n	R^2
3-month bill – EFFR	no	−3.00	(1.86)	30	0.05
3-month bill – EFFR	yes	−1.15	(2.11)	26	0.68
90-day – 30-day financial CP	no	−1.56	(1.52)	30	0.02
90-day – 30-day financial CP	yes	−0.24	(1.38)	26	0.67
90-day financial CP – 3-month bill	no	+0.47	(0.81)	30	0.01
90-day financial CP – 3-month bill	yes	+0.05	(0.54)	26	0.53

Appendix E Ownership, market structure and other instruments

The results of this appendix are derived in the kinked benchmark of Appendix F, in which a bank pays a constant cost per dollar below its requirement and per dollar below its actual stress need, and the issuer chooses its quantity. The propositions use the region of that benchmark in which the requirement does not bind; the numbers in Figures E.1 and E.2 and in Section E.9 are those of the benchmark and share its fragility, which Appendix F explains.

The main text takes the stock of coins as demand-determined and lets the issuer choose only its reserve mix. This appendix lets an issuer that faces no binding demand constraint choose its quantity as well, with a convex issuance cost $cs^2/2$ and the convenience value $v \ln(1 + s)$ used in an earlier version of the calibration, and asks how the choice depends on who owns the issuer. As in the main text, s counts issuance beyond today’s stock. The parameters are $v = 10$, $c = 2.7 \times 10^{-5}$ (so that a deposit-backed issuer paired with a large bank issues about \$1.5 trillion under laissez-faire, near the 2028 projection) and a failure cost $F_B = \$50$ billion; they are not targeted to data, and the reserve ratio is held at 0.4 rather than optimized as in Section 6.5. The direction of the ownership results does not depend on these values; the magnitudes do, and the quantities are what an issuer would issue if demand did not bind rather than equilibrium stocks: on the demand curve of Section 6, \$1.5 trillion of coins would require an adoption level about five times today’s. The question is practical. The GENIUS Act allows bank subsidiaries to issue, the European framework routes e-money tokens through credit institutions, and a bank-owned issuer may belong to one bank or to a consortium. Each design changes the weight the issuer places on bank profits, and through that weight the amount it issues, where it holds reserves and how exposed its partner bank becomes.

E.1 The issuer's problem

The issuer chooses issuance s and the bond share of reserves α to maximize

$$\Phi(s, \alpha; \omega) = \Pi_S(s, \alpha) + \omega \Delta\Pi_B(s, \alpha), \quad (\text{E.1})$$

where Π_S is seigniorage, $s[\alpha(1/q_0 - 1) + (1 - \alpha)(R_D^S - 1)]$, minus expected run losses $\rho \phi_{\text{run}} s$ and a convex issuance cost $cs^2/2$, and $\Delta\Pi_B$ is the change in the owning banks' profits relative to no issuance. The share ω runs from zero (a non-bank issuer) to one (a bank-owned issuer). Two ownership structures matter. In a consortium all banks own the issuer in proportion to their size, and the issuer's deposits are spread across them; the issuer internalizes the industry's profit. Under single ownership one bank owns the issuer and receives all of its deposits; the issuer internalizes that bank's profit only.

E.2 Consortium ownership reduces issuance

A bank's profit changes with issuance through three channels: it loses household deposits (a franchise loss of $R - R_D$ per dollar), it gains issuer deposits (a wholesale margin of $R - R_D^S$ per dollar at the receiving bank), and in the constrained regime it bears the collateral repricing and liquidation costs of Proposition 4.1. For the consortium the first two net to

$$\frac{\partial \Delta\Pi_B}{\partial s} = -(R - R_D) + (1 - \alpha)(R - R_D^S) \leq 0 \quad \text{whenever } R_D^S \geq R_D, \quad (\text{E.2})$$

in the slack regime, with the collateral terms adding further negative contributions beyond s_1 .

Proposition E.1 (consortium ownership). *If $R_D^S \geq R_D$, the consortium's profit change is decreasing in issuance at every (s, α) in the slack regime. Hence, for a given bond share α , Φ has increasing differences in $(s, -\omega)$ on that regime, and the issuer's optimal issuance is nonincreasing in ω for any seigniorage function and any issuance cost. When the issuer chooses α as well, the statement is conditional on α ; in the US calibration it holds for the jointly optimal choice (Figure E.1), but we have no general proof.*

The result uses only the envelope theorem on the banks' side and monotone comparative statics on the issuer's; it needs neither concavity of Φ nor an interior solution. The condition says that wholesale deposits cost banks at least what retail deposits cost, which holds whenever issuer deposits are uninsured and priced at or above money-market rates. The intuition is that a consortium cannot capture deposits from rivals—every dollar of issuer deposits is

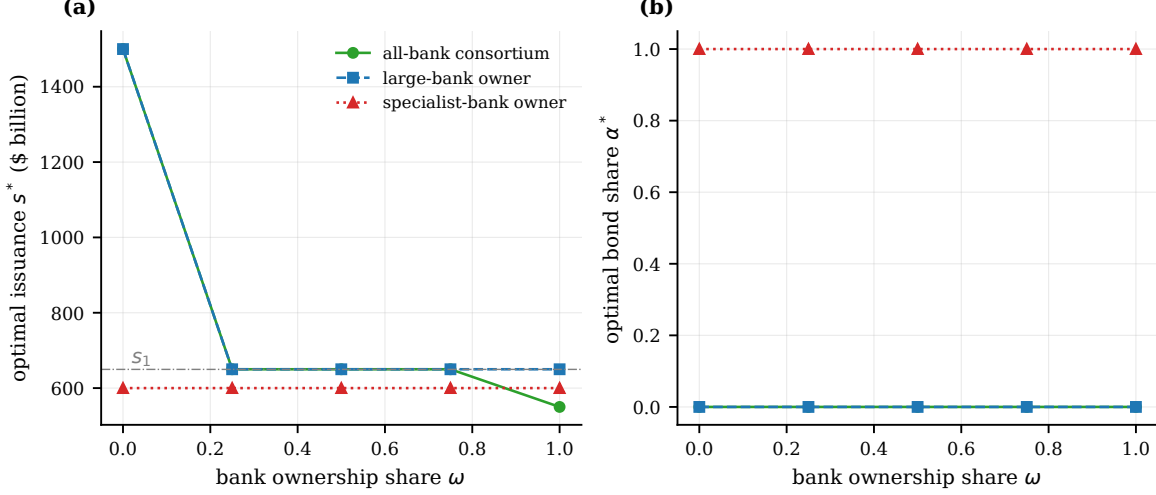


Figure E.1: Optimal issuance and bond share against the ownership share. US calibration. (a) Optimal issuance $s^*(\omega)$ for an all-bank consortium, a large-bank owner and a specialist-bank owner; the horizontal line is s_1 . (b) Optimal bond share $\alpha^*(\omega)$.

a dollar of lost household deposits somewhere in the consortium—so it sees issuance as a franchise loss and issues less the more it owns.

Figure E.1 shows the US calibration. Where the unconstrained optimum (\$1.5 trillion) exceeds s_1 , consortium issuance falls from \$1,500 billion to \$550 billion as ω rises from 0 to 1, a 63 percent reduction. The reduction is a transfer internalized, not a social cost removed; Proposition E.6 returns to this point.

E.3 Single ownership can increase issuance

Proposition E.2 (single ownership and deposit capture). *Under single ownership with N symmetric banks, the owner's profit change is increasing in issuance in the slack regime if and only if*

$$(1 - \alpha)(R - R_D^S) > \frac{R - R_D}{N}, \quad (\text{E.3})$$

and optimal issuance can then be increasing in ω . The capture incentive disappears once the collateral constraint binds.

A single owner receives all of the issuer's deposits but loses only its $1/N$ share of the household deposits that fund the coins; if the wholesale margin on captured deposits exceeds the diluted franchise loss, ownership raises issuance. Condition (E.3) is more likely to hold when the issuer holds its reserves as deposits (α low), the wholesale spread is wide and the owner is small. In the US calibration the large-bank owner stops at s_1 for every $\omega > 0$, because the capture incentive is bounded by the point at which further deposits reprice

collateral, while the specialist-bank owner holds bills (next subsection) and stops at s_1 for the reason of Proposition E.5. Without bank ownership ($\omega = 0$) an issuer partnered with a large bank holds deposits and issues past s_1 into the constrained regime (Figure E.1a), which is the laissez-faire benchmark of the policy map in Section E.9; one partnered with a specialist bank holds bills and stops near s_1 . This is the formal content of the distinction between a consortium rule and a one-bank-one-issuer arrangement: the first restrains issuance, the second may not.

Proposition E.3 (concentration of redemption). *(i) The run threshold falls with redemption concentration, $\partial\pi^*/\partial K = -1/(2\mathcal{H}) < 0$, while the secondary-market price impact $K\lambda$ rises with it.*
(ii) The marginal reduction in run risk from concentration, $1/(2\mathcal{H})$, is increasing in each step value of ϕ .
(iii) Hence, if the issuer chooses K to trade off run losses against a cost of price instability that is increasing in K , and the prior on the fundamental has a nondecreasing density on the relevant range, the optimal concentration is nondecreasing in each step value of ϕ : an issuer whose deposits sit at a weak partner, where $\phi_1 = \alpha\bar{h} + (1 - \alpha)\text{LGD}$ applies beyond $\hat{\lambda}$, concentrates redemption more.

Part (iii) is the mechanism of Ma et al. (2026, Proposition 3) under a different sufficient condition: they fix the demand elasticity, we fix the shape of the prior. The calibrated prior of Section 6 satisfies the condition.

E.4 Partial backstops raise run risk

A bank that owns an issuer may promise to buy coins at par in a run.

Proposition E.4 (the backstop trap). *Suppose the partner bank stands ready to buy up to a share H of the coins at par in the secondary market and holds them. (i) If $H \leq \min\{\hat{\lambda}, 1 - \phi_0\}$, the run threshold rises by $KH^2/(2\mathcal{H})$: every feasible partial backstop raises run risk. (ii) The bank's capacity is bounded by its stress liquidity beyond its buffer, $H \leq H_{\max} \equiv \max\{0, \theta q_1 b_1 - \gamma_{\max} D_1^H - N_{0,1} - \lambda_{\text{reg}}(1 - \alpha)s\}/s$, which falls to zero as issuance approaches the collateral threshold; a backstop that eliminates runs requires $H \geq 1 - \phi$, which exceeds $H_{\max}$ whenever the partner's slack is smaller than the issuer's size.*

The reason, as in Ma et al. (2026, Appendix C.2), is that a par bid removes the price discount that discourages early redemption without removing the insolvency that makes late redemption costly; holders redeem earlier. The full backstop that would stabilize is the one

the bank cannot afford precisely when it is needed, because its slack is the same collateral that the run consumes.

E.5 Self-limiting bond-backed issuance

Let $y_I = 1/q_I - 1$ and $y_{III} = 1/q_{III} - 1$ be the bond yields in the slack and fully constrained regimes of the two-state stress model, let $s_2 > s_1$ be the issuance at which the fully constrained regime begins, and let $\hat{c} \equiv (y_I - \rho\bar{h})/s_1$ be the issuance cost at which the monopolist's unconstrained optimum is exactly s_1 .

Proposition E.5 (self-limiting issuance). *Consider a monopolist issuer that holds bonds ($\alpha = 1$) under a two-state stress distribution with $\theta > \gamma_{\max}$. (i) Its profit is strictly decreasing in s throughout the transition region $(s_1, s_2]$, with slope $\gamma_{\max}/\theta - 1 - \rho\bar{h} - cs < 0$. (ii) If $c \leq \hat{c}$, so that the unconstrained optimum lies beyond s_1 , the issuer stops exactly at s_1 provided $c \geq \underline{c} \equiv [(y_I - \rho\bar{h}) - \sqrt{(y_I - \rho\bar{h})^2 - (y_{III} - \rho\bar{h})^2}]/s_1$; for $c < \underline{c}$ it jumps past the transition into the constrained regime.*

Proposition E.5 is why, in Figure E.1 and in the policy map of Section E.9, bond-backed issuers cluster at s_1 , as does a deposit-backed issuer whose owning bank bears the collateral repricing, while a non-bank deposit-backed issuer and price-taking competitive issuers go beyond it. A monopolist that holds bills internalizes the collateral price externality through its own margin; a deposit-backed issuer does not, because its reserves do not trade in the market it congests. The threshold therefore binds behaviorally, not only technologically, when reserves are bonds and issuance is concentrated. In the US calibration $\underline{c}$ is an order of magnitude below the calibrated cost and $\hat{c}$ is above it, so the bond-backed monopolist stops at s_1 .

E.6 Summary of the ownership results

Ownership affects issuance through a transfer (the franchise loss) and through the partner bank's exposure (deposit concentration and the backstop trap). The transfer is internalized fully by a consortium and partly, with the opposite sign, by a single owner; the exposure is created by deposit reserves and concentrated at the owner. Neither effect is a social cost in itself; both change the location of the two externalities of Section 4. That is the question Section 5 answers for a demand-determined stock and Sections E.7 and E.9 answer for an issuer that chooses its quantity: which instrument addresses which externality, and whether ownership rules are the right instrument at all.

E.7 Ownership regulation, the bonds-only rule and market structure

The main text compares instruments at a demand-determined stock. For an issuer that chooses its quantity, three further results hold; they use the welfare function of Section 5 and, in addition to (C1), the participation condition (C2): the issuer participates at the planner's quantity, $(1 - \mu)r_B + \mu r_\mu \geq c s^{SP}$.

Proposition E.6 (ownership regulation). *(i) Welfare depends on the ownership share only through the allocation it induces: ω is a transfer. A consortium floor internalizes the franchise loss, which is not a social cost, and lowers issuance below the planner's level whenever the planner's marginal convenience value exceeds the issuer's marginal seigniorage at the planner's quantity.*

(ii) Under (C1), for any consortium floor inducing $(s^\omega, \alpha^\omega)$, the pair consisting of a cap $\bar{s} = s^\omega$ and a reserve ratio $\hat{\mu}$ that shuts the loop attains at least the floor's welfare for every F_B , provided the issuer's bond share under the pair is no larger than α^ω ; in particular whenever both are at the deposit corner.

Proposition E.7 (the bonds-only rule). *Let (s^{LF}, α^{LF}) be the laissez-faire allocation and s^B the allocation under a bonds-only rule. (i) If the partner cannot fail under laissez-faire ($\hat{\lambda} \geq 1$), the rule raises the liquidation cost at every redemption level, raises the run threshold and the fire-sale exposure, and lowers welfare unless the quantity change compensates; for a monopolist it also triggers Proposition E.5. (ii) If the partner can fail, the rule dominates laissez-faire if and only if $F_B \geq \bar{F}_B$, where $\bar{F}_B$ is the fire-sale and quantity loss of the rule divided by the run probability under laissez-faire; $\bar{F}_B = \infty$ in case (i).*

Proposition E.8 (market structure). *Price-taking issuers expand until per-coin seigniorage net of run losses equals the marginal issuance cost and therefore enter the constrained regime whenever $y_I - \rho \bar{h} > cs_1$; a bond-backed monopolist stops at s_1 (Proposition E.5). Market power is a substitute for the cap.*

E.8 Consolidation of a bank-subsidiary issuer

Question 59 also asks whether a bank holding company should count the HQLA of a stablecoin-issuing subsidiary in its own LCR, and whether the subsidiary's outflows should be counted with it. The threshold formula answers this directly, because the treatment changes what a coin consumes.

Corollary E.1 (consolidation of a bank-subsidiary issuer). *Let the issuer be a subsidiary of a bank holding company and let λ_c be the run-off rate the LCR assigns to the coins if they are consolidated. The headroom a coin consumes, the denominator of (4.4), is*

- (i) $\lambda_c - \gamma_{\max}$ if the subsidiary's reserves and the coins' outflows are both consolidated, whatever the reserve mix α ;*
 - (ii) $\theta\alpha/q_0 + \lambda_c - \gamma_{\max}$ if the reserves are excluded but the outflows are counted;*
 - (iii) $\theta\alpha/q_0 + \lambda_{\text{reg}}(1 - \alpha) - \gamma_{\max}$ if both are excluded, as for a non-bank issuer.*
- Mixed treatment (ii) gives the lowest threshold; under (i) the reserve choice no longer matters and the threshold is set by λ_c alone.*

At today's values the room for further issuance by an issuer holding bills is \$650 billion under (iii), \$678 billion under (i) with $\lambda_c = 1$ (\$1,910 billion with $\lambda_c = 0.4$), and \$320 billion under (ii). Excluding the subsidiary's HQLA while counting its outflows, one of the options the FDIC raises, would halve the distance to the threshold for bank-issued coins. Consolidation keeps the run channel inside the group: under (i) a coin run is a run on the parent, and the capital condition of Proposition 5.3(iv) applies to the parent with $D^S = s$.

E.9 A policy map for an issuer that chooses its quantity

Figure E.2 compares the instruments at the baseline calibration for the two partner types. To check the ranking, we repeat the comparison on a grid of the spread between wholesale deposit rates and bill yields (-40 , 0 and $+40$ basis points), the elasticity of safe-asset supply (0 , 2 and 10) and the two partner types, for a non-bank issuer ($\omega = 0$). The ranking of the main text survives the change of framework in most cells. The recycled reserve ratio is the best instrument in 16 of the 18 cells, and the cap at the planner's quantity is within \$4.6 billion of the best instrument in every cell. The two exceptions have a large partner and a fixed supply of safe assets, the case of Figure E.2. There a deposit-backed issuer goes past s_1 , but the shortfall it creates sits at the partner's requirement, whose cost is a transfer in the benchmark (Appendix F), so its issuance has almost no real cost: the planner would issue \$2,200 billion against \$1,500 billion under laissez-faire, the cap does not bind, the recycled ratio changes welfare by -0.9 billion dollars, and a 50 percent run-off rate, which relaxes the partner's requirement, changes it by $+4.2$ billion dollars, the most of any instrument. The cap's gain is \$0–26 billion: zero where a specialist-partnered issuer holds bills, which stops it near s_1 (Proposition E.5), and in the cells just described, and positive where the issuer goes past s_1 at a real cost, issuing \$700–1,600 billion. In those cells the recycled ratio gains \$2.7–4.6 billion more than the cap, at a larger quantity, \$900–1,000 billion, because it removes part of the collateral each coin absorbs rather than the coins. The reserve ratio

without recycling lowers welfare in 15 of the 18 cells (changes of -24 to -6.4 billion dollars) and is never first. The bonds-only rule changes welfare by -5.6 to $+24.4$ billion dollars, the gain coming from the cells in which it moves the issuer out of deposits and so stops it at s_1 , the loss from the cells with a large partner and a fixed supply, in which deposit-backed issuance past s_1 had no real cost. The consortium, which spreads the issuer's deposits across all banks, stops near s_1 on its own when supply is inelastic (Proposition E.1) and then comes within \$5.6 billion of the cap; it is the worst instrument when supply is elastic (-65 to -41 billion dollars over *laissez-faire*): an issuer owned by all banks then issues \$2,200 billion, because with elastic supply the collateral channel contracts lending without lowering bond returns, the loan rate rises along the demand curve, and aggregate bank profits increase with the contraction. Ownership then turns the collateral externality into a cartel instrument; the dominance result of Proposition E.6 is its inelastic-supply case, and the cartel case depends on the slope of loan demand.⁴ The reserve choice follows Proposition 4.4 with the run-loss differential visible: an issuer partnered with a specialist bank holds bonds at -40 and 0 basis points and deposits only at $+40$, while one partnered with a large bank holds deposits at every spread, because deposits at a partner that cannot fail carry no run loss while bonds carry the fire-sale discount. Lowering the run-off rate to 50 percent leaves a specialist-partnered issuer in bonds and a large-bank-partnered issuer at \$1,500 billion of deposits, and raises the room for deposit-backed issuance from \$678 billion to \$1,465 billion: the collateral face of Proposition 5.3. For a bank-subsidiary issuer ($\omega = 1$) the map keeps the same ranking: the recycled ratio is first in 18 of the 18 cells, the cap within \$4.0 billion of it, the ratio without recycling is never first, and the consortium is the worst instrument in all 6 cells with elastic supply; the cap's gain is smaller, \$0–9 billion, because the owned issuer already restrains itself.

The European rule is the mirror image of the bonds-only rule: MiCA requires issuers of e-money tokens to hold at least 30 percent of reserves as bank deposits, and 60 percent if they are significant. On the same grid, for a non-bank issuer, such a floor beats the cap in none of the 18 cells, at either 30 or 60 percent. With a large partner it is neutral in every cell: the issuer already holds deposits there, so the floor does not bind. With a specialist partner it does harm in the 7 cells where the issuer would otherwise hold bills: it places at least 30 percent of reserves in a bank that a full redemption would fail, opening the run–bank loop that the bond reserve had closed, at a cost of up to \$23.6 billion, the same at 30 and 60 percent because the loss is the opening of the loop rather than the size of the deposit; where

⁴With linear loan demand, bank revenue $R(L)L$ is increasing in L at the calibrated point, so the gain comes from the funding side: with elastic supply the price of the collateral the issuer absorbs does not rise, bond returns are unchanged, and the lending contraction raises the loan rate on the inframarginal stock. A steeper or concave loan demand reduces the gain.

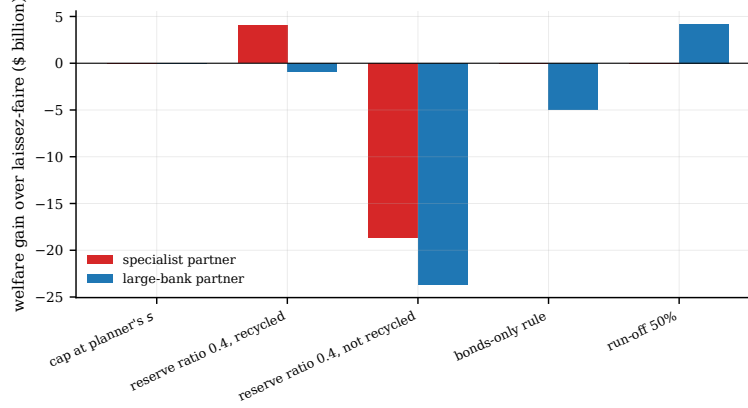


Figure E.2: Policy instruments for an issuer that chooses its quantity (appendix framework). Non-bank issuer ($\omega = 0$), baseline calibration. Welfare gain over laissez-faire by instrument for an issuer partnered with a specialist bank and with a large bank. The reserve ratio is 0.4; the run-off rule sets the regulatory rate on issuer deposits at 50 percent.

the issuer already held deposits the floor does not bind. A deposit floor is a liquidity rule for the issuer that becomes a solvency risk for a small partner, which is Proposition 5.3(iii) seen from the issuer's side: a floor on deposits should be paired with a bank-relative limit on where they are placed.

Appendix F The linear benchmark

The earlier version of this paper modelled the cost of a requirement shortfall as linear, with no daily shocks. The version reported here corrects it so that the bank's objective, its first-order condition, its profit and welfare contain the same two costs: a constant regulatory cost $\tau_r = p\varphi\theta$ per dollar below the requirement REQ_j , a shadow price that is a transfer in welfare, and the expected liquidation cost $p\varphi\theta$ per dollar below the actual stress need $NEED_j$. The existing stock enters the initial balance sheets with its observed reserves, including \$14.5 billion of issuer deposits, whose regulatory outflow is already in the disclosed net outflows. The benchmark has the structure of the limit $\sigma \rightarrow 0$ of the model of Section 3 (Proposition 4.1(iv)), with the cover cost c replaced by τ_r , no discount-window coverage of stress shortfalls and no limit on the capacity to cover the requirement. It has up to four regimes: an asset-swap regime below s_1 ; a transition in which banks hold exactly their buffers and the price of collateral rations the residual; a plateau at the premium τ_r in which only the requirement of the bank whose actual need lies below it is short; and a constrained regime in which every bank is below its actual need. With deposit reserves the economy moves from the asset-swap regime to the plateau at s_1 , and the multiplicity band of Proposition 5.2(iv) appears near the threshold. Figure F.1 shows lending along these regimes.

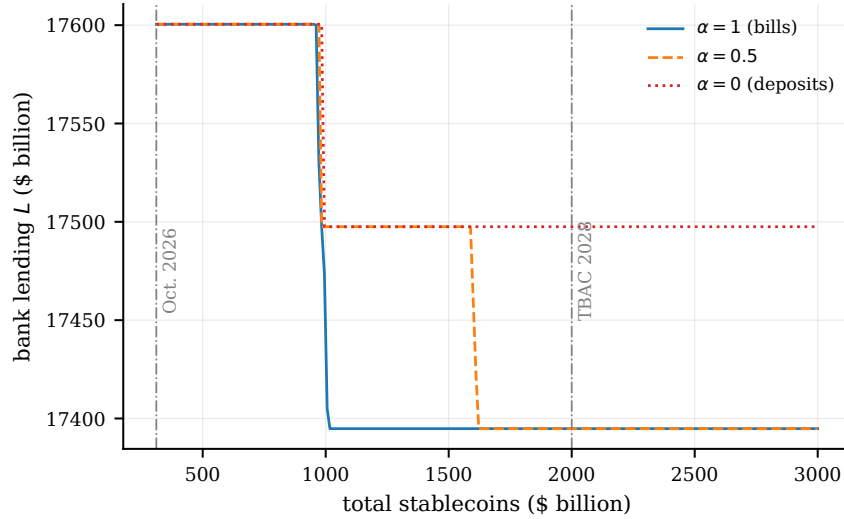


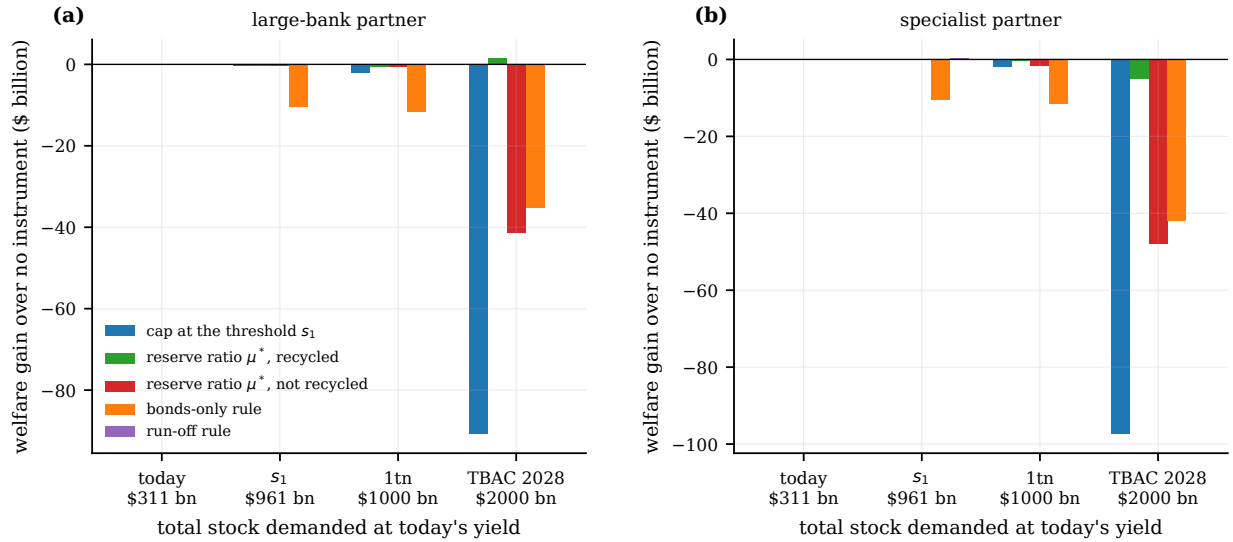
Figure F.1: Linear benchmark: lending against the stock of stablecoins. US calibration: bank lending against total stablecoins by the reserve share α of new issuance, the existing reserves at their observed mix; vertical lines at the October 2026 stock and the 2028 projection.

The linear benchmark is fragile in a way that the regulation itself explains. With a constant cost per dollar of shortfall, a bank whose actual stress need is below its requirement—the bank that receives issuer deposits, whose regulatory run-off of 100 percent exceeds their actual run-off of 0.13—is the cheapest place to hold the system’s regulatory shortfall, and in equilibrium it does: at the \$2 trillion demand level the issuer’s partner runs a requirement shortfall of \$1,256 billion under laissez-faire, and because that shortfall costs only a transfer, deposit-backed issuance past s_1 has almost no real cost. The daily compliance and remediation rules of Board of Governors of the Federal Reserve System (2026) make such a position impossible, and the narrow banking of the partner banks in the data (Section 7.2) is the opposite of it. The model of Section 3 replaces the constant cost with the probabilistic, capacity-bounded cost the rules imply. Tables F.1–F.3 and Figure F.2 report the policy comparison in the benchmark against laissez-faire, for reference. Its rankings differ from those of Section 6.5 for the same reason. At the \$1 trillion demand level a cap at s_1 changes welfare by -2.2 billion dollars, the recycled reserve ratio by -0.5 and the bonds-only rule, which moves the shortfall from the partner’s requirement into actual liquidation losses at other banks, by -11.7 ; at \$2 trillion the recycled ratio changes it by $+1.5$ billion dollars, while the cap loses \$91 billion and the bonds-only rule \$35 billion. Because the benchmark lets the partner hold a shortfall that the rules would force it to cover, it understates the cost of deposit-backed issuance and the value of the instruments that relieve it.

Table F.1: Linear benchmark: instruments against laissez-faire, large-bank partner.

US calibration, welfare gains over laissez-faire in \$ billion. Demand is the total stock demanded at today's bill yield; issuance solves the demand curve at the model's own yield under each instrument, and "LF stock" gives the laissez-faire total stock and regime. The cap is at s_1 . The threshold level is \$961 billion; there laissez-faire has two stable equilibria, constrained and slack (Proposition 5.2). Λ is the lending and liquidation loss of the constrained regime, ΔV the convenience value of the coins a cap at s_1 removes; the run-off column gives the optimal set of regulatory rates. Fire-sale discounts are transfers; F_B equals the partner's equity. The specialist partner is in Table F.2.

Demand (\$ bn)	LF stock	Λ	ΔV	Cap	Ratio (μ^*)		Bonds only	Run-off (λ_{reg})
					recycled	not recycled		
311 (today)	311, I	0.0	0	+0.0	+0.0 (0.00)	+0.0	+0.0	+0.0 (0.00–1.00)
961, constrained	1,124, III	4.2	5	−0.3	−0.3 (0.00)	−0.3	−10.5	+0.0 (1.00)
961, slack	961, I	0.0	0	+0.0	+0.0 (0.00)	+0.0	−10.2	+0.0 (0.00–1.00)
1,000	1,170, III	4.2	6	−2.2	−0.5 (0.00)	−0.5	−11.7	+0.0 (1.00)
2,000	2,339, III	4.2	102	−90.9	+1.5 (0.46)	−41.3	−35.3	+0.0 (0.40–1.00)

**Figure F.2: Linear benchmark: instruments against laissez-faire, US calibration.**

Welfare gain over laissez-faire by instrument at four levels of demand, (a) with a large-bank partner and (b) with a specialist partner; constrained equilibrium where two exist. The reserve ratio μ^* is the smallest that keeps the system in the slack regime.

Table F.2: Linear benchmark: specialist partner. As Table F.1.

Demand (\$ bn)	LF stock	Λ	ΔV	Cap	Ratio, recycled (μ^*)	Ratio, not recycled	Bonds only	Run-off (λ_{reg})
311 (today)	311, I	0.0	0	+0.0	+0.0 (0.00)	+0.0	+0.0	+0.0 (0.00–1.00)
961, constrained	1,124, III	4.2	5	−0.2	−0.2 (0.00)	−0.2	−10.4	+0.3 (0.15)
961, slack	961, I	0.0	0	+0.0	+0.0 (0.00)	−0.0	−10.2	+0.0 (0.00–1.00)
1,000	1,170, III	4.2	6	−2.0	−0.4 (0.04)	−1.8	−11.6	+0.0 (1.00)
2,000	2,339, III	4.2	102	−97.5	−5.1 (0.48)	−47.9	−41.9	+0.0 (0.40–1.00)

Table F.3: Linear benchmark: sensitivity. Large-bank partner; welfare gains over laissez-faire in \$ billion at the \$1 trillion and \$2 trillion demand levels; constrained equilibrium where two exist. “Fixed quantity” holds issuance at the level label whatever the yield.

Variant	Demand (\$ bn)	LF stock	Λ	ΔV	Cap	Ratio, recycled (μ^*)	Ratio, not recycled	Bonds only	Run-off (λ_{reg})
baseline	1,000	1,170	4.2	6	-2.2	-0.5 (0.00)	-0.5	-11.7	+0.0 (1.00)
	2,000	2,339	4.2	102	-90.9	+1.5 (0.46)	-41.3	-35.3	+0.0 (0.40-1.00)
$\varepsilon = 2.0$	1,000	1,049	4.2	3	+0.9	+2.8 (0.00)	+2.8	-6.4	+2.8 (0.00-0.80)
	2,000	2,098	4.2	211	-207.3	+1.5 (0.46)	-41.5	-31.5	+1.5 (0.40)
$\varepsilon = 10.1$ (weekly)	1,000	1,272	4.2	9	-4.9	-3.3 (0.00)	-3.3	-16.6	+0.0 (0.80-1.00)
	2,000	2,545	4.2	90	-74.8	-0.2 (0.46)	-43.0	-41.2	+0.0 (0.40-1.00)
$\varepsilon = 20.3$ (2023-26)	1,000	1,623	4.2	18	-14.2	-12.7 (0.00)	-12.7	-36.2	+0.0 (0.60-1.00)
	2,000	3,247	4.2	92	-56.2	+1.6 (0.46)	-40.9	-59.8	+0.0 (0.30-1.00)
$\lambda_a = 0.10$	1,000	1,170	4.2	6	-2.2	-0.5 (0.00)	-0.5	-11.7	+0.0 (1.00)
	2,000	2,339	4.2	102	-90.9	+1.5 (0.46)	-41.3	-35.3	+0.0 (0.40-1.00)
$\lambda_a = 0.25$	1,000	1,170	4.2	6	-2.2	-0.5 (0.00)	-0.5	-11.7	+0.0 (1.00)
	2,000	2,339	4.2	102	-90.9	+1.5 (0.46)	-41.3	-35.3	+0.0 (0.40-1.00)
$\lambda_a = 0.40$	1,000	1,170	4.2	6	-2.2	-0.5 (0.00)	-0.5	-11.7	+0.0 (1.00)
	2,000	2,717	12.4	104	-60.2	+32.1 (0.46)	-10.7	-4.7	+29.9 (0.00-0.30)
$\lambda_a = 1$	1,000	1,358	17.1	7	+9.7	+11.4 (0.00)	+11.4	+0.2	+7.7 (0.00-0.60)
	2,000	2,717	48.7	104	-55.1	+37.2 (0.46)	-5.6	+0.4	+4.3 (0.00-0.30)
$\tau_r \times 0.5$	1,000	1,082	2.1	4	-2.4	-0.7 (0.00)	-0.7	-7.9	+0.0 (1.00)
	2,000	2,165	2.1	98	-90.8	+1.6 (0.46)	-41.3	-29.4	+1.6 (0.40)
$\tau_r \times 2$	1,000	1,358	8.5	7	+1.1	+2.8 (0.00)	+2.8	-17.0	+2.8 (0.00-0.60)
	2,000	2,717	8.5	104	-64.1	+28.3 (0.46)	-14.6	-21.1	+22.9 (0.30)
fixed quantity	1,000	1,000			+2.5	+4.2	+4.2	-4.5	+4.2
	2,000	2,000			-88.2	+4.2	-38.8	-27.5	+4.2

Appendix G Sensitivity of the policy comparison

Table G.1 repeats the comparison of Table 6 for a large partner at the \$1 trillion and \$2 trillion demand levels under each variant of the calibration: the daily shock’s standard deviation (0.05 and 0.10 of the requirement), a crisis cover cost of 136 basis points, discount-window coverage of stress shortfalls of 0 and 0.9, compliance costs treated as real, the demand semi-elasticities 2.0, 10.1 and 20.3, an actual run-off of 0.25, reserves held as bank deposits in the baseline, an internal run-off of 1 for banks outside the ratio, and a bill supply that responds to the price. The cap lowers welfare in every variant. The recycled Federal Reserve deposit share is never harmful, and it is the best instrument in every calibration with a normal equilibrium at \$2 trillion but one: when reserves are already held as bank deposits, the concentration limit gains slightly more. Where the discount window already covers 90 percent of stress shortfalls, the Federal Reserve deposit share and discount-window readiness gain little or nothing. Discount-window readiness raises both the capacity κ and the coverage ω ; raising only ω , as an earlier version of this paper did, gives +1.29 billion dollars at the \$1 trillion level and +11.7 at \$2 trillion with $\varepsilon_S = 2$. Where no normal equilibrium exists at \$2 trillion the table says so: the fold of Section 6.4 lies below that level in those calibrations.

Recycling and the ranking. Proposition 5.1 rests on condition (C1). Table G.2(a) relaxes it. The Federal Reserve lends back only a share r of the deposits it receives; in the second column for each level, every recycled dollar is also secured by a dollar of loans taken out of

Table G.1: Instruments against current rules: sensitivity. Large partner; welfare gains over current rules in \$ billion a year at the \$1 trillion and \$2 trillion demand levels. Columns as in Table 6; “Fed deposits” is the recycled share.

Variant	Demand	Cap	Fed deposits	DW ready	Deposits	Bonds only	Requirement
Baseline	1,000	−3.0	+1.3	+1.3	+1.2	−0.4	0.0
	2,000	no normal equilibrium (beyond the fold)					
$\sigma = 0.05$	1,000	−2.9	+0.9	+0.9	+0.8	−0.5	0.0
	2,000	−83.6	+12.5	+10.4	+12.2	−1.3	0.0
$\sigma = 0.10$	1,000	−3.1	+1.8	+1.7	+1.4	−0.3	0.0
	2,000	no normal equilibrium (beyond the fold)					
$c = 136$ bp (crisis)	1,000	−3.4	+1.6	+1.8	+1.2	−0.5	+0.1
	2,000	no normal equilibrium (beyond the fold)					
$\omega = 0$	1,000	−2.9	+3.0	+3.0	+2.8	−0.8	0.0
	2,000	no normal equilibrium (beyond the fold)					
$\omega = 0.9$	1,000	−2.6	+0.1	0.0	+0.1	−0.1	0.0
	2,000	−93.2	+1.4	−0.3	+1.3	−1.2	0.0
Compliance cost real	1,000	−2.7	+2.3	+1.5	+0.9	−0.4	+0.2
	2,000	no normal equilibrium (beyond the fold)					
$\varepsilon = 2.0$	1,000	−2.0	+2.6	+1.9	+1.6	−0.4	0.0
	2,000	−196.7	+14.3	+11.4	+10.3	−1.3	+0.1
$\varepsilon = 10.1$	1,000	−4.2	+0.6	+0.8	+0.6	−0.4	0.0
	2,000	no normal equilibrium (beyond the fold)					
$\varepsilon = 20.3$	1,000	−8.8	+0.3	−1.7	−1.9	−0.6	0.0
	2,000	no normal equilibrium (beyond the fold)					
$\lambda_a = 0.25$	1,000	−3.0	+1.4	+1.4	+1.2	−0.3	0.0
	2,000	no normal equilibrium (beyond the fold)					
Reserves as deposits	1,000	−1.7	+0.2	+0.2	0.0	−1.5	+0.2
	2,000	−88.2	+2.7	+2.7	+1.1	n/a	0.0
$\lambda_{\text{int}} = 1$	1,000	−3.0	+1.3	+1.3	+1.2	−0.4	0.0
	2,000	no normal equilibrium (beyond the fold)					
$\varepsilon_S = 2$	1,000	−2.3	+3.6	+2.3	+1.9	−0.5	+0.1
	2,000	−81.3	+15.2	+12.3	+11.3	−1.5	+0.2
$\varepsilon_S = 2, \omega = 0$	1,000	−2.1	+5.5	+4.3	+3.7	−0.8	+0.1
	2,000	−69.2	+29.0	+26.2	+22.5	−2.1	+1.0

the collateral pledged at the discount window, which lowers κ and ω . The deposit share is worth using only if r is at least 90 percent at the \$1 trillion level and 70 percent at \$2 trillion, and it beats discount-window readiness only above 99.96 and 94.6 percent. Every unrecycled dollar is a dollar of bank funding withdrawn, which costs the loan spread, while the collateral the share frees is worth a premium of basis points. If holding the deposits costs the Federal Reserve k a year per dollar, through credit risk on the recycled lending or operating costs, and the share is re-optimized under that cost, it stops beating discount-window readiness at $k = 0.09$ basis points at the \$1 trillion level and $k = 21$ at \$2 trillion. Securing the recycled funds with window collateral moves these thresholds little (99.97 and 94.9 percent), because at these stocks the deleveraging and stress terms that the window collateral covers are small. Table G.2(b) varies the demand semi-elasticity from 0.5 to 20.3 and the cover cost from 27

Table G.2: Recycling and the ranking of instruments. Large partner; welfare gains over current rules, \$ billion a year. (a) Federal Reserve deposit share at its best value μ^* (in parentheses) when a share r of the deposits is lent back to banks; “encumbered”: each recycled dollar is secured by a dollar of loans out of the collateral pledged at the window. (b) Gain of the recycled deposit share minus that of discount-window readiness, by demand semi-elasticity ε (rows) and cover cost c in basis points (columns); n.e.: no normal equilibrium. \$1 trillion with a fixed bill supply; \$2 trillion with $\varepsilon_S = 2$.

(a) Recycling								
Recycled share r	\$1 trillion				\$2 trillion, $\varepsilon_S = 2$			
	Fed deposits		encumbered		Fed deposits		encumbered	
0.0	0.0	(0.0)	0.0	(0.0)	0.0	(0.0)	0.0	(0.0)
0.5	0.0	(0.0)	0.0	(0.0)	0.0	(0.0)	0.0	(0.0)
0.6	0.0	(0.0)	0.0	(0.0)	0.0	(0.0)	0.0	(0.0)
0.7	0.0	(0.0)	0.0	(0.0)	+1.6	(0.4)	+0.5	(0.4)
0.8	0.0	(0.0)	0.0	(0.0)	+5.4	(0.5)	+4.8	(0.5)
0.9	+0.3	(0.1)	+0.2	(0.1)	+9.9	(0.6)	+9.6	(0.6)
1.0	+1.3	(0.5)	+1.3	(0.6)	+15.2	(1.0)	+15.2	(1.0)
Discount-window readiness			+1.3				+12.3	
Deposit reserves			+1.2				+11.3	

(b) Fed deposits minus discount-window readiness								
ε	\$1 trillion, c (bp)				\$2 trillion, $\varepsilon_S = 2$, c (bp)			
	27	54	95	136	27	54	95	136
0.5	+0.7	+1.0	+1.3	+1.6	+4.8	+5.9	+7.6	+9.3
1.0	+0.7	+0.9	+1.2	+1.5	+4.6	+5.6	+7.2	+8.9
2.0	+0.6	+0.7	+1.0	+1.2	+4.3	+5.1	+6.5	+8.0
4.0	+0.4	+0.4	+0.5	+0.6	+3.6	+4.1	+5.1	+6.2
6.6	+0.1	0.0	−0.1	−0.2	+2.7	+2.8	+3.3	+3.9
10.1	0.0	−0.2	−0.4	−0.6	+1.8	+1.6	n.e.	n.e.
15.0	+0.7	+0.6	+0.3	−0.3	n.e.	n.e.	n.e.	n.e.
20.3	+2.1	+2.1	+1.6	+0.1	n.e.	n.e.	n.e.	n.e.

to 136 basis points, with full recycling. The cap lowers welfare in every cell, by \$1.9–10.0 billion at the \$1 trillion level and \$63–695 billion at \$2 trillion, where the loss grows sharply as demand becomes less elastic. At \$2 trillion the Federal Reserve deposit share is the best instrument in all twenty-two cells with a normal equilibrium. At \$1 trillion discount-window readiness overtakes it in six of the thirty-two cells, all with a semi-elasticity at or above the baseline and most with a cover cost above it, by at most \$0.6 billion.

The issuer’s reserve mix and the coin holders’ alternative. Table G.3 repeats the comparison with the reserve mix chosen as in Section 6.4: the rate on issuer deposits is the partner bank’s marginal value of them, $R(L) - \partial \mathcal{C}_1 / \partial D_1^S$; the bill share responds to the bill-minus-deposit spread with the semi-elasticity η , applied to the whole stock; and

Table G.3: Instruments when the issuer chooses its reserve mix. Large partner; welfare gains over current rules with the same reserve-mix rule, \$ billion a year. η : percentage points of the bill share per basis point of the spread between the bill return and the rate on issuer deposits (Section 6.4). Stock in \$ billion; premium in basis points; the share μ of the Federal Reserve deposit share in parentheses.

Reserve mix	Demand	Stock	Bill share	Premium	Cap	Fed deposits (μ)	DW ready	Deposits
fixed (observed)	\$1tn	1,074	0.95	112	-3.0	+1.3 (0.5)	+1.3	+1.2
	\$2tn	no normal equilibrium (beyond the fold)						
chosen, $\eta = 0.35$	\$2tn, $\varepsilon_S = 2$	2,244	0.95	204	-81.3	+15.2 (1.0)	+12.3	+11.3
	\$1tn	1,058	0.85	88	-3.1	+0.7 (0.5)	+0.7	+0.6
	\$2tn	2,273	0.77	203	-86.6	+9.2 (0.5)	+8.0	+7.6
chosen, $\eta = 0.7$	\$2tn, $\varepsilon_S = 2$	2,228	0.78	191	-84.1	+11.9 (1.0)	+9.6	+8.1
	\$1tn	1,051	0.81	79	-3.1	+0.6 (0.5)	+0.6	+0.4
	\$2tn	2,258	0.63	193	-88.0	+7.7 (0.4)	+6.9	+6.0
	\$2tn, $\varepsilon_S = 2$	2,225	0.64	189	-85.5	+10.4 (1.0)	+8.4	+6.6

every instrument is evaluated at the issuer’s response to it, except the deposit-reserve and bonds-only rules, which fix the mix. The fold under these assumptions is in Table 5. For the coin holders’ alternative we write the opportunity cost as $r = y + w [(r_D - r_{D,0}) - (y - y_0)]$, with r_D the competitive household deposit rate $R(L) - \partial \mathcal{C}_2 / \partial D_2^H$ and the estimated semi-elasticity applied to r ; $w = 0$ is the baseline.

A finite willingness to pay. The logarithmic inverse demand gives the first coins an unbounded marginal valuation, which matters for a cap that rations many of them. Table G.4 caps the willingness to pay at 8 and 12 percent a year, two and three times today’s bill yield, the return at which holders would switch to the next-best payment service; equilibria do not change, because equilibrium opportunity costs lie far below the cap, and only the convenience value of the coins a cap removes does. At \$1 trillion the cap’s loss hardly changes. At \$2 trillion with $\varepsilon_S = 2$ it falls from \$81 billion to \$62–78 billion, and with $\varepsilon = 0.5$ from \$694 billion to \$67–107 billion; the cap remains the worst instrument. The Federal Reserve deposit share and discount-window readiness do not ration coins, and their gains are unchanged.

Table G.4: A finite willingness to pay. Large partner, fixed reserve mix; welfare gains over current rules, \$ billion a year, with the logarithmic inverse demand and with the willingness to pay capped at $r_{\max}$, two and three times today's bill yield.

Semi-elasticity	Demand	Cap			Fed deposits	
		log	$r_{\max} = 2r_0$	$r_{\max} = 3r_0$	log	$r_{\max} = 2r_0$
$\varepsilon = 6.6$ (baseline)	\$1tn	-3.0	-3.0	-3.0	+1.3	+1.3
$\varepsilon = 6.6$ (baseline)	\$2tn, $\varepsilon_S = 2$	-81.3	-62.3	-77.9	+15.2	+15.2
$\varepsilon = 0.5$	\$1tn	-2.7	-2.4	-2.7	+3.0	+3.0
$\varepsilon = 0.5$	\$2tn, $\varepsilon_S = 2$	-694.2	-67.5	-106.5	+18.8	+18.8
$\varepsilon = 1.0$	\$1tn	-2.1	-2.1	-2.1	+2.9	+2.9
$\varepsilon = 1.0$	\$2tn, $\varepsilon_S = 2$	-360.8	-67.0	-103.8	+18.5	+18.5
$\varepsilon = 2.0$	\$1tn	-2.0	-2.0	-2.0	+2.6	+2.6
$\varepsilon = 2.0$	\$2tn, $\varepsilon_S = 2$	-194.6	-66.1	-98.5	+17.9	+17.9